Variational and Optimization-based Approaches for Event-Based Vision Applications

From unobserved modes to well-posed estimators

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Dissertation Defense ▪ Fall 2026

SLIDE 1

Title

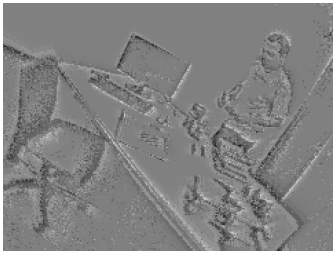
Thank you, Professor Antil, for your guidance throughout this work, and thank you, Professors Griva, Warma, and Löhner, for serving on my committee and for your consideration and feedback on this work.

My dissertation develops variational and optimization methods for event cameras, sensors that report only changes in brightness.

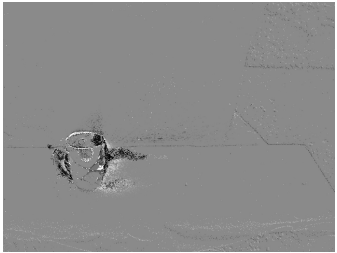
Let me start with what these methods produce.

[Advance.]

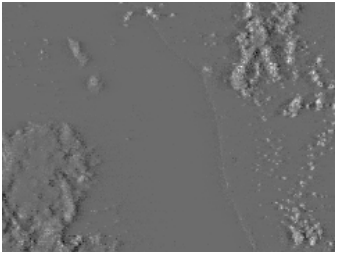
Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Office, handheld



Bullet hits a cup



Earth from the ISS (Falcon Neuro)

What can brightness changes reveal about images, motion, and depth—and at what computational cost?

SLIDE 2

Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit

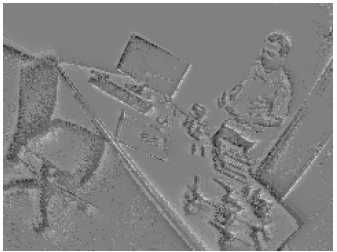
[Pause so the committee can look at all three panels.]

These three images were reconstructed from event-camera data alone, with no ordinary frames, by the method in my second chapter.

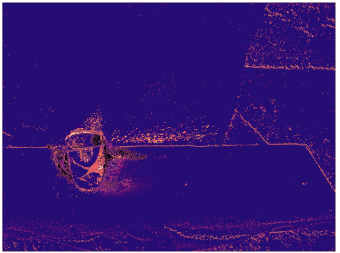
On the left is a handheld recording in an office; the desk, the chair, and the person are recognizable. In the middle, a bullet strikes a ceramic cup. The events resolve the fracture and the spray of fragments, which an ordinary camera at a standard frame rate would blur. On the right is the Earth seen from the International Space Station by the Falcon Neuro event cameras: the Honduran coastline and clouds, from a slowly changing scene with sparse events.

[Advance to the false-color version.]

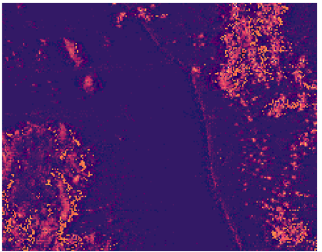
Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Office, handheld



Bullet hits a cup



Earth from the ISS (Falcon Neuro)

What can brightness changes reveal about images, motion, and depth—and at what computational cost?

SLIDE 3

Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit

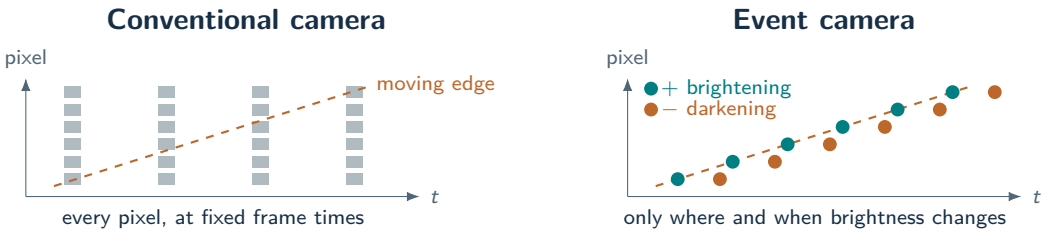
These are the same reconstructions of the cup and the ISS, now in false color; color and grayscale are only display choices.

Why does this matter? An event camera reports each pixel's brightness changes with microsecond timing and low power. That suits fast scenes, difficult lighting, and platforms where power and computation are scarce, such as drones and spacecraft like this one. On those platforms the estimator also has to be cheap and predictable.

So the dissertation asks: from changes alone, what can we recover, with what guarantees, and at what cost? I'll answer for images, motion, and depth.

[If asked whether the reconstruction runs in real time: slides 12 and 13, and Backup 07, page 32.]

Why an event camera? Each pixel reports its own changes, with microsecond timing and 140 dB range



- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].

SLIDE 4

Why an event camera? Each pixel reports its own changes, with microsecond timing and 140 dB range

[Trace the moving bar through the two diagrams.]

A conventional camera samples the whole image at fixed frame times. In the left diagram, each vertical column is one frame across all the pixels. The dashed diagonal line is an edge moving across the scene the pixels are viewing: time runs left to right, and the pixels are stacked vertically.

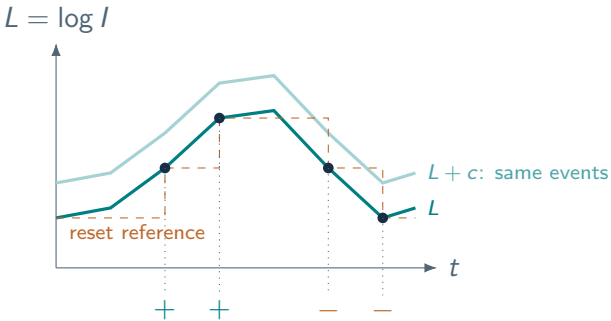
In an event camera, each pixel works on its own. When the leading edge of this bright bar reaches a pixel, the pixel gets brighter and fires a positive event. When the trailing edge passes, it gets darker and fires a negative event.

The output is a stream of locations, timestamps, and signs. That's why event displays tend to look like moving outlines. Under steady lighting, a pixel's brightness changes when a boundary crosses it. The inside of a uniform region produces little activity even while it moves, and so does a straight edge sliding along itself, because the local pattern stays the same. Changes in lighting trigger events too, since the sensor responds to any change in brightness over time.

Event sensors also have microsecond timing and high dynamic range, so they suit fast motion and difficult lighting.

For the mathematics, the key feature is that each pixel measures change relative to its own reference. Let me make that rule precise.

Each event marks one threshold of log-brightness change, so the baseline level stays free



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

SLIDE 5

Each event marks one threshold of log-brightness change, so the baseline level stays free

[Follow the solid curve and point to one positive and one negative event.]

The solid curve is the log intensity at one pixel. The dashed curve is the reference the pixel compares against.

When the log intensity rises one threshold above that reference, the pixel reports a positive event and resets its reference. A drop of one threshold gives a negative event. Every pixel resets on its own schedule, which is why event times are asynchronous.

The equation on the right states this ideal rule. An event carries a pixel location, a time, and a polarity, and the change since the last reset is the polarity times the threshold.

Now shift the whole log-intensity curve up by a constant. Every change stays the same, so the events stay the same. That's the first ambiguity in the talk: the events leave the baseline level free.

Real sensors add threshold variation and other errors. We use the ideal rule to motivate the observation model, and then pose reconstruction as a regularized fit to the data.

[If asked about the continuous-time version of the event rule: Backup 03, page 28. If asked whether the baseline is all the sensor loses: Backup 34, page 59.]

From events to images, motion, and depth: each estimate has its own unobserved quantity

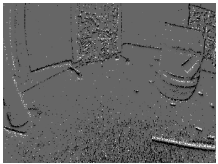
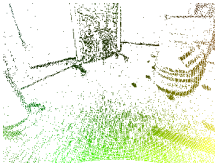


Image reconstruction (Ch. 2)

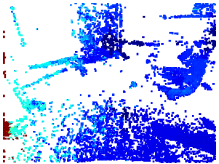
Unobserved: the brightness baseline



Optical flow (Ch. 3)

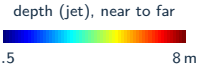
Unobserved: motion along an edge





Stereo depth (Ch. 4)

Unobserved: which points correspond



Data: MVSEC [35].

SLIDE 6

From events to images, motion, and depth: each estimate has its own unobserved quantity

[Indicate the three panels from left to right.]

These three estimates come from MVSEC, the Multi-Vehicle Stereo Event Camera dataset. It records a synchronized pair of event cameras on a drone flying indoors and on a car driving outdoors. The cameras also capture ordinary frames, which we use as reference images, and the dataset includes lidar-derived ground-truth depth, which we'll use for stereo. All three estimates here come from one indoor flight, at nearly the same moment. Each method uses its own observation window, and the depth estimate uses two cameras.

On the left is an image reconstructed by the temporal model. In the middle is optical flow, the apparent motion in the image plane. Color shows motion direction and magnitude over the observation window; white marks pixels with no events. For example, the yellow-green floor at the bottom is moving down the image. On the right is depth from the two event cameras, with the color scale running from half a meter to eight meters.

Each problem has its own difficulty. For reconstruction, the brightness baseline is unknown. For flow, a local edge reveals the motion across it and leaves the motion along it free. For stereo, we have to decide which points in the two views are the same point in the scene.

Here's what the talk shows for each of the three.

[If asked which methods produced these panels: a B-spline temporal kernel, contrast maximization, and weighted least squares. Contrast maximization has a nonconvex objective outside the quadratic flow family analyzed later; its existence result is Backup 46, page 71. If asked about the spline-parameterized flow regularizer under refinement: Backup 68, page 93. If asked what the stereo method is, in one slide: Backup 48, page 73. If asked about the quality of MVSEC's ground truth: Backup 60, page 85, for depth; Backup 26, page 51, for flow.]

Three estimators, each well posed

	Reconstruction (Ch. 2)	**Optical flow** (Ch. 3)	**Stereo depth** (Ch. 4)
Model supplies	the baseline, through the penalty	the motion along edges, through smoothness	the match, through one temporal profile shared by both views
Guarantee	unique and stable; one tridiagonal solve per pixel	unique when edges span two directions	unique inner solve at each candidate depth

The common question: which quantities do the events determine, and which does the model supply?

SLIDE 7

Three estimators, each well posed

[Move down each column.]

Each column answers the open quantity from the last slide.

For reconstruction, the penalty supplies the brightness baseline. Each pixel's history comes from one tridiagonal linear solve, and the estimate is unique and stable for every choice of positive weights. That's the largest part of the talk, including what it costs to compute.

For optical flow, smoothness supplies the motion along each edge. Three different ways of turning events into motion constraints fit into one quadratic variational problem, and the flow is unique as soon as the scene has edges in two different directions.

For stereo depth, the model supplies the match through one temporal profile shared by both views. The inner problem at each candidate depth has a unique solution, and on the MVSEC benchmark the method is competitive in accuracy and scores the most points.

The common question behind all three: which quantities do the events determine, and which does the model supply? Let's start with a single pixel.

[If asked which prior work this builds on: Backup 72, page 97, and Backup 32, page 57. If asked what is mathematically new: Backup 31, page 56.]

Events fix each pixel's brightness increments and leave one constant free

Observation at one pixel

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t),$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

Increments of the unknown history

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix},$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}, \qquad \ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

SLIDE 8

Events fix each pixel's brightness increments and leave one constant free

Here's the temporal model in scene terms. The unknown is each pixel's log-brightness history over the recording. The events tell us how much that brightness rose or fell in each interval. Each pixel is reconstructed on its own, and the model picks the reference level that the events leave open. A pixel with no events stays at that reference level, however bright it really is.

Now the details. Fix a pixel and take its events over a time window. At each event time, place a kernel and multiply it by the polarity. Adding these up gives the represented temporal signal on the left.

Then we integrate over the reconstruction intervals. Each entry of y is the signed event contribution in one interval. The algorithm chooses the reconstruction times.

[Point to the first row of A.]

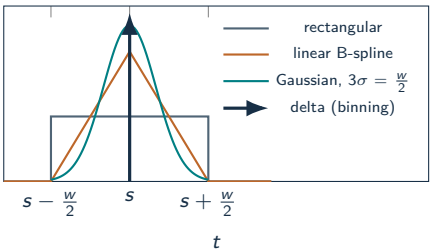
The matrix A takes differences of consecutive samples of the unknown profile. Its first row subtracts the first sample from the second, and each later row does the same one step further on. So A f should approximately equal y. These are plain increments; dividing by the time step would turn them into difference quotients.

If A f is zero, every consecutive pair is equal, so f is constant. And every constant vector gives zero differences. So the null space is exactly the constants.

In short, the data constrain the increments, and we need one more rule to choose the level.

[If asked whether the constant is all the information the sensor loses: Backup 34, page 59. If asked which choices fix the level: Backup 02, page 27; for a mean-zero formulation: Backup 17, page 42. If asked what space the model lives in: the theorems are in R^N, one coordinate per reconstruction time, with a weighted l two norm. The continuous-time version is posed in H one for a regular kernel, and in the space of bounded-variation functions in the delta limit, Backup 03, page 28.]

Each kernel spreads an event's sign over nearby bins and conserves its total



$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt,$$

$$\sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

SLIDE 9

Each kernel spreads an event's sign over nearby bins and conserves its total

[Point to the kernel curves, then to the bin-integral formula.]

The kernel controls how an event's contribution spreads over time. Delta binning puts the whole event into its own bin. The rectangular, spline, and truncated Gaussian kernels spread it over a short interval.

Delta binning is natural when we trust event timing at our reconstruction scale. Gaussian weighting is motivated by uncertainty concentrated near the timestamp, while rectangular weighting treats a bounded neighborhood equally. Splines provide local averaging with controlled smoothness. These interpretations motivate the alternatives; our experiments analyze which trade-off works better for the evaluated scenes.

Each kernel has unit mass, so an event contributes its full signed amount whenever its whole support lies inside the observation window. The sum on the right expresses that conservation across the bins. Events whose support crosses the edge of the window need special handling at the boundary.

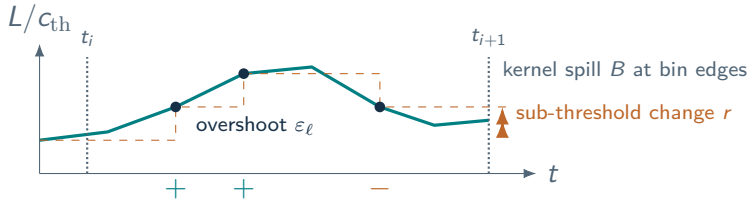
A wider kernel spreads the contribution further. That can reduce sensitivity to timing jitter, and it also blurs fast changes. The curves here share one width for illustration; the experiments use different widths for different kernels.

The difference matrix A stays the same. The kernel changes y, and through y it can also change the weights. That will matter when we get to conditioning.

Next: when does this integrated observation match a physical brightness change?

[If asked why the kernel changes conditioning when A stays fixed, or which kernel is best: Backup 05, page 30.]

The event sum matches the log-brightness change up to three sensor error terms



Exactly, for any times $a < b$ at one pixel:

$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill at bin edges}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a < s_\ell \le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}} .$$

SLIDE 10

The event sum matches the log-brightness change up to three sensor error terms

So far, y is something we compute from the events.

[Point to the diagram.]

In the ideal sensor, each event marks a change in log brightness of one contrast threshold. So if we add up the signed events between two reconstruction times, we get, approximately, the change in log brightness divided by the threshold.

[Point to the exact identity.]

Below the diagram is the exact version, from Chapter 2. Three terms separate the two sides. The boundary term, B, is kernel mass that spills across a bin boundary. Epsilon is the overshoot or threshold mismatch at each event; a missed or spurious event makes it as large as the threshold itself. And r is change since the pixel's last event that stayed below threshold, so the sensor never reported it. In the diagram, that's the gap at the right end.

When those terms are small, A f approximately equal to y is a model of brightness changes. The events measure change; the absolute level has to come from the model. Next is how the model chooses it.

[If asked about the continuous-time limit: Backup 03, the jump primitive. If asked whether sub-threshold signals can disappear entirely: Backup 34.]

A unique, stable estimate for every positive weight

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \tfrac{1}{2}\,\|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \tfrac{1}{2}\,\|\mathbf{\Lambda}\mathbf{f}\|^2, \qquad \mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succeq \lambda_{\mathrm{min}}^2\mathbf{I}.$$

Unique	$\mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}$ has one solution	
Stable	$\|\delta\mathbf{f}^*\| \le (2/\lambda_{\mathrm{min}}^2)\,\|\delta\mathbf{y}\|$	one spurious event moves a history by at most $2/\lambda_{\mathrm{min}}^2$
Reference level	$\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$	quiet intervals carry the largest weights and set the level

SLIDE 11

A unique, stable estimate for every positive weight

The objective has two terms. The first fits the reconstructed increments to y. The second penalizes the size of the profile, weighted by a positive diagonal matrix, Lambda, held fixed for the theorem.

[Point to the matrix inequality.]

Setting the gradient to zero gives the normal equations, with matrix A transpose A plus Lambda transpose Lambda. Its quadratic form is the squared norm of A z plus the squared norm of Lambda z, and the second term alone is at least the smallest weight squared times the squared norm of z. So the matrix is positive definite, and the minimizer is unique.

[Point to the stability line.]

Now compare two data vectors with the same weights. The difference of the solutions is the inverse matrix times A transpose times the difference of the data, and the norm of A is at most two. One spurious or missed event changes the data by at most one in norm, because each kernel splits a single unit of polarity across the bins. So one noise event moves a pixel's history by at most two over the smallest weight squared.

[Point to the reference-level identity.]

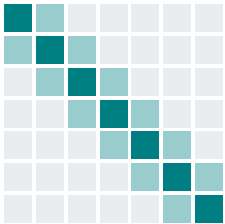
Take the inner product of the normal equations with the constant vector. A sends constants to zero, so the weighted average of the solution is zero. In scene terms, the quiet intervals carry the largest weights, and they set each pixel's reference level. The penalty also adjusts the fitted increments, because the solution balances both terms.

In the experiments, the weights come from the events: a sigmoid of each interval's net activity, so quiet intervals get large weights and busy intervals small ones, with a positive floor.

Next: what this solve costs.

[If asked for the derivation or the condition-number bound: Backup 74, page 99. If asked how the weights were chosen: Backup 75, page 100. If asked whether every weight must be positive: Backup 02, page 27. If asked how lambda should scale when the time grid is refined: Backup 37, page 62. If asked about a mean-zero formulation: Backup 17, page 42. If asked how close the estimate is to the true signal: Backup 33, page 58. If asked about recomputed weights: Backup 65, page 90.]

Each pixel costs one tridiagonal sweep; streaming adds constant work per new frame



$$\mathbf{K}_{\text{time}} = \mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda}$$

diagonal $(1, 2, \ldots, 2, 1) + \lambda_i^2$

off-diagonals -1

$O(N)$ per pixel	Thomas sweep, no pivoting; tridiagonal and diagonally dominant
Independent pixels	a pixel without events returns $\mathbf{f} = 0$
Newest frame	the forward sweep alone gives it: constant work per pixel for each new interval

MVSEC indoor flying 1: dense solve 1.5 s → tridiagonal 0.03 s; agrees with MATLAB within 10^{-15}.

SLIDE 12

Each pixel costs one tridiagonal sweep; streaming adds constant work per new frame

At the pre-defense, the committee asked what this costs, and whether it could run on small, low-power hardware. The structure of the matrix answers the cost question.

[Point to the matrix.]

A takes consecutive differences, so A transpose A has twos on the diagonal, ones at the ends, and minus ones beside the diagonal. Adding the squared positive weights preserves that structure. One forward sweep and one backward sweep solve each pixel's system in time proportional to the number of samples. Across the image, that is O(H W N_t). A pixel whose whole observation vector is zero has the exact zero solution and can skip the solve.

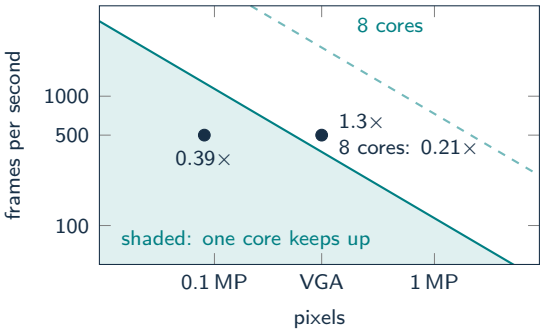
If only the newest frame is needed, the forward recurrence returns the endpoint of the batch solution on the history accumulated so far. It keeps three recurrence numbers plus an output value per pixel. Each new interval costs constant work per pixel; older frames would need backward substitution to be revised.

[Point to the bottom line.]

On an MVSEC recording, the tridiagonal solve takes three hundredths of a second where the dense solve takes one and a half, and it agrees with the MATLAB reference to rounding error.

Next, what that structure costs in time.

One CPU core streams about 114 million pixel updates per second, with 16 bytes of state per pixel



$$\text{pixels} \times \text{frames/s} \;\leq\; 1.14 \times 10^{8} \text{ per core}$$

8.8 ns per pixel update; 16 B of state per pixel

Batch work follows scene activity: skipping pixels without events cuts it 10× at 6.8% active (0.43 → 0.04)

Points: two recordings, a frame every 2 ms, float32; labels give processing time ÷ recording time. 8-core line: measured at VGA.

SLIDE 13

One CPU core streams about 114 million pixel updates per second, with 16 bytes of state per pixel

[Point to the plot.]

Because each streaming step does constant work per pixel, the cost comes down to one rate. On one CPU core, the closest stand-in for a small board, the method performs about 114 million pixel updates per second, just under nine nanoseconds each, and keeps sixteen bytes of state per pixel.

The two dots are two recordings, each producing a new frame every two milliseconds. They differ in resolution and by a factor of nine in event rate, and they give the same rate to within a third of a percent. The labels are processing time divided by recording time, so anything below one keeps up. The 346 by 260 recording streams at 0.39 of real time; the 640 by 480 recording streams at 1.3 on one core and 0.21 on eight.

So for any sensor, the budget is pixels times frames per second, below 114 million per core. At 500 frames per second that's about 229 thousand pixels per core; at 100 frames per second, about 1.1 million.

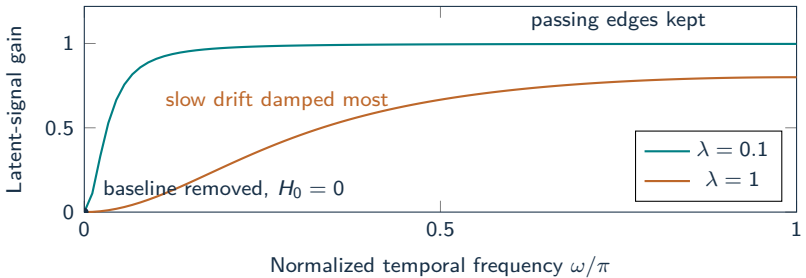
Batch reconstruction also follows the scene: in a typical window only about seven percent of pixels have events, and skipping the rest cuts the work tenfold.

[For multithread timings: Backup 07, page 32. For problem size, memory, and low-power strategies: Backup 83, page 108.]

The penalty removes the baseline and damps slow drift most; passing edges remain

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\text{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



SLIDE 14

The penalty removes the baseline and damps slow drift most; passing edges remain

To interpret the penalty, take a single scalar weight, so Lambda is lambda times the identity. And suppose the observations are exact differences of a latent signal.

The eigenvectors of A transpose A are temporal modes. In that basis, reconstruction multiplies each mode of the latent signal by g over g plus lambda squared, where g is the mode's eigenvalue.

[Point first to zero frequency, then along the two curves.]

The constant mode has eigenvalue zero, so reconstruction removes it entirely. Among the remaining modes, the slowest are damped the most. In the images, slow brightness drift fades and passing edges remain, which is why the reconstructions look edge-enhanced. A larger lambda damps more.

That's what the penalty does to the latent signal. Kernel averaging is a separate step, and it can blur fast changes before the solve.

The curves connect the finite-grid gains so the trend is visible. A general positive diagonal still gives the uniqueness theorem, but its effect on the modes needs a fuller matrix analysis.

Next, how these reconstructions compare with the bilevel approach on MVSEC.

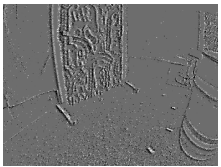
[If asked what the penalty does to measurement noise: Backup 04, page 29. If asked about bias in the damped modes: Backup 33, page 58. If asked what happens as lambda goes to zero: Backup 69, page 94.]

Events alone give an inner system about 1,500 times better conditioned than frame-assisted bilevel

MVSEC [35] indoor flying 2



Reference



Event only

	Bilevel [2]	Events
Inputs	events + frames	events
PSNR indoor (dB)	15.37	10.85
PSNR outdoor (dB)	8.41	8.71
CN indoor	39755	25.51
CN outdoor	39755	27.10

SLIDE 15

Events alone give an inner system about 1,500 times better conditioned than frame-assisted bilevel

[Point to the two images, then to the table.]

On the left is an indoor frame from MVSEC beside the event-only reconstruction. The structure of the scene is there; the overall brightness is the level the model selected, since the events constrain only changes.

On the right is the relation to the Antil–Sayre bilevel approach. In this chapter, the events set activity-based weights, and then we do one fixed convex solve over time. The bilevel approach uses events together with intensity frames, and its outer optimization adjusts z, a parameter of the event-based blur kernel that links each blurry frame to the sharp latent image. So their outer variable shapes how events explain the frames, while my rule chooses the magnitude weights of the temporal profile.

The table compares complete methods with different inputs. Indoors, the bilevel approach, which has the frames, reaches higher PSNR; outdoors, the event-only approach is slightly higher. PSNR compares with a reference frame, so it also scores the absolute brightness, which the events leave to the model. The condition numbers are for each method's inner linear system: about forty thousand for the bilevel lower level with its uniform penalty, and about twenty-five to twenty-seven with the activity-based weights.

The guarantees answer different questions too. The bilevel analysis proves existence, and local uniqueness under additional conditions; the result I presented gives a unique global minimizer for every fixed positive diagonal.

Last in this chapter: what changes when the pixels share one scene.

[If asked how the kernels compare: Backup 76, page 101, and Backup 05, page 30. If asked how PSNR and condition number are computed: Backup 06, page 31. If asked which prior work this builds on: Backup 32, page 57.]

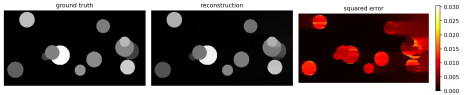
Known motion lets every view constrain one scene

Model

$$B_{\mathrm{SAR}} = (A \otimes I)\, P$$

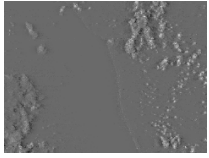
θ scene · P known motion · A differences between views

Simulated, known truth (v2e [46]); 1 px shift per view

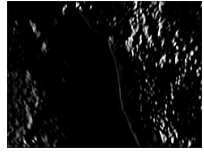


scene · shared-scene estimate · squared error, concentrated on rows constant along the motion

Falcon Neuro on the ISS [38], near-nadir pass



each pixel on its own



all views, one scene (ESAR [45])

SLIDE 16

Known motion lets every view constrain one scene

On the International Space Station, motion is the enemy of a frame camera. For an event camera with a known orbit, motion is what's measured.

The station moves at about seven point seven kilometers per second, so any exposure long enough to gather light smears the ground. An event pixel timestamps each change to the microsecond, covers about 140 decibels, and sends only changes, which suits a payload with little power. And because the orbit is known, every view sees the same ground through a known motion.

In joint work with Professor Antil and Dr. Sayre, called ESAR, we use exactly that: the same event constraints, now describing one static scene seen by a moving camera.

[Top lane: the operator equation.]

Theta is the scene. P holds the known motion: it maps the scene into each view. A takes differences between consecutive views, as before. The name borrows from synthetic aperture radar, which combines a moving sensor's views into one image.

We fit the scene to all the event counts at once: least squares plus a penalty, as for a single pixel. With a quadratic penalty, it's one linear system, and the dissertation proves the solution is unique and bounds how far noise can move it.

[Middle lane: the simulated disks.]

The simulated disks show the mathematics. The error concentrates along rows through the disks, because a pattern that is constant along the motion looks the same in every view. It's the same lesson as the baseline: find what the observations control, and let regularization supply the rest.

[Bottom lane: the orbital pair.]

This is a real Falcon Neuro pass looking straight down. On the left, each pixel is reconstructed on its own, and the structure is faint and noisy. On the right, every view constrains one shared scene, and the coastline and the land texture come out. The motion that blurs a frame camera gives the event camera more views of the same ground.

Here the motion was known. In optical flow, the motion is the unknown.

[If asked what regularization supplies when the observations leave directions free: Backup 64, page 89. If asked about related prior work: Backup 32, page 57.]

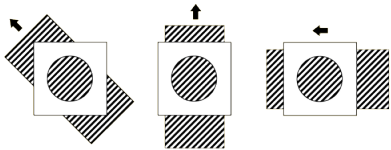
Three different motions look identical through a small window

Through the windows



The visible stripe motions look the same.

Reveal the full patterns



The patterns translate in different directions.

Publisher animations, unchanged: Wolfe et al. [109], Activity 8.2, Movies 5–6.

SLIDE 17

Three different motions look identical through a small window

A drone that stabilizes itself, dodges an obstacle, or tracks its own motion needs to know how the image is moving, right now. A frame camera infers motion by comparing two pictures. At thirty or sixty frames a second, a fast edge jumps many pixels between them and blurs within them, and the answer arrives a frame late. An event camera timestamps the edge as it crosses each pixel, to the microsecond. Those crossing times are the motion measurement: their slope gives the edge's speed. The catch is that a single edge reveals only part of its motion.

Optical flow is the apparent velocity in the image plane. A local view can leave part of that velocity free. This is the aperture problem; an aperture here just means a limited window.

[Play the embedded animation in the left panel.]

Look at the stripes through these three circular windows. They all seem to move the same way. Each window shows only the inside of a straight striped pattern, so all we can see is how fast the stripes move across their own edges.

[Play the embedded animation in the right panel. Pause.]

With the full patterns, we see three different motions: diagonal, upward, and leftward. All three share the same motion perpendicular to the stripes. They differ in the motion along the stripes, and the window hides that part.

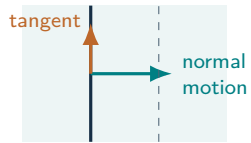
The animations show the geometry, and the same ambiguity holds for an event camera. Under steady lighting, these ideal patterns produce the same brightness changes inside each window, and so the same events, given the same thresholds and starting references.

Next, I'll draw this ambiguity for a single edge and state what the flow model assumes.

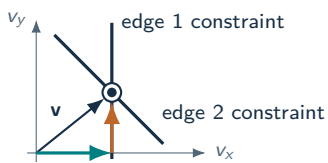
[If asked to derive the aperture problem from the event-generation model: Backup 63, page 88. If asked when the data do determine the motion: Backup 35, page 60.]

One edge fixes its normal speed; a second orientation fixes the rest

Image: one moving edge



Velocity: what the events allow



normal: fixed by edge 1 tangent: fixed by edge 2, if it shares the motion

SLIDE 18

One edge fixes its normal speed; a second orientation fixes the rest

[Point to the left diagram: the edge, the teal arrow, then the amber arrow.]

First, what the camera sees. An edge in the scene is where the light changes sharply: a dark object's outline against a bright wall, or a painted stripe. As this edge sweeps over a pixel, the pixel fires, so the firing times trace where the edge is.

The teal arrow is the edge's motion perpendicular to itself; the dashed line is where it will be a moment later. The amber arrow is motion along the edge. Sliding a straight edge along itself, with its ends out of view, leaves every pixel's brightness unchanged, so the events fix the teal part and leave the amber part free.

[Point to the right diagram.]

The right panel plots possible velocities. Each line is every velocity that fits one edge's events. Edge 1 is vertical, so its events fix the horizontal speed, the teal arrow, and leave the vertical speed free.

A second edge, elsewhere in the image and oriented differently, gives the diagonal line. If both edges move with the same velocity, the crossing picks it out: the ringed dot, v. It's the teal part edge 1 measured, plus the amber part edge 2 fixed.

The two edges can be anywhere in the window. The model assumes neighboring pixels move alike, and that carries each edge's information across the image. The crossing is the true velocity when the edges really move together: two sides of one object translating across the view, or a camera pan. That's a modeling assumption. Non-parallel lines always cross, so if the edges belong to objects moving differently, the model still returns a velocity, and it's wrong for both.

So the flow model is: the unknown is the image velocity at every pixel; events give each edge's perpendicular speed; neighbors moving alike fill in the rest. The three representations differ in how they get the edge measurement: crossing times, transported event activity, or gradients pooled over a small window.

The condition to remember is edges in two different orientations, anywhere in the recording: their constraint lines cross, and the crossing fixes the velocity. On slide 20 it decides whether the answer is unique. Next, the three event representations that supply each edge's measurement.

[If asked whether two orientations anywhere suffice: slide 20; necessity, Backup 35, page 60. If asked what links distant edges: with exact data and one shared velocity, the constant field fits every edge and has no gradient, so it is the unique minimizer at any alpha and separation. If objects move differently but each has a corner or texture: the events contradict one velocity, and smoothing blends the motions across the boundary, Backup 27, page 52, and Backup 70, page 95.]

Three representations, one quadratic variational problem

$$E_{\mathbf{B},b}(\mathbf{v}) = \underbrace{\frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v})}_{\text{directions the edges measure}} + \underbrace{\frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2}_{\text{neighbors move alike}} .$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s\nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

SLIDE 19

Three representations, one quadratic variational problem

[Point to the energy, then briefly indicate the three table rows.]

This keeps the classical data-plus-smoothness structure, with the event representation supplying the coefficients.

The first integral measures agreement with the motion constraints. B is the data matrix: it records which motion directions the local edges measure, and how strongly. The forcing b carries the measured values. The second integral penalizes spatial variation of the velocity, with positive strength alpha squared.

For the time surface, B is the outer product of the time-surface gradient, from following an edge's crossing times. For voxel projection, it's the outer product of the activity gradient, from transporting event activity with the scene. For the structure tensor, it's the pooled spatial block, from gradients over a small neighborhood.

The natural space is H one on the image domain, for both velocity components: square-integrable velocities with square-integrable weak derivatives.

The advantage is that we prove one result for a fixed data matrix B and forcing b, and then apply it to each representation once we've checked its hypotheses.

The question now is whether the data and the smoothness term, together, control every direction in that space.

[If asked for the derivations: time surface, Backup 77, page 102; voxel projection, Backup 78, page 103; structure tensor, Backup 79, page 104. If asked why H one rather than total variation: Backup 27, page 52. If asked which boundary condition the model uses: Backup 08, page 33. If asked about direction-dependent smoothing: Backup 13, page 38, and Backup 14, page 39. If asked how the model is solved on a grid: Backup 11, page 36, and Backup 12, page 37.]

Integrated directional information gives a unique flow in H^1

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

Nonparallel edges control constant velocities; smoothness supplies motion where $\mathbf{B} = 0$.

SLIDE 20

Integrated directional information gives a unique flow in H^1

This result specializes the classical well-posedness framework to the event-derived coefficients, under these hypotheses.

[Explain each hypothesis by its role.]

The domain is bounded, connected, and Lipschitz, and the smoothness weight is positive. The data matrix B is bounded, symmetric, and positive semidefinite, and the forcing b is square-integrable.

The key assumption is that the integral of B over the domain is positive definite. Under these hypotheses, the energy has a unique minimizer in H one.

What does that integrated condition mean? Take any nonzero constant velocity. Summed over the domain, the observations must give positive energy in that direction. In scene terms, the recording has edges in at least two orientations, possibly at different pixels: the two crossing lines from the velocity diagram.

This is the requirement from slide 18: the one hypothesis the scene has to supply, while the others come from how we set up the problem. If every edge is parallel, as inside a stripe pattern, a uniform slide along the stripes costs nothing, and many answers fit equally well. The theorem promises one answer; whether it's the true motion depends on slide 18's assumption that neighbors move alike, and slide 21 measures how close it comes.

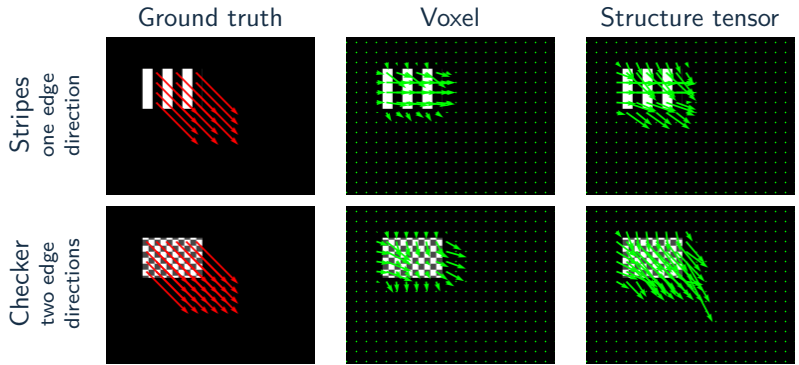
Individual locations may give only one direction. Regions with no texture or no events give zero data curvature, and their neighbors supply the motion there.

Connectedness matters because it makes any field with zero gradient a single constant vector over the whole domain. The integrated matrix controls that remaining freedom.

The proof follows the picture. Split the velocity into its mean and a zero-mean fluctuation. Smoothness controls the fluctuation, by Poincaré's inequality; the integrated data matrix controls the mean, the uniform drift that smoothness can't see. Together they give coercivity, and Lax–Milgram gives the unique minimizer. Next, two examples where we know the true motion.

[If asked for the proof: the idea is Backup 80, page 105; the full proof is Backup 09, page 34. If asked what the estimate does where there are no events: it is the harmonic fill-in of the surrounding motion, Backup 70, page 95. If asked how to check the condition on data: the discrete version is the smallest eigenvalue of the two-by-two sum of sampled data blocks, Backup 11, page 36; its size is the margin m. If asked about a disconnected domain: each component carries its own constant drift and needs its own integrated condition. If asked whether the integrated condition is necessary: Backup 35, page 60. If asked whether the convergence proof covers the solvers used in the experiments: Backup 12, page 37. If asked what happens as alpha goes to zero or grows large: Backup 69, page 94, and Backup 38, page 63.]

Two edge directions (checkerboard) resolve more of the motion than one (stripes)



SLIDE 21

Two edge directions (checkerboard) resolve more of the motion than one (stripes)

[Give the audience a moment to compare the two rows.]

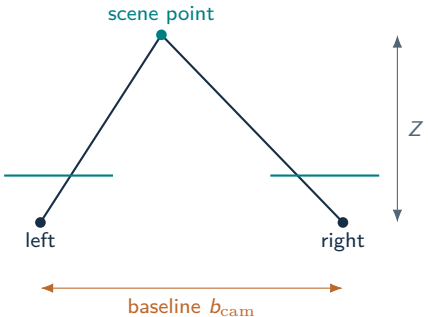
These selected simulation windows illustrate the directional-information argument, using a known diagonal motion. The top row has a stripe texture, and the bottom row a checkerboard. The true-motion arrows and the estimates share a common scale.

Inside a vertical stripe region, the strong gradients are horizontal, so they mainly constrain the normal motion. In the theorem's terms, this texture gives the integrated matrix little energy along the stripes. In the velocity diagram from earlier, the stripe interior gives only one line. The checkerboard has edges along both axes, so its lines cross, which reduces the aperture ambiguity. Object boundaries can add directional information as well. How accurate the estimate is also depends on the representation and the regularization.

Last, a short summary of stereo depth.

[If asked about other methods: the contrast-maximization comparison on these scenes is Backup 43; "why analyze the quadratic family" is Backup 45. If asked how the examples were selected and scored: Backup 44, page 69. If asked about real-data flow: Backup 26, page 51. If asked whether contrast maximization has a global minimizer: Backup 46, page 71. If asked whether the spline regularizer loses coercivity under refinement: Backup 68, page 93.]

Two event cameras match by timing: one shared temporal profile per candidate depth



$$Z = \frac{f_{\mathrm{cam}}\, b_{\mathrm{cam}}}{d}, \qquad d = x_L - x_R > 0.$$

For fixed d and frozen weights, the inner solve is unique.

SLIDE 22

Two event cameras match by timing: one shared temporal profile per candidate depth

I'll close the technical part with stereo depth.

A vehicle needs metric depth: how far away the obstacle is. One camera leaves the scale free; two cameras fix it. Frame stereo matches brightness patterns, and blur, glare, and darkness corrupt those patterns. Two event cameras watching the same edge see it change at the same moment, so time becomes the matching signal.

[Point to the diagram.]

Two calibrated cameras see the same scene point. After rectification the image rows line up, so matching is a search along one row, and the horizontal offset is the disparity. By similar triangles, depth is the focal length times the baseline, divided by the disparity.

For each pixel and each candidate disparity, the model fits a temporal profile to each view, ties the two together softly, and scores how well one shared explanation fits both. We call it WLS, for weighted least squares. The weights treat busy intervals as noisier, following a Poisson count model; they're updated between solves and frozen within each one. For a fixed candidate with frozen weights, the inner problem is the temporal argument with one more nonnegative term, so its solution is unique. Choosing the disparity is a separate comparison across candidates, after semi-global matching aggregates the costs.

The challenge is that only edges fire, so the hard part is how much of the scene gets a depth at all. That is where WLS leads.

[If asked why the inner solve is unique: Backup 51, page 76. If asked whether a unique solve gives the right disparity: Backup 54, page 79; the conditions that do are Backup 55, page 80, and Backup 56, page 81. If asked whether IRLS converges: Backup 59, page 84. If asked why the weights treat busy intervals as noisier: Backup 58, page 83. For the full stereo argument: Backup 47, page 72; in one slide, Backup 48, page 73.]

WLS scores the most points, 2.66M, at competitive depth error

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	**2.66**
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20



MVSEC [35] indoor flying 1–4; each method scored on its own estimates in $[0.5, 8]$ m.
Three more methods in the backup table.

SLIDE 23

WLS scores the most points, 2.66M, at competitive depth error

[Point to the table, then the bars.]

On the MVSEC indoor flights, against lidar-based ground truth, WLS has the lowest error on flight two and the lowest log RMSE. The adaptation of Piatkowska and colleagues' method is lower on the other three flights, by one and a half to six centimeters. WLS scores two point six six million points, the most of the five methods and more than twice Piatkowska's. Each method is scored on its own valid pixels, so read accuracy and coverage together.

[If asked for the full table: Backup 81, page 106. If asked whether the comparison is fair: Backup 60, page 85.]

Each next step starts from a result in the dissertation

Sensor modeling

From: the error terms of slide 10

Measure them on real sensors; the stability bounds turn them into estimate bounds.

Learned weights

From: a smooth solution map

Learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{f}^*(\mathbf{\Lambda})$ is smooth.

Shared scenes

From: the null space of B_{SAR}

Find its motion-invariant modes for a trajectory (ESAR [45]); apply to orbital streams.

Edge hardware

From: the tridiagonal structure

Update only pixels with events; measure time and power on a Raspberry Pi 5 and edge sensors.

SLIDE 24

Each next step starts from a result in the dissertation

[Move through the four boxes, left to right.]

First, event-sensor modeling. Slide ten separated the measured increments from brightness changes by three error terms. Measuring those terms on a real sensor, together with timestamp and binning error, would let the stability bounds translate sensor error into estimate error. The GMU Neuromorphic Imaging and Digital Twins Lab makes those controlled acquisitions possible.

Second, learned temporal regularization, as a possible extension. When all the weights are positive, the fixed-weight solution is a smooth function of those weights. That gives a well-defined lower-level map for an outer learning problem, in the spirit of Antil, Di, and Khatri.

Third, shared-scene and orbital sensing. For a given trajectory, which scene patterns does the motion leave invisible? The null space of B sub SAR answers that question, and sparse orbital streams like the ISS data are a natural application.

Fourth, the edge implementation. The streaming form already updates each pixel in constant time. The next steps are updating only the pixels that receive events, and measuring time and power on a Raspberry Pi 5 and with sensors designed for the edge.

Each direction starts from something the dissertation already controls: an observation error, a solution map, a null space, or the structure of the matrix.

[If asked whether the learned-weight problem is well posed: Backup 67, page 92. If asked how the results extend to other regularizers: Backup 64, page 89, and Backup 27, page 52. If asked how the parameters are chosen now: Backup 66, page 91. If asked how sensor error reaches the estimate: Backup 36, page 61, and Backup 34, page 59.]

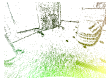
What the data constrain and what the models prove



Reconstruction

Events constrain: brightness increments

Model proves: a unique history for each fixed positive weight matrix, with perturbation bounds



Optical flow

Events constrain: directional motion constraints

Model proves: a unique H^1 minimizer under integrated identifiability



Stereo depth

Events constrain: paired temporal event features

Model proves: a unique frozen-weight inner solve; a separate comparison of candidates selects the disparity

Takeaway. Under the stated hypotheses, regularization makes each estimator determinate; accuracy also depends on the representation.

Which directions do the observations control, and which are selected by the model?

Thank you.

SLIDE 25

What the data constrain and what the models prove

[Read down each column: what the events constrain, then what the model proves.]

For temporal reconstruction, the events constrain brightness increments. For each fixed positive weight matrix, the regularized history is unique, with perturbation bounds, and the penalty selects the baseline.

[If asked about recomputed weights: Backup 65, page 90, gives a local Lipschitz bound beyond the dissertation. If asked how the theorems relate to the complete stereo estimator: Backup 30, page 55. If asked what is mathematically new: Backup 31, page 56, and Backup 64, page 89. If asked how close a stable estimate is to the truth: Backup 33, page 58. If asked what the events lose beyond the baseline: Backup 34, page 59. If asked whether the flow result is stable in the raw events: Backup 36, page 61. If asked whether a unique inner solve gives the right disparity: Backup 54, page 79.]

For flow, the events give directional constraints on the motion. Smoothness and integrated directional information together give a unique H one minimizer.

For stereo, the two cameras give paired event-feature profiles. At a fixed candidate disparity, with the weights frozen, the inner solve is unique. Choosing the disparity is a separate comparison across candidates.

The proof tools are classical. The contribution is the analysis of these event-derived models under explicit hypotheses.

Across all three, the common question is which directions the observations control, and which the model chooses.

Thank you. I'm happy to take your questions.

[Stop. Leave the closing slide visible. For questions: the discussion guide is Backup 01, page 26; flow detail starts at Backup 41, page 66; the stereo arc at Backup 47, page 72; references at page 109.]

Discussion guide

Further detail: time-surface regression and contrast maximization; the full stereo arc (Chapter 4); references.

Observation and temporal inversion

- Minimal uniqueness condition / reference level
- Measure-valued continuous-time limit
- Signal gain versus noise gain
- Metrics; real time; problem size
- Kernels; $\mathbf{\Lambda(y)}$ stability

Flow analysis

- Weak Euler equation and boundary condition
- Full proof, stability, and discrete sweeps
- Exact block-Jacobi spectral argument
- Anisotropy; $\lambda, \alpha \to 0$; event-free regions
- Regression, projection, evaluation; splines
- Straight-edge aperture model

Stereo analysis

- Chapter 4 results summary
- Mean-zero conditioning
- Cost ellipsoid and observation perturbations
- IRLS contraction proof
- Fixed points versus likelihood stationarity
- Implemented scoring and aggregation

Evidence and interpretation

- Depth coverage; alignment; sensitivity
- One lemma behind the uniqueness results
- Contribution; parameters; learning
- Literature anchors and source map

BACKUP 01 / DECK 26

Discussion guide

[Navigation only. Choose the relevant technical question from this index; no spoken introduction is needed.]

Must every temporal penalty entry be positive?

For arbitrary $\boldsymbol{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \boldsymbol{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\boldsymbol{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. Any choice of reference level resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

BACKUP 02 / DECK 27

Must every temporal penalty entry be positive?

Likely question: Do you really need a positive penalty at every temporal sample?

[Point first to the kernel intersection, then to the connected-chain null space.]

Answer

No. The requirement is weaker: the observation operator and the penalty must not share a nonzero null vector. On the slide, that's the kernel intersection. Here A is the first-difference operator and Lambda is the penalty. The quadratic form of the reconstruction matrix is the squared norm of A times a vector, plus the squared norm of Lambda times that vector.

On a connected temporal chain, the null space of A is the constant profiles. So a single nonzero penalty entry is enough to remove the ambiguity. Fixing one reference value, or requiring zero mean, also picks out a unique answer. We use a fully positive diagonal in the theorem because it gives a simple quantitative lower bound along with uniqueness. A measured reference would go further: it would supply actual physical information about the brightness baseline.

If asked for more

Selecting the constant is different from regularizing the whole profile. A zero-mean constraint fixes only the unobserved offset. A magnitude penalty also damps the nonconstant modes and changes how they respond to noise. With a single anchoring entry, the matrix is still positive definite, but the lower bound based on the smallest diagonal entry has to be replaced by a bound for that anchored matrix.

The event measure has a jump primitive

At one pixel, set $\mu = \sum_j p_j \delta_{s_j}$, and choose a consistent right-continuous primitive

$$g(t) = f_0 + \mu((t_0, t]), \qquad g(b) - g(a) = \mu((a, b]).$$

For a standard smooth compactly supported unit-mass kernel,

$$e_h = \mu * \phi_h, \qquad g_h(t) = f_0 + \int_{t_0}^{t} e_h(\xi)\, d\xi.$$

For finitely many interior events, $g_h \to g$ in L^1 as $h \downarrow 0$.

- g belongs to BV; a nonzero interior jump prevents $g \in H^1$.
- The displayed g is an unregularized cumulative-event surrogate.

Takeaway. Keep interval conventions consistent and use increments in $\mathbf{Af} \approx \mathbf{y}$.

Dissertation: Continuous-time formulation (p. 21); endpoint convention made explicit here.

BACKUP 03 / DECK 28

The event measure has a jump primitive

Likely question: What is the precise continuous-time interpretation of accumulating events?

[Indicate the right-closed interval convention, then the two primitives.]

Answer

At one pixel, we write the events as a signed measure: a point mass at each timestamp, weighted by its polarity. Its cumulative primitive is a step function. Start from a chosen initial value, and add the signed mass of the events up to the current time. Counting the right endpoint of each interval makes the primitive right-continuous, and its increments then match the event measure exactly.

With finitely many events, this primitive has bounded variation. Smoothing the measure with a standard kernel gives a regular temporal driver, and integrating that driver gives a smooth approximation of the staircase. As the kernel width shrinks, these smooth primitives converge in L one, for finitely many interior events. That describes the cumulative-event surrogate. The regularized reconstruction then makes its own, separate choice about fitting those increments and controlling the size of the profile.

If asked for more

A nonzero jump has a singular distributional derivative, so the unsmoothed primitive naturally lives in bounded variation, outside H one. At any positive kernel width, the kernels here give an L two driver and an H one primitive. On the discrete side, the first-difference matrix computes increments; dividing by the time step would instead approximate a derivative. Keeping that scaling and the interval endpoints consistent is essential when comparing the continuous and discrete descriptions. A joint limit in the grid and the regularization would need its own argument, beyond convergence of these primitives.

Latent-signal gain and observation-noise gain are different

For the unweighted monocular model with scalar $\lambda > 0$,

$$\mathbf{f}^* = (\mathbf{A}^\top \mathbf{A} + \lambda^2 \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{y}.$$

Input	Output modal gain	Interpretation
Latent signal	$g_k/(g_k + \lambda^2)$	Low modes attenuated most
Difference observation	$\sqrt{g_k}/(g_k + \lambda^2)$	Noise amplitude transfer
White difference noise	$\sigma^2 g_k/(g_k + \lambda^2)^2$	Reconstructed modal variance

$$\max_{g \geq 0} \frac{g}{(g + \lambda^2)^2} = \frac{1}{4\lambda^2}.$$

Dissertation: Theorem 4.6.21, Proposition 4.6.23.

BACKUP 04 / DECK 29

Latent-signal gain and observation-noise gain are different

Likely question: What does regularization do to the signal and to measurement noise?

[Contrast the first and second table rows before discussing the variance row.]

Answer

They're two related but different transfer functions. On this slide, the penalty is a single positive scalar, lambda, and the observations are unweighted. Let g be an eigenvalue of A transpose A.

First, the signal. Take an underlying temporal profile, apply A to get ideal differences, and reconstruct. Each profile mode comes back multiplied by g over g plus lambda squared. The low modes are damped most, and the constant mode is removed.

Second, the noise. If instead we perturb the difference observations, the amplitude gain is the square root of g, over g plus lambda squared. For independent white noise with variance sigma squared, squaring that gain gives the reconstructed variance in each mode.

If asked for more

The singular value decomposition makes this explicit. A maps the right singular modes of the profile to the left singular modes of the observations, and the inverse maps back with the gain shown here. Treating g as continuous, the variance factor peaks at g equal to lambda squared, where it equals one over four lambda squared. On a finite chain the maximum can be smaller, because the eigenvalues are discrete. These exact formulas hold for the scalar, unweighted model; general diagonal or adaptive weights need the corresponding matrix analysis.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Conditioning and reconstruction bias are separate properties.
- Reference-image scores also measure the baseline the model selects.

Dissertation: Table 2.4, Table 2.5.

BACKUP 05 / DECK 30

Kernel performance depends on the scene and adaptive weights

Likely question: Why does changing the temporal kernel change both accuracy and conditioning?

[Point across the indoor and outdoor columns to show the change in ranking.]

Answer

Because the kernel changes the data, and the adaptive penalty is computed from the data. The first-difference operator stays the same. But the represented observations change with the kernel, so the penalty entries can change too. That changes the reconstruction matrix, and with it the condition number.

The table also shows that the accuracy ranking depends on the scene. Indoors, the B-spline gives a PSNR of 8.86 with condition number 22.16; the Gaussian is very close, at 8.84, with condition number 20.22. Outdoors, delta binning has the highest PSNR of these four, 7.97, with condition number 53.34. So the representation should be chosen with the scene and the weighting rule in mind. Conditioning measures how sensitive the linear solve is; accuracy also depends on the data, and on the bias that regularization introduces.

If asked for more

If A and Lambda were both held fixed, changing only the right-hand side would leave the condition number unchanged. The positive-definiteness theorem applies to every fixed positive penalty this produces. It guarantees the solve, while the table compares actual reconstructions. Those are scored against reference images, and the events still leave the absolute temporal offset to the regularization or the reference convention.

What the reconstruction scores measure

Quantity	What it compares	What it establishes
PSNR / SSIM	Reconstruction and reference images	Image agreement under the scoring convention
TC / TG	Temporal changes in the output	Smoothness of the returned sequence
ERC	Magnitudes of reconstructed changes and represented events	Correlation with event activity
CN	Sensitivity of a specified matrix	A numerical conditioning diagnostic

A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

Takeaway. Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

BACKUP 06 / DECK 31

What the reconstruction scores measure

Likely question: How is each reconstruction metric computed, and what do the scores establish?

Answer

Briefly, metric by metric.

PSNR starts from the mean squared pixel error against a reference image. It's ten times the base-ten log of the peak value squared over that error. **SSIM** compares local means, variances, and covariance, which captures luminance, contrast, and structure. Both are averaged over frames.

EPI correlates the gradient magnitude of the mean reconstruction with the normalized event-count map. **ERC** correlates the absolute change between consecutive reconstructed frames with the absolute bump-integrated event data.

TC takes the root-mean-square change between consecutive frames, divides it by the reconstruction's intensity range, and exponentiates the negative. **TG** averages the absolute frame-to-frame changes over pixels and frame pairs. **TV** averages the sum of absolute horizontal and vertical gradients over pixels and frames.

SC is the fraction of each frame's Fourier power inside a central low-frequency disk, averaged over frames. **CN** is the ratio of the evaluated matrix's largest singular value to its smallest.

These measure different things. Under these conventions, a constant sequence gets the best possible TC, one, and an ERC of zero. Smoothness scores reward a static output, so we read reference fidelity, event consistency, and conditioning together.

If asked for more

For ERC, both sequences are truncated to the same number of slices, flattened, and compared by Pearson correlation. EPI also uses Pearson correlation. Inputs with zero variance get zero correlation. In SC, an all-zero frame contributes zero power fraction, and the disk radius is one eighth of the smaller image dimension.

PSNR and SSIM depend on the normalization used for scoring, and CN describes the specified diagnostic matrix. The baseline choice and the regularization affect image agreement as well as smoothness.

Can the reconstruction run in real time?

CPU measurements (one HPC CPU node; float32; processing time ÷ recording time)

Data	Pixels	Events/s	Batch: 100 ms, $N = 50$ 1 thr.	Batch: 100 ms, $N = 50$ 8 thr.	Streaming: 2 ms steps 1 thr.	Streaming: 2 ms steps 8 thr.
DAVIS346 pendulum	90k	0.25 M	0.04	0.008	0.39	0.06
Cup	307k	2.3 M	1.0	0.19	1.3	0.21

MVSEC: tridiagonal 0.03 s vs dense 1.5 s; agrees with MATLAB, and streaming with batch, within 10^{-15}.

Why it is fast

- $O(N)$ tridiagonal solve per pixel.
- Skipping pixels without events: 10× less work on DAVIS.

Small and portable

- State: 16 bytes per pixel (1.4 MB DAVIS, 4.9 MB cup).
- NumPy and Numba: the code targets ARM boards.

Takeaway. One core streams DAVIS data at 0.39 of real time with 1.4 MB of state.

Source: edge implementation, October 2026 CPU timings and MATLAB comparison; problem size and deployment options: Backup 83.

BACKUP 07 / DECK 32

Can the reconstruction run in real time?

Likely question: Can the reconstruction run in real time, and on what hardware?

Answer

Yes, on an ordinary CPU. The table reports processing time divided by recording time, including binning, for the Chapter 2 estimator on one CPU node. For DAVIS data, one thread takes about four milliseconds per hundred-millisecond window, and the streaming solver returns a new frame every two milliseconds at 0.39 of real time with 1.4 megabytes of state. The 640 by 480 cup recording, at 2.3 million events per second, runs at real time on one thread for batch windows and at 1.3 for streaming; eight threads bring both to about a fifth of real time.

The code solves the Chapter 2 normal equations, with the final interval weight repeated at the terminal sample, and it agrees with the MATLAB reference to rounding error. It needs only NumPy and Numba, so the same code targets ARM boards such as a Raspberry Pi. Backup 83 covers memory and low-power choices.

If asked for more

The streaming recurrence uses the positive terminal diagonal, one plus the final interval weight squared. It matches the newest endpoint of the full mathematical batch solve on a growing prefix, with finalized interval observations. The batch code skips pixels whose net event sum is zero, and a nonzero history can have zero net sum, so compare streaming and batch on the same pixel set.

A moving window changes the left boundary and needs additional handling. The streaming path uses sigmoid weights and delta binning; symmetric kernels would require look-ahead to finalize bins. Quiet pixels with earlier activity still evolve under the recurrence. I also built a C++ and CUDA module in iniVation's DV software for live reconstruction using a general sparse solver.

[For operation counts, memory, and deployment tradeoffs: Backup 83, page 108.]

The weak equation supplies the natural boundary condition

For every $\boldsymbol{\eta} \in H^1(\Omega;\mathbb{R}^2)$,

$$\int_\Omega (\mathbf{B}\mathbf{v}^* - b)\cdot\boldsymbol{\eta} + \alpha^2 \int_\Omega \nabla\mathbf{v}^* : \nabla\boldsymbol{\eta} = 0.$$

Interior tests give the distributional equation

$$-\alpha^2\Delta\mathbf{v}^* + \mathbf{B}\mathbf{v}^* = b.$$

Since the coefficients imply $\Delta\mathbf{v}^* \in L^2$, Green's identity gives

$$\partial_n\mathbf{v}^* = 0 \quad \text{in } H^{-1/2}(\partial\Omega;\mathbb{R}^2).$$

Takeaway. The theorem uses H^1 fields and weak traces on a Lipschitz domain.

Dissertation: Theorem 3.3.9, Theorem 3.4.2, and Theorem 3.4.3.

BACKUP 08 / DECK 33

The weak equation supplies the natural boundary condition

Likely question: Which boundary condition does the flow model use?

[Move from the variational equation to the interior equation and then the weak boundary trace.]

Answer

The natural one: homogeneous Neumann. We minimize over all of H one, so the boundary values are free. Taking the first variation gives the weak equation shown here, for every H one test field. B is the data coefficient matrix, b is the data forcing, and alpha sets the spatial smoothing.

The argument has two steps. First, test fields supported inside the domain give the differential equation in the distributional sense. Under the coefficient hypotheses, B times the solution and b are both in L two, so the solution's distributional Laplacian is in L two as well. That's enough regularity to define a weak normal trace of the gradient. Second, going back to general test fields shows that this normal derivative is zero on the boundary, in that weak trace sense. That's the right boundary statement on the Lipschitz domains the theorem uses.

If asked for more

The last step is the generalized Green's identity. Its boundary pairing must vanish against the trace of every H one test field. The trace map reaches the whole boundary trace space, so the normal trace is zero. For a fixed anisotropic diffusion tensor, the same argument applies to the flux: the normal component of D times the gradient is zero. That's a conormal boundary condition. The weak formulation uses the bounded tensor directly, so D never needs a classical derivative.

Proof: control the mean and the fluctuation separately

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

$$\|\mathbf{w}\|_{L^2} \le C_P \, \|\nabla \mathbf{v}\|_{L^2} .$$

Let $\beta_B = \|\mathbf{B}\|_{L^\infty}$ and $m = \lambda_{\min}(\int_\Omega \mathbf{B}) > 0$ (how well the edges cover every direction).

$$m|\bar{\mathbf{v}}|^2 \le 2 \int_\Omega \mathbf{v}^\top \mathbf{B} \mathbf{v} + 2\beta_B C_P^2 \, \|\nabla \mathbf{v}\|_{L^2}^2 .$$

Therefore the bilinear form satisfies

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) = \int \mathbf{v}^\top \mathbf{B} \mathbf{v} + \alpha^2 \int |\nabla \mathbf{v}|^2 \ge c_0 \, \|\mathbf{v}\|_{H^1}^2 , \qquad c_0 = c_0(\Omega, \alpha, \beta_B, m).$$

Takeaway. Poincaré controls fluctuations, data control the mean, and Lax–Milgram supplies the unique weak solution.

Dissertation: proof of Theorem 3.6.1; Lax–Milgram [72].

BACKUP 09 / DECK 34

Proof: control the mean and the fluctuation separately

Likely question: How exactly does the proof control the mean?

[Point to each line as its role is explained.]

Answer

Through m, the smallest eigenvalue of the integrated data matrix B. Write the velocity as its spatial mean plus a zero-mean fluctuation, w. The gradient of w is the same as the gradient of the full field, so Poincaré's inequality bounds the L two norm of w by the gradient norm. Smoothness controls the fluctuation.

Smoothness charges nothing for a uniform pan, so the data have to fix the mean. Applying the integrated quadratic form to the mean gives a lower bound of m times the mean's squared size. To connect this to the energy of the full field, write the mean as v minus w. Apply the square root of B and use the squared triangle inequality. That bounds the mean term by twice the data energy of v, plus twice a B-weighted fluctuation term. The upper bound on B and Poincaré's inequality control that last term, which gives the displayed inequality.

Now the energy controls the mean, the fluctuation, and the gradient, so it bounds the full H one norm from below. The bilinear form is bounded and coercive, so Lax–Milgram gives a unique weak solution. The weak equation is the first-order condition of this strictly convex energy, so its solution is the unique minimizer.

If asked for more

An admissible constant is c zero equals the smaller of two numbers: m over twice the domain's area, and alpha squared over one plus C P squared plus twice the area times beta B times C P squared over m. It comes from writing the squared H one norm as the area times the mean's squared size, plus the squared L two norm of the fluctuation, plus the squared gradient norm. For fixed alpha and beta B, this bound decreases linearly as m decreases.

Stability follows by subtracting weak equations

Suppose two coefficient pairs share bounds

$$\|\mathbf{B}_i\|_{L^\infty} \le \beta_B, \qquad \lambda_{\min}\left(\int \mathbf{B}_i\right) \ge m > 0, \qquad \|b_i\|_{L^2} \le R_b.$$

Here $c_0 = c_0(\Omega, \alpha, \beta_B, m)$ is independent of b_i. For $\mathbf{z} = \mathbf{v}_1^* - \mathbf{v}_2^*$,

$$\mathcal{B}_1(\mathbf{z}, \mathbf{z}) = \int (b_1 - b_2) \cdot \mathbf{z} + \int \mathbf{z}^\top (\mathbf{B}_2 - \mathbf{B}_1)\mathbf{v}_2^*, \quad \text{hence}$$

$$\|\mathbf{z}\|_{H^1} \le c_0^{-1} \|b_1 - b_2\|_{L^2} + c_0^{-2} R_b \|\mathbf{B}_1 - \mathbf{B}_2\|_{L^\infty}.$$

In scene terms: jitter, binning, and noise events change the coefficients; constants grow as m shrinks (stripes, slide 21). **Accuracy:** the bound concerns the minimizer of the chosen energy; its distance from the true motion depends on coefficient accuracy and on H^1 smoothing across motion boundaries (known-motion AEE, page 79).

Takeaway. The estimate is Lipschitz in its coefficients under uniform identifiability.

Dissertation: Theorem 3.6.1, Eq. 3.61.

BACKUP 10 / DECK 35

Stability follows by subtracting weak equations

Likely question: How do you get the stability bound?

[Point to the two terms in the bound.]

Answer

By subtracting the two weak equations and reusing one coercivity constant. Take two sets of coefficients with common bounds on the data matrix B and the forcing b, and a common positive lower bound m on integrated identifiability. Those shared bounds give one coercivity constant, c zero, that works for both problems.

Subtract the two weak equations, and test with the difference of the solutions, z. The left side controls c zero times the squared H one norm of z. The right side has two parts. One comes from the change in b. The other comes from the change in B, acting on the second solution. Cauchy–Schwarz, together with the bound R b over c zero on that solution, gives the final estimate.

So small changes in the coefficients produce controlled changes in the flow, as long as identifiability stays uniformly positive. The coefficients change because of timestamp jitter, binning choices, and noise events. The constants grow as m shrinks; the stripe scene on slide 21 comes close to that case.

If asked whether a stable estimate is close to the true motion

The theorem is about the minimizer of the chosen energy. How close that minimizer is to the true motion depends on two things: how accurately the represented coefficients describe the motion, and how much the H one penalty smooths across motion boundaries. The known-motion experiments measure that distance directly. In the stripe interior, the estimate is unique but mostly horizontal, and the normal-flow reference there is 2.12 pixels from the true diagonal motion.

If asked for more

Linking the bound to raw sensor errors needs an additional model of the representation error. If a density-dependent diffusion tensor is recomputed, the regularizer changes too, and its L infinity perturbation must enter the bound as well.

Discrete flow: connectivity and sampled directions matter

Fix $\alpha > 0$, $q_i > 0$, and $\mathbf{B}_i = \mathbf{B}_i^\top \succeq 0$ on a connected graph with $w_{ij} = w_{ji} \geq 0$ and degree $d_i = \sum_j w_{ij} > 0$.

One simultaneous block-Jacobi sweep

$$\underbrace{(q_i \mathbf{B}_i + \alpha^2 d_i \mathbf{I}_2)}_{\text{local } 2\times 2 \text{ matrix}} \mathbf{v}_i^{(k+1)} = q_i b_i + \alpha^2 \sum_j w_{ij} \mathbf{v}_j^{(k)}.$$

- At each node, solve for two velocity components using local data and the previous neighbor velocities.
- Update every node from the same old field; then repeat the sweep.
- Scene terms: each pixel blends its edge measurement with neighbors' motion.

If $\sum_i q_i \mathbf{B}_i \succ 0$, the discrete energy has a unique minimizer, and these sweeps converge to it from every initial field.

Dissertation: Theorem 3.7.1 and Corollary 3.7.2; extension to positive node factors q_i.

BACKUP 11 / DECK 36

Discrete flow: connectivity and sampled directions matter

Likely question: How is the flow computed, and when does the iteration converge?

[Point to the local two-by-two matrix, then to the two terms on the right.]

Answer

By simultaneous block-Jacobi sweeps on a pixel graph. They converge to the unique discrete minimizer from any starting field when the weighted sum of the sampled data blocks is positive definite.

Each node stores two velocity components. Its data matrix B describes the measured directions, and b carries the corresponding observations. A positive node factor, q, weights that local contribution. At node i, the matrix combines the local data with alpha squared times the weighted degree: the total weight linking the node to its neighbors. Solving this two-by-two system blends the pixel's own edge measurement with its neighbors' previous velocities.

Every node uses the same old field. Once all the new velocities are computed, we replace the field and repeat. Each sweep carries information from image edges one more pixel into textureless regions.

The theorem assumes the graph is connected, its edge weights are symmetric and nonnegative, its degrees and node factors are positive, and each data block is positive semidefinite.

If asked for more

The reason mirrors the continuous argument: the graph controls variation, and the sampled data control the remaining constant direction. The graph structure also gives the spectral condition this particular iteration needs; the next backup has that argument. The theorem applies when a representation's stencil and update schedule match these hypotheses.

If asked whether this is a finite-element scheme

For the voxel solver, exactly. Its three-by-three box mean is the bilinear Q1 stiffness stencil, which weights all eight neighbors equally, and evaluating the data blocks at pixels is nodal quadrature, a lumped mass matrix. Its reflected border padding reproduces the Q1 natural Neumann rows as well, so that scheme is Q1 finite elements with lumped mass, boundary included. The structure-tensor solver's four-neighbor stencil is the P1 stencil on a uniform grid of right triangles in the interior. Its replicated border halves the inward coupling at boundary pixels, which is the cell-centered finite-volume Neumann condition instead. The continuum coefficients in Appendix A.1.3, alpha squared h squared over three and over four, are these correspondences.

Why block-Jacobi converges for this graph model

Absorb q_i into $\widehat{\mathbf{B}}_i = q_i \mathbf{B}_i \succeq 0$. Set

$$D_i = \alpha^2 d_i \mathbf{I} + \widehat{\mathbf{B}}_i, \qquad R_{ij} = \alpha^2 w_{ij} \mathbf{I}.$$

For symmetric nonnegative edge weights,

$$\mathbf{z}^\top (D \mp R)\mathbf{z} = \sum_i \mathbf{z}_i^\top \widehat{\mathbf{B}}_i \mathbf{z}_i + \alpha^2 \sum_{\{i,j\}} w_{ij} \left\| \mathbf{z}_i \mp \mathbf{z}_j \right\|^2 .$$

Connectedness and $\sum_i \widehat{\mathbf{B}}_i \succ 0$ eliminate both constant and alternating constant zero modes (the latter on bipartite graphs).

$$D - R \succ 0, \quad D + R \succ 0 \quad \Rightarrow \quad -\mathbf{I} \prec D^{-1/2} R D^{-1/2} \prec \mathbf{I}.$$

Takeaway. The Jacobi iteration matrix is similar to a symmetric matrix with spectral radius strictly below one.

Dissertation: extension of Theorem 3.7.1 and Corollary 3.7.2 by absorbing q_i into $\widehat{\mathbf{B}}_i$. $\alpha > 0$, $d_i > 0$.

BACKUP 12 / DECK 37

Why block-Jacobi converges for this graph model

Likely question: Why does block-Jacobi converge for your discrete flow system?

[Explain the two signs in the energy identity before pointing to the spectral interval.]

Answer

The proof uses the graph structure of this particular system. The graph is connected, its edge weights are symmetric and nonnegative, and the sampled data blocks are positive semidefinite with a positive-definite sum. D holds each node's weighted degree and data block; R holds the couplings between neighbors. Positive alpha and positive degrees make D positive definite.

The key step is showing that both D minus R and D plus R are positive definite. Their quadratic forms are sums of nonnegative data terms plus squared differences, or squared sums, across edges. Connectedness narrows their possible zero modes to constant fields or alternating constant fields, and the data blocks rule those out. After scaling by D to the minus one-half, these two inequalities put every eigenvalue of the symmetric coupling matrix strictly between minus one and one. That's exactly what simultaneous Jacobi sweeps need in order to converge.

If asked for more

For D minus R, a zero contribution on every edge forces neighbors to agree, so on a connected graph the field is one constant vector. For D plus R, neighbors must be opposites. A bipartite graph allows an alternating copy of one vector; an odd cycle forces that vector to zero. Either way, the positive-definite sum of data blocks rules out any surviving nonzero vector. The Jacobi matrix is similar to the scaled symmetric matrix, so the two share a spectrum. This proof covers the simultaneous update schedule.

If asked how fast it converges, or why not CG or multigrid

Slowly on large textureless regions. Where the data blocks vanish, a sweep is Jacobi for the graph Laplacian; on a region N pixels wide its spectral radius is close to one minus pi squared over two N squared, so the number of sweeps grows like N squared. Jacobi was used because each sweep is an independent two-by-two solve at every pixel: matrix-free, fully parallel, and the classical Horn–Schunck iteration. The theorem makes the system symmetric positive definite, so conjugate gradients applies directly, with iterations growing like N rather than N squared; Chapter 3 also notes that Gauss–Seidel and SOR converge by Ostrowski–Reich. Multigrid is the standard fast solver for this class; Bruhn and colleagues use it for real-time variational flow. The time-surface code sweeps in place, Gauss–Seidel style, while the theorem analyzes simultaneous sweeps.

Anisotropy needs a uniform ellipticity floor

For the normalized Nagel–Enkelmann tensor [67], write $g = \nabla \rho_{\mathrm{evt}}$ and $\gamma_{\mathrm{NE}} > 0$:

$$\mathbf{D} = \frac{g^{\perp}(g^{\perp})^{\top} + \gamma_{\mathrm{NE}}^2 \mathbf{I}}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

Its tangent and normal eigenvalues are

$$1, \qquad \frac{\gamma_{\mathrm{NE}}^2}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

If $g \in L^{\infty}$,

$$\theta \mathbf{I} \preceq \mathbf{D} \preceq \mathbf{I}, \qquad \theta = \frac{\gamma_{\mathrm{NE}}^2}{\|g\|_{L^{\infty}}^2 + \gamma_{\mathrm{NE}}^2} > 0.$$

Takeaway. Diffusion across event-density edges is reduced, while coercivity survives with $\alpha^2\theta$.

Dissertation: normalized variant of Eq. 3.46, Lemma 3.4.6, Corollary 3.6.4.

BACKUP 13 / DECK 38

Anisotropy needs a uniform ellipticity floor

Likely question: How can directional smoothing preserve the well-posedness theorem?

[Use the two eigenvalues to distinguish tangent smoothing from normal smoothing.]

Answer

The diffusion tensor needs a uniform positive lower bound on its eigenvalues. We use the normalized Nagel–Enkelmann form, built from the spatial gradient of event density. That gradient points across a density edge, and its perpendicular points along the edge. The tensor has eigenvalue one along the edge and a smaller positive eigenvalue across it.

So it penalizes flow changes across a density edge less, while still smoothing fully along it. The positive parameter gamma provides the floor. When the density gradient is bounded, the formula gives one positive lower bound, theta, over the whole domain. The coercivity proof then goes through with alpha squared times theta in place of alpha squared. So the directional model keeps the same existence and uniqueness argument under the integrated identifiability condition, with constants that depend explicitly on that floor.

If asked for more

A pointwise positive eigenvalue could still approach zero somewhere in the domain, which is why the proof needs a uniform floor. If the tensor is recomputed when the data change, the stability estimate must also account for the change in D, measured in the essential-supremum norm. The tensor follows event-density edges; treating those as motion boundaries is a modeling choice. The theorem guarantees solvability for the specified tensor.

A variable diffusion tensor must stay inside the divergence

For differentiable $\mathbf{D}$,

$$\begin{aligned}\nabla\cdot(\mathbf{D}\nabla u) = {} & D_{11}u_{xx} + 2D_{12}u_{xy} + D_{22}u_{yy} \\ & + (\partial_x D_{11} + \partial_y D_{12})u_x + (\partial_x D_{12} + \partial_y D_{22})u_y.\end{aligned}$$

Take $\rho_{\mathrm{evt}} = x^2/2$, $\gamma_{\mathrm{NE}} = 1$, and $u = x$:

$$\mathbf{D} = \operatorname{diag}\left(\frac{1}{1+x^2}, 1\right), \qquad \nabla\cdot(\mathbf{D}\nabla u) = \frac{-2x}{(1+x^2)^2} \not\equiv 0.$$

All second derivatives of u vanish, so the principal part alone gives the wrong answer.

Dissertation: Anisotropic numerical discretization (p. 83).

BACKUP 14 / DECK 39

A variable diffusion tensor must stay inside the divergence

Likely question: Can the anisotropic operator be implemented by weighting the second derivatives?

[Point to the coefficient-derivative terms, then to the affine example.]

Answer

Only when the extra coefficient terms vanish, for example when the tensor is constant. For a spatially varying tensor, the divergence form contains both second derivatives of the field and first derivatives of the tensor coefficients; the product rule gives all the terms on this slide. Keeping only the tensor-weighted Hessian drops the coefficient terms.

The example makes this concrete. Take event density x squared over two, gamma equal to one, and the test field u equal to x. The tensor is diagonal, with first entry one over one plus x squared. The field has zero second derivatives, yet its flux changes with x, because that tensor entry changes. Differentiating the flux gives minus two x over the square of one plus x squared. A discretization has to capture that changing flux to approximate the stated model.

If asked for more

The weak formulation is useful when D is only bounded and measurable. It integrates the test-field gradient against D times the solution gradient, so coefficient derivatives never appear. A flux-based discretization can approximate this divergence structure directly.

The regression radius formula is a heuristic

For full disks on a square lattice of spacing h, with r/h large,

$$Q \sim \frac{\pi r^4}{4h^2}\mathbf{I}, \qquad U_h(r) = \frac{4}{81}L_H^2 r^4 + \frac{8}{\pi}\sigma^2 h^2 r^{-4}.$$

The minimum of this leading expression, when all parameters are positive, is

$$r^* = (162/\pi)^{1/8}(\sigma h/L_H)^{1/4}.$$

- The full disk must lie in a region with the asserted smoothness bound.
- U_h is an asymptotic expression for the finite-design bound.
- Affine time surfaces have zero curvature bias; this model supplies no finite balancing radius there.

Dissertation: Proposition 3.3.7 and Radius selection as a heuristic.

BACKUP 15 / DECK 40

The regression radius formula is a heuristic

Likely question: Is the regression-radius formula an optimal choice of neighborhood?

[Identify the increasing bias term and decreasing variance term before discussing the radius.]

Answer

Only in a leading-order sense. It optimizes the leading approximation to the exact error bound for the actual lattice disk of sample points. The assumptions are a complete disk on a square lattice, a radius large compared with the lattice spacing, and a time surface that's smooth throughout the disk. L sub H measures how fast the Hessian can change, sigma is the timestamp-noise standard deviation, and h is the lattice spacing.

The leading squared-bias term grows like the radius to the fourth, and the variance term shrinks like its reciprocal. Balancing them gives the radius shown. Noisier data favor more averaging; faster-changing curvature favors a smaller neighborhood. We use that scaling as a principled guide. For an actual finite neighborhood, the exact lattice sums give the appropriate bound. And the whole fitting disk has to stay inside the region where the smoothness assumptions hold.

If asked for more

Minimizing one constant times r to the fourth plus another constant times r to the minus fourth makes r to the eighth equal to the ratio of the second constant to the first. Substituting the coefficients gives the factor shown. The asymptotic remainder can have either sign, so the leading expression approximates the finite-design bound from either side. And for an affine time surface the curvature bias is zero, so more samples only reduce variance, and this model has no finite balancing radius.

Uniform and OLS weights agree through quadratics and differ on cubics

The continuous OLS temporal slope over $[s, s+T]$ is

$$\widehat{\beta} = \int_s^{s+T} w_s(t) V_t(t)\, dt, \qquad w_s(t) = \frac{6(t-s)(s+T-t)}{T^3}.$$

Its transport-consistent spatial partner is the same w_s-weighted projection. Pairing it with a uniform projection gives residual

$$\mathbf{v} \cdot \nabla \int_s^{s+T} (T^{-1} - w_s(t)) V(\mathbf{x}, t)\, dt.$$

For $V(x,t) = (2 + x - t)^3$, $\mathbf{v} = (1, 0)$, $s = 0$, $T = 1$, this residual is $1/10$.

Takeaway. Derivative consistency needs its own check, beyond well-posedness of a fixed-coefficient energy.

Dissertation: Section 3.4.1, consistent derivative-operator discussion.

BACKUP 16 / DECK 41

Uniform and OLS weights agree through quadratics and differ on cubics

Likely question: Why must the temporal slope and spatial projection use compatible weights?

[Point to the common temporal weight and then the cubic residual.]

Answer

Because they enter the same transport equation, and that equation stays exact only if both use the same averaging. The ordinary least-squares temporal slope can be written as a weighted average of the time derivative. Its weight is the parabola shown here: zero at the endpoints and largest in the middle of the interval.

If we integrate the transport equation against that weight, the spatial term must use the same weighted projection. Pairing the OLS slope with a uniform temporal projection instead leaves the residual shown. For the transported cubic, that residual is one tenth, even though the underlying field satisfies transport exactly. So there are two separate questions: whether the coefficient fields give a well-posed problem, and whether they represent the physical transport relation exactly. The flow theorem answers the first, for any coefficients that meet its hypotheses.

If asked for more

Integrating by parts in the OLS formula produces the parabolic weight. Uniform averaging and that weight agree on the zeroth and first temporal moments, so low-degree tests reveal little. For transported polynomials up to degree two, the mismatch is zero, which makes the cubic the natural explicit example. If the resulting coefficient error is controlled in the norms of the stability theorem, and identifiability stays positive, that theorem bounds its effect on the flow.

Mean-zero conditioning removes the unobserved mode

In the unweighted scalar model, restrict to

$$\mathcal{V}_0 = \{(\mathbf{f}_L, \mathbf{f}_R) : \mathbf{1}^\top \mathbf{f}_L = \mathbf{1}^\top \mathbf{f}_R = 0\}.$$

The sum–difference transform preserves this space. Its remaining eigenvalues are

$$g_k + \lambda^2, \qquad g_k + \lambda^2 + 2\gamma, \qquad k = 1, \dots, n-1.$$

Thus

$$\kappa_2(\mathbf{M}|_{\mathcal{V}_0}) = \frac{g_{\max} + \lambda^2 + 2\gamma}{g_1 + \lambda^2} \le \frac{g_{\max} + 2\gamma}{g_1}, \qquad g_1 = 4\sin^2 \frac{\pi}{2n}.$$

Takeaway. The restriction is SPD even at $\lambda = 0$. The uniform-in-λ bound depends on the temporal grid; full-space perturbations remain different.

Dissertation: Corollary 4.6.17; Proposition 4.6.18 (notes).

BACKUP 17 / DECK 42

Mean-zero conditioning removes the unobserved mode

Likely question: Can a mean-zero formulation avoid poor conditioning as lambda becomes small?

[Point to the excluded k equals zero mode and the new smallest eigenvalue.]

Answer

Yes, in the scalar, unweighted model. Removing the constant temporal modes gives a condition-number bound that stays finite as lambda goes to zero. The first-difference operator sends constants to zero, so the right-hand side has zero mean in each view. The scalar system preserves the mean-zero subspace, so for positive lambda the solution already lies in it.

On that subspace, the smallest temporal eigenvalue is g one instead of zero. The sum and difference blocks have eigenvalues g k plus lambda squared, and g k plus lambda squared plus two gamma, with the constant mode left out. Their ratio gives the exact restricted condition number, and its denominator stays positive even at lambda equal to zero. So a mean-zero parameterization helps the algebraic solve, provided we state the convention for the unobserved brightness baseline.

If asked for more

The bound is uniform in lambda for a fixed temporal grid and fixed coupling. It still depends on the grid: g one behaves like pi squared over n squared for the unscaled chain, so the lambda-uniform bound grows quadratically with the number of samples. At lambda equal to zero the full-space matrix still has a constant null mode, so to use the restricted guarantee, a numerical implementation has to actually enforce the mean-zero restriction. General diagonal penalties or unequal weights need their own matrix bounds.

The selection theorem has an exact ellipsoid interpretation

Let

$$\mathbf{S} = \tfrac{1}{2}(\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top), \quad \mathbf{T} = \tfrac{1}{4}\mathbf{A}[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}]\mathbf{A}^\top, \quad \mathbf{P} = \mathbf{S} + \mathbf{T} \succ 0.$$

Assume the ideal match $\mathbf{y}_{d_*} = \mathbf{y}_L$. For $\mathbf{e}_d = \mathbf{y}_L - \mathbf{y}_d$,

$$C(d) - C(d_*) = \|\mathbf{e}_d\|_{\mathbf{P}}^2 - 2\mathbf{y}_L^\top \mathbf{S}\mathbf{e}_d = \|\mathbf{y}_d - \mathbf{c}\|_{\mathbf{P}}^2 - \rho^2,$$

$$\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L, \qquad \rho^2 = \mathbf{y}_L^\top \mathbf{S}\mathbf{P}^{-1}\mathbf{S}\mathbf{y}_L.$$

Takeaway. The ideal match wins if every competitor is outside the ellipsoid. In particular, $\|\mathbf{e}_d\|_{\mathbf{P}} > 2\rho$ is sufficient.

Dissertation: Proposition 4.6.8(a), (b2). Candidate-local unweighted model.

BACKUP 18 / DECK 43

The selection theorem has an exact ellipsoid interpretation

Likely question: Can you characterize exactly which competing stereo profiles beat the ideal match?

[Explain inside, boundary, and outside using the completed-square expression.]

Answer

Yes. For the candidate-local unweighted model, the competitors that beat the ideal match fill an ellipsoid in observation space. We hold the positive regularizer and the coupling fixed across candidates, and assume the ideal candidate's right profile equals the left observation. S is the monocular cost matrix, T is the coupling-discrepancy matrix, and P is their sum.

Completing the square writes the cost gap as the squared P-distance from the candidate profile to a center c, minus rho squared. P defines the metric, so the surface of equal cost is an ellipsoid, and the ideal match lies on its boundary. A competitor inside the ellipsoid costs less than the ideal match; one outside costs more. That's an exact geometric picture of the attenuation effect. It also gives a simple sufficient rule: any mismatch larger than two rho in the P-norm loses to the ideal match.

If asked for more

Let the mismatch be the left observation minus the competing right observation. The cost gap is its squared P-norm minus a linear term involving S. Cauchy–Schwarz in the P inner product bounds that linear term by two rho times the mismatch norm, so the gap is positive whenever the mismatch norm exceeds two rho. The exact ellipsoid test is sharper, because it uses direction as well as distance. Both statements concern the fixed, unweighted candidate cost before spatial aggregation.

An observation perturbation bound can protect a disparity winner

Keep $\mathbf{y}_L$, the candidate set, $\mathbf{A}$, $\mathbf{\Lambda}$, and γ fixed. Perturb each right observation by $\boldsymbol{\xi}_d$. The quadratic candidate cost has

$$\nabla_{\mathbf{y}_d} C = 2\mathbf{P}(\mathbf{y}_d - \mathbf{c}), \qquad \nabla^2_{\mathbf{y}_d} C = 2\mathbf{P}, \qquad \|\mathbf{P}\| \leq \frac{1}{2}.$$

Consequently a uniform bound is

$$\eta = \max_d \left(\|\mathbf{y}_d - \mathbf{c}\| \, \|\boldsymbol{\xi}_d\| + \frac{1}{2} \|\boldsymbol{\xi}_d\|^2 \right).$$

If $\eta < \Delta/2$, the unique winning disparity is preserved.

Takeaway. The ideal correct match may be perturbed too. Changing the left data or adaptive parameters needs additional terms.

Dissertation: Corollary 4.6.10.

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An observation perturbation bound can protect a disparity winner

Likely question: How much observation noise can the stereo decision tolerate?

[Indicate that the maximum includes the winner, then the half-margin condition.]

Answer

With a positive winning margin, we can put a number on it, for a fixed candidate list and fixed model parameters. We hold the left observation, A, Lambda, and gamma fixed, and perturb each right candidate profile. The cost is quadratic in that right observation: its gradient is two P times the displacement from c, and its Hessian is two P.

The operator norm of P is at most one half. Expanding the quadratic, each cost changes by at most the candidate's distance from c times the size of the perturbation, plus one half the perturbation size squared. Eta is the largest such bound over all candidates, including the original winner. If eta is less than half the original winning margin, every competitor stays more expensive. That protects the decision even though the formerly ideal right profile may be perturbed too.

If asked for more

The factor of two in the margin condition covers two changes at once: the winner's cost can rise by eta while a competitor's falls by eta. So the perturbed gap is at least the original margin minus two eta. For this observation bound, the quadratic expansion is exact. If the preprocessing, the left data, or adaptive parameters also change, their contributions must go into the score-perturbation estimate. The same margin argument then applies to whatever uniform score bound results.

How the reciprocal IRLS contraction bound is obtained

Write $\mathbf{B}_{\text{time}} = \operatorname{diag}(\mathbf{A}, \mathbf{A})$, $\mathbf{f} = \Phi(\mathbf{g})$, $\mathbf{f}' = \Phi(\mathbf{g}')$, and $\mathbf{r}' = \mathbf{y} - \mathbf{B}_{\text{time}}\mathbf{f}'$. Set $\mathbf{s} = \mathbf{B}_{\text{time}}\mathbf{g}$ and $\mathbf{s}' = \mathbf{B}_{\text{time}}\mathbf{g}'$. Subtract the frozen systems:

$$\mathbf{f} - \mathbf{f}' = \mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\left[\mathbf{W}^{-1/2}(\mathbf{W} - \mathbf{W}')\mathbf{r}'\right].$$

Use

$$\left\|\mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\right\| \le \ell^{-1/2}, \qquad \left\|\mathbf{W}'^{1/2}\mathbf{r}'\right\| \le \left\|\mathbf{W}'^{1/2}\mathbf{y}\right\|.$$

For reciprocal weights, $|w_i - w_i'| \le |s_i - s_i'|w_i w_i'$ and $w_i, w_i' \le \varepsilon^{-1}$.

Takeaway. Combining operator, residual, and weight-increment bounds gives $q = \|\mathbf{A}\|_2 \|\mathbf{y}\| / (\sqrt{\ell}\,\varepsilon^{3/2})$.

Dissertation: proof of Proposition 4.6.13; unclipped reciprocal map.

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How the reciprocal IRLS contraction bound is obtained

Likely question: Where does the sufficient IRLS contraction constant come from?

[Follow the difference identity through the operator, residual, and weight estimates.]

Answer

We compare two frozen-weight solves generated by two input states. B time is the block-diagonal temporal difference operator for the two views. Subtracting the normal equations expresses the difference of the solutions as an inverse-system operator applied to a weight difference times a residual.

Three estimates control that expression. First, the positive regularizer bounds the inverse operator in terms of ell, the square of the smallest penalty entry. Second, comparing the frozen optimum with zero bounds its weighted residual by the weighted data norm. Third, reciprocal weights change in a controlled way: the change in an absolute value is at most the change in its argument, and every weight is at most one over epsilon. Combining the three gives q on the slide. When q is below one, the frozen-solve map is a contraction, and its iterates converge geometrically to a unique fixed point.

If asked for more

The weight estimate pairs with the weighted residual, and that's what produces the epsilon to the minus three-halves. The change in predicted temporal differences is at most the norm of A times the change in the state, which supplies the remaining factor. This is a sufficient global bound for the unclipped reciprocal map. It certifies contraction whenever that constant is below one.

The frozen-weight fixed point and the likelihood stationary point differ

Consider the scalar Gaussian proxy with $y > 0$ and $\varepsilon > 0$, **without a regularizer**:

$$v(\theta) = |\theta| + \varepsilon, \qquad Q_k(\theta) = \frac{(y-\theta)^2}{2v(\theta_k)}.$$

Every frozen quadratic is minimized at $\theta = y$: the IRLS map has that fixed point. The state-dependent negative log-likelihood is

$$\mathcal{L}(\theta) = \frac{(y-\theta)^2}{2v(\theta)} + \frac{1}{2}\log v(\theta).$$

Yet $\mathcal{L}'(y) = 1/[2(y+\varepsilon)] > 0$.

Takeaway. The frozen-weight fixed point differs from the stationary point of this proxy likelihood.

Dissertation: Proposition 4.6.2. Scalar logical illustration outside the positive-Λ standing assumptions.

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The frozen-weight fixed point and the likelihood stationary point differ

Likely question: Does convergence of the reweighted iteration imply a maximum-likelihood estimate?

[At the fixed point, show that the residual term vanishes while the log-variance derivative remains.]

Answer

No. When the variance depends on the unknown state, a frozen-weight fixed point and a likelihood optimum solve different derivative equations. A scalar example shows this directly. Take a positive observation y, with variance equal to the absolute prediction plus a positive epsilon. With the variance frozen, every least-squares step is minimized at the prediction y, so the iteration reaches that fixed point immediately.

The full Gaussian negative log-likelihood is different: the variance moves with the prediction, which adds a log-variance term. At the prediction y the residual is zero, but the derivative of the log-variance term is positive: one over two times the quantity y plus epsilon. So lowering the prediction slightly lowers the negative log-likelihood: its minimizer lies just below the fixed point.

If asked for more

This scalar example deliberately has no regularizer, to keep the mechanism visible. It isolates the weighting mechanism from the positive-penalty temporal theorem. The dissertation also gives a two-sample temporal example with positive regularization, where a fixed point differs from a stationary point of the final-state-weighted score, even before the Gaussian log term is added. The empirical estimator is evaluated as the specified surrogate procedure, with its own objective and stopping rules.

What the implemented IRLS procedure returns

At every candidate disparity:

1. Initialize from the unweighted exact solve.
2. Recompute predicted-rate weights, **clip for the solve**, and freeze them.
3. Solve the SPD system and damp toward its minimizer.
4. Recompute **unclipped scoring weights**; apply acceptance and stopping rules.
5. Return the last accepted iterate and its score.

The score omits the log-variance term. Masking, averaging, margin normalization, and SGM [95] then change candidate comparisons.

Takeaway. Brouwer and the contraction theorem describe fixed maps; early stopping and aggregation require their own analysis.

Dissertation: Section 4.5, Section 4.6.1; Section A.2.

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What the implemented IRLS procedure returns

Likely question: What does one candidate-level IRLS call actually return?

[Walk through the five steps, emphasizing the different solve and scoring weights.]

Answer

It returns the last accepted latent iterate and its computed score. The procedure starts from the exact unweighted solve. At each iteration it forms predicted-rate weights, clips them for the linear solve, and freezes them. That system is positive definite. The algorithm solves it and takes a damped step toward the frozen minimizer.

For scoring, it recomputes the weights at the proposed state with the unclipped rule, then applies the acceptance and stopping tests. The score uses the weighted residual and the regularization terms, without a Gaussian log-variance term. So clipping, damping, and stopping are all part of the estimator being evaluated. After these candidate-level steps, masking, averaging, margin normalization, and semi-global matching adjust the candidate comparisons. The existence and contraction results describe the specified reweighting map; the experiments assess the complete implemented procedure.

If asked for more

A stopping rule can return an accepted iterate before it reaches a fixed point. Later, finite penalty values used for guidance or rejection go into averaging, so they act as soft preferences. The final winner is the exact minimizer of its computed score list, and path aggregation supplies the spatial context for that list.

Depth evaluation distinguishes returned coverage and scored coverage

1. Ground truth is finite and in the reporting depth band.
2. The returned estimate is finite.
3. The estimate also lies in the reporting band.

Configuration	Returned coverage	In-band retention	Scored points
Control	50.17%	51.54%	1.43 M
WLS	49.48%	96.40%	2.66 M

The scoring band is $[0.5, 8]$ m. Out-of-band finite depths are excluded by benchmark convention.

Dissertation: Table A.4, Weighting comparison (p. 182). Complete configurations; method-specific populations.

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Depth evaluation distinguishes returned coverage and scored coverage

Likely question: Does WLS achieve its better scores by returning estimates at many more pixels?

[Contrast the first two numeric columns before discussing the scored-point totals.]

Answer

No. Both methods return estimates at about half the eligible pixels; the difference is in how many of those estimates fall inside the reporting band. The table keeps returned coverage separate from the population used to score accuracy. First, the ground truth must be finite and inside the band. Returned coverage counts finite estimates at those locations. Scoring also requires the estimate itself to fall between 0.5 and 8 meters.

The control returns finite estimates at 50.17 percent of eligible locations, and WLS at 49.48 percent. The in-band share of those estimates is where they differ: 51.54 percent for the control, and 96.40 percent for WLS. That gives 1.43 million scored points for the control, and 2.66 million for WLS. So WLS has a much larger scored population with comparable returned coverage. These are complete configurations, so the comparison includes their preprocessing, weighting, and final masking.

If asked for more

A finite depth outside the band is excluded by the evaluation convention, though it may still be physically possible. That's why returned coverage, in-band retention, and accuracy answer different questions. The benchmark's raw outputs include interpolation around the finest-level score minimum, and leave out the separate posterior refinement step. Reading the errors alongside these counts keeps the interpretation precise. A common-population comparison would score both methods on the same eligible pixels.

Exact zeros alone produce large pooled kurtosis

Suppose a residual distribution mixes a point mass at zero with a zero-mean Gaussian of variance $\sigma^2 > 0$, with zero-mass probability $0 \le \pi_0 < 1$.

$$\mathbb{E}X^2 = (1-\pi_0)\sigma^2, \qquad \mathbb{E}X^4 = 3(1-\pi_0)\sigma^4.$$

Its population excess kurtosis is

$$\frac{\mathbb{E}X^4}{(\mathbb{E}X^2)^2} - 3 = \frac{3}{1-\pi_0} - 3.$$

- Event-free pixels can have exactly zero fitted transport residual.
- Regularization, unequal variances, and temporal dependence affect diagnostics.
- Test event-active and variance-standardized fitted residuals before choosing a tail model.

Dissertation: Section 3.8.1; Model Mismatch in the Residuals.

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Exact zeros alone produce large pooled kurtosis

Likely question: Does the large pooled residual kurtosis justify a heavy-tailed noise model?

[Point to the single active-probability factor in both moments and its effect in their ratio.]

Answer

On its own, it's a reason to look more closely at the residuals. This simple mixture shows why. Suppose a fraction of pixels have exactly zero residual, and the rest are zero-mean Gaussian with variance sigma squared. The Gaussian part has ordinary Gaussian tails. But pooling it with the point mass at zero gives an excess kurtosis of three over the active fraction, minus three.

So the more zeros, the larger the pooled kurtosis. Event-free pixels can have exactly zero fitted transport residuals, so sparsity alone can create the effect. Regularized fitting, unequal variances, and temporal dependence also shape the residual distribution. So I'd read the pooled diagnostic alongside the event-active residuals and how their variance scales. That separates concentration at zero from the tails of the active errors, and gives a better basis for choosing the observation model.

If asked for more

The moment calculation is direct. Only the active Gaussian part contributes to the second and fourth moments, so each picks up one factor of the active probability. Dividing the fourth moment by the square of the second leaves the reciprocal of that probability. A useful follow-up would report the zero fraction, standardize the active residuals by their variance, and compare with synthetic residuals passed through the same fitting procedure. Robust penalties remain an option, but choosing one needs evidence beyond this pooled moment.

Adaptive temporal weights are fixed in the theorem

For event activity $a_i = |dc_i|$, Chapter 2 uses these positive rules:

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}} \qquad \text{(sigmoid)}$$

$$\lambda_i = \max\left(10^{-3}, \min\left(\frac{100}{10^{-3} + a_i}, 100\right)\right) \qquad \text{(inverse proportional)}$$

$$\lambda_i = 10^{-3} + 10e^{-\min(a_i, 10)} \qquad \text{(exponential decay)}$$

Chapter 4 instead uses $\lambda_{xy} = \max\{\lambda_{\text{floor}}, \lambda_0 \|\mathbf{b}\| / (\|\mathbf{b}\| + 2)\}$.

Takeaway. Each fixed $\mathbf{\Lambda}$ is covered by the theorem; for the Chapter 2 rules, the data-dependent map $\mathbf{y} \mapsto \mathbf{\Lambda}(\mathbf{y}) \mapsto \mathbf{f}^*$ is locally Lipschitz (p. 98).

Dissertation: Adaptive temporal weights (p. 38); Section 4.5 (regularization terms); Theorem 2.2.5.

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Adaptive temporal weights are fixed in the theorem

Likely question: How do the adaptive temporal penalties relate to the fixed-weight theorem?

[Contrast the activity-decreasing Chapter 2 rules with the energy-increasing Chapter 4 rule.]

Answer

The theorem applies once the adaptive rule has produced a positive penalty matrix. In Chapter 2, the sigmoid, inverse-proportional, and exponential rules all lower the magnitude penalty as event activity grows. That allows more temporal variation where there's activity, while a positive floor keeps every weight above zero.

Chapter 4 uses a different empirical rule, based on energy before normalization. Its weight increases with that energy, and it also has a positive floor. So the two chapters make different design choices. For every fixed positive matrix either rule produces, the quadratic solve is unique. But when the data change and the rule recomputes Lambda, both the data term and the penalty change. The sensitivity of the full adaptive reconstruction has to include both contributions, beyond the fixed-penalty data bound.

If asked for more

Where the adaptive rule is differentiable, differentiating the normal equations gives two forcing terms: A transpose times the change in the observations, and minus the change in Lambda squared applied to the current solution. Multiplying by the inverse reconstruction matrix turns those into the first-order change in the solution. Absolute values and clipping introduce kinks, but every rule is still Lipschitz, so the local bound on page 90 needs no derivatives. The positive-floor theorem is still useful, because it controls every individual solve the rule produces.

A real MVSEC flow example is qualitative

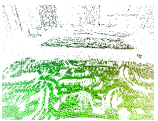
MVSEC indoor_flying2, frame 0340.



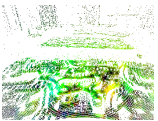
hue: direction
brightness: size, scaled within each panel (95th percentile); white: no flow value



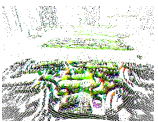
Reference



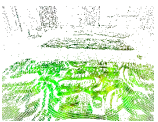
Ground truth



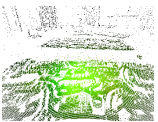
Event-accumulation baseline



Time surface



Voxel



Structure tensor voxel

Takeaway. One qualitative example; Chapter 3's AEE values use known-motion scenes.

Dissertation: Figure 3.2.

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A real MVSEC flow example is qualitative

Likely question: What does the real-data flow comparison establish?

[Identify the reference and ground truth first; compare methods only through visible features.]

Answer

It's a qualitative comparison on one frame: MVSEC indoor flying 2, frame 0340. The reference image shows the scene, and the ground-truth flow is the visual target. The other panels show the event-accumulation baseline and the time-surface, voxel, and structure-tensor voxel estimates.

Those last three are the representations covered by the common quadratic analysis. They build different coefficient fields from the same kind of event input, so the panels show how those choices change the flow pattern in a real scene. The color legend is the one from slide 6: hue is direction and brightness is size, with each panel scaled to its own ninety-fifth-percentile magnitude, so compare directions across panels and sizes only within a panel. The accumulation baseline is a separate comparison method. Chapter 3's endpoint errors come from the known-motion scenes, so here I describe what's visible in this frame.

If asked for more

Average endpoint error would measure the Euclidean distance between estimated and true velocity at a specified set of valid locations, then aggregate over a stated population. The theorem answers a different question: for coefficient fields meeting its assumptions, the regularized estimate exists, is unique, and depends continuously on the coefficients. Accuracy against true motion comes from evaluation, reported with its own valid set and aggregation rule.

The ground-truth flow has known limits too. It is generated from a Velodyne lidar with a 3D reconstruction, so it carries estimation and alignment error, and accumulating points over frames accumulates odometry drift that can make objects look inflated or shifted against the event data. The reference motion is also small: roughly 95 percent of it is under about three pixels, so it says little about fast motion.

Quadratic H^1 regularization and TV answer different questions

The flow energy uses a quadratic gradient penalty:

$$E(\mathbf{v}) = \tfrac{1}{2}\int_\Omega \left(\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}\right) + \tfrac{\alpha^2}{2}\,\|\nabla \mathbf{v}\|_{L^2}^2\,, \qquad \mathbf{v} \in H^1(\Omega;\mathbb{R}^2).$$

- Lax–Milgram gives existence, uniqueness, and coefficient stability under the stated hypotheses.
- TV [41] can preserve jumps, but needs a separate BV existence argument and its own uniqueness and stability assumptions.
- Normalized NE diffusion gives directional quadratic smoothing with a uniform ellipticity floor.

Takeaway. The integrated full-rank condition supports the quadratic H^1 proof; TV needs its own existence argument.

Dissertation: Theorem 3.6.1, function-space remark, Eq. 3.46.

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Quadratic H^1 regularization and TV answer different questions

Likely question: Why choose a quadratic H-one flow model rather than total variation?

[Point to the quadratic gradient term, then state its smoothing consequence before discussing TV.]

Answer

The quadratic model gives one explicit well-posedness analysis that covers all the event-derived representations. Its gradient penalty defines a bilinear form, and the integrated directional-information condition controls the constant velocity component. Together with the gradient term and Poincaré's inequality, that gives H one coercivity. Lax–Milgram then gives a unique solution, and subtracting weak equations gives the coefficient-stability estimates.

The tradeoff is that quadratic smoothing spreads out sharp motion boundaries. Total variation is attractive when preserving jumps matters most, and its natural setting is the space of functions of bounded variation. It would need its own existence, uniqueness, and stability analysis. Within the quadratic framework, the normalized directional tensor is another option: it reduces smoothing across selected density edges, while keeping a positive ellipticity floor and the same coercivity proof.

If asked for more

The integrated full-rank condition is sufficient for this particular quadratic H one argument. Replacing the gradient penalty with total variation changes how the functional grows, and which compactness and coercivity arguments are available. A squared-residual fidelity can be studied with TV, but the theorem has to address that functional directly. For the signed-linear energy written here, simply substituting TV can even make the energy unbounded below. Stronger fidelity assumptions can give strict convexity and uniqueness in a TV model.

Frozen weights match the value; an MM majorizer also matches the derivative

For a scalar prediction s, let $v(s) = |s| + \varepsilon$, $\varepsilon > 0$, and

$$S(s) = \frac{(y-s)^2}{2v(s)} + R(s), \qquad Q_k(s) = \frac{(y-s)^2}{2v(s_k)} + R(s).$$

The frozen surrogate matches S at s_k, but away from $s = 0$ its derivative omits

$$-\frac{v'(s)(y-s)^2}{2v(s)^2}.$$

Takeaway. A differentiable tangent majorizer must match the derivative at the current iterate; the frozen quadratic matches the value.

Dissertation: Section 4.6.1, Proposition 4.6.2; derivative consequence of the proposition.

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Frozen weights match the value; an MM majorizer also matches the derivative

Likely question: Is the reciprocal-weight iteration a majorization–minimization algorithm?

[Point to the omitted denominator derivative and explain the derivative-contact requirement.]

Answer

Only when derivatives also match. For majorization–minimization, the frozen objective would have to lie above the state-dependent score and touch it at the current state. For differentiable functions, touching also means matching derivatives. Here the variance proxy depends on the predicted signal, so the derivative of the full score has an extra term from the changing denominator.

The frozen quadratic matches the score's value at the current state, but its derivative treats that denominator as constant. So the term shown remains in the full derivative, and it's generally nonzero. That's why I analyze the method through its frozen-solve map: I prove that a fixed point exists, and that the iteration converges under a sufficient contraction condition. Those are precise properties of the map. A monotone-descent claim for a state-dependent score would need a separate surrogate inequality, or an acceptance rule that actually enforces descent.

If asked for more

If a differentiable surrogate majorizes the score and touches it at an interior point, their difference has a local minimum of zero there. So the derivative of the difference vanishes, and the derivatives must match. The extra variance derivative vanishes in special cases, such as a zero residual. The implemented acceptance rule is a different mechanism and can allow small cost increases, so describe its behavior through its actual thresholds.

A Gaussian model gives an evidence interpretation of attenuation

For common fixed $\mathbf{\Lambda} \succ 0$, $\gamma \geq 0$, and independent unit Gaussian noise, the candidate conditional is

$$\mathbf{y}_R \mid \mathbf{y}_L \sim \mathcal{N}(\mathbf{c}, (2\mathbf{P})^{-1}), \qquad \mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L.$$

Coupled versus independent evidence satisfies

$$-\log \frac{p_\gamma(\mathbf{y}_L, \mathbf{y}_R)}{p_0(\mathbf{y}_L, \mathbf{y}_R)} = D(\mathbf{y}_L - \mathbf{y}_R) + \text{constant}.$$

Takeaway. The conditional mean explains attenuation; the discrepancy alone is an unevaluated alternative score.

Dissertation: Supplementary Gaussian derivation from Lemma 4.6.6, Proposition 4.6.8.

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A Gaussian model gives an evidence interpretation of attenuation

Likely question: Is there a probabilistic interpretation of the attenuation in the stereo cost?

[Connect the conditional mean c to the ellipsoid center, then discuss the evidence ratio.]

Answer

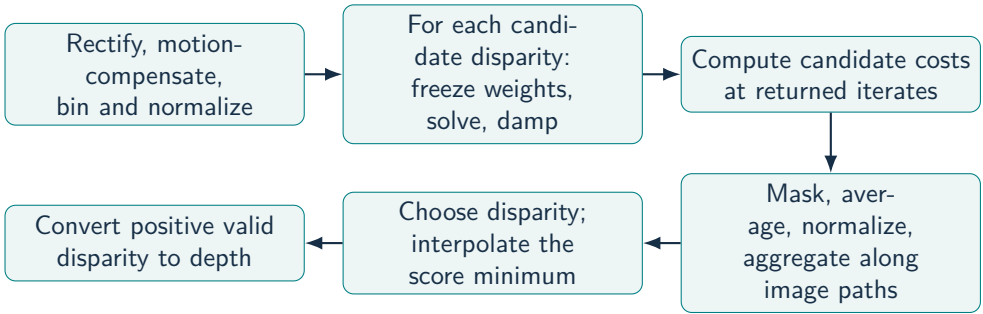
Yes, for the Gaussian model with fixed common parameters. Read the magnitude and coupling penalties as a Gaussian prior on the two latent profiles, and assume independent unit-variance Gaussian observation noise. After integrating out the profiles, the right observation given the left is Gaussian. Its mean is c, the same shifted center as in the cost ellipsoid, and its covariance is the inverse of two P.

So the model predicts an attenuated right observation. For a scalar regularizer, that attenuation is diagonal in the temporal observation modes. Comparing the coupled model's evidence with the independent model's leaves the discrepancy term plus a constant that is the same for every candidate. Under these Gaussian assumptions, that explains both the shifted prediction and a possible discrepancy-based comparison.

If asked for more

Integrating a quadratic Gaussian model produces determinant terms as well as the minimized quadratic cost. With fixed common parameters, the determinants are the same for every candidate, which is what allows this comparison. The prediction operator has eigenvalues between zero and one and is a contraction in the P metric; a Euclidean contraction would need extra structure. The discrepancy alone would favor an ideal equal-profile candidate over distinct competitors, when the coupling is positive and A has full row rank. That's an alternative score, separate from the evaluated stereo method.

The implemented estimator has several distinct layers



Theorems describe specific layers; empirical evaluation describes the complete estimator.

Takeaway. The exact minimum of a computed score list certifies the best listed candidate for that score.

Dissertation: Section 4.5; Section A.2.

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The implemented estimator has several distinct layers

Likely question: How do the individual theorems relate to the complete stereo estimator?

[Follow the diagram in order; attach each guarantee to its specific box.]

Answer

Each guarantee attaches to one layer of the estimator. The estimator first rectifies and motion-compensates the events, then bins and normalizes them. For each candidate disparity, it runs the frozen-weight solve with damping and computes a score at the returned iterate. Spatial masking, averaging, normalization, and semi-global matching then adjust the candidate costs.

The final step picks a disparity from that score list, interpolating around the finest-level minimum, and converts a positive valid disparity to depth. Semi-global matching encourages spatial coherence by penalizing disparity changes along several image paths. The inner uniqueness theorem covers the frozen quadratic solve. The selection and margin theorems cover candidate decisions under their hypotheses. The benchmark tests all the stages together, so its results describe the complete estimator and its scored population.

If asked for more

A few distinctions inside those layers. Solve weights are clipped; scoring weights are recomputed without clipping. Finite guidance penalties go into averaging, so the preference they create is soft. Picking the minimum of the final finite list is exact relative to that list; whether it's the physically correct match is a separate question. Likewise, path aggregation is its own specified spatial algorithm. The reported benchmark includes score-minimum interpolation and leaves out the separate hole-filling and refinement stage.

What does the dissertation contribute mathematically?

- A transparent temporal inversion with an explicit observation null space and quantitative fixed-weight guarantees.
- One well-posedness and stability theorem for several event-derived quadratic flow energies, with representation-specific hypotheses.
- A distinction between a coercive spline regularizer and a nonconvex contrast objective, plus full-objective existence.
- Stereo block structure, profiled-cost bias, ideal selection and robustness criteria, and a scoped adaptive-weight analysis.

Takeaway. Classical proof tools connect the models; the event-specific formulation, hypotheses, consequences, and limitations are the research content.

Dissertation: Section 5.3 and Table 5.1.

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What does the dissertation contribute mathematically?

Likely question: What is the mathematical contribution if the proof tools are classical?

[Use one specific consequence, such as attenuation or integrated directional information, after listing the contributions.]

Answer

The contribution is building event-based inverse models and analyzing them with precise assumptions and consequences. For temporal reconstruction, I identify the observation null space, set up a transparent regularized inverse, and derive quantitative guarantees for the fixed-weight solution. For flow, I put several event-derived coefficient constructions into one quadratic framework, and prove well-posedness and stability under explicit directional-information conditions.

For contrast maximization, I separate the coercivity that the spline regularizer supplies from the nonconvex focus term, and prove existence for the full finite-dimensional objective. For stereo, the block structure leads to a profiled-cost decomposition, an explicit attenuation mechanism, selection and robustness conditions, and a separate analysis of the adaptive weights. Classical tools make those arguments possible.

If asked for more

Stereo is a good example of why this matters. A unique latent reconstruction at each candidate tells us how the inner solve behaves. The cost decomposition then shows that the candidate comparison also carries monocular regularization bias. The attenuation example exposes that effect, and the separation and margin conditions are sufficient for picking and keeping the right candidate. Similarly, the common flow theorem makes explicit which directional information each event representation must supply, so that one global regularizer determines both velocity components.

Literature anchors and where to find the claims

Image reconstruction

- Rebecq et al. [12]
  Events-to-Video.
- Zhang, Yezzi & Gallego [6]
  Image reconstruction as a linear inverse problem using optical flow.
- Antil–Sayre [2]
  Bilevel inverse problems in neuromorphic imaging.

Sensor overview

Gallego et al. [1]: *Event-Based Vision: A Survey.*

Flow models and analysis

- Horn–Schunck [16]
  Determining Optical Flow.
- Benosman et al. [50]
  Event-Based Visual Flow.
- Weickert–Schnörr [57]
  Convex regularizers for image motion.
- Gallego, Rebecq & Scaramuzza [3]
  A Unifying Contrast Maximization Framework.

Evaluation data

Zhu et al. [35]: *The Multivehicle Stereo Event Camera Dataset.*

Dissertation: Chapter 2; Section 3.2; Section A.2. Numbers match the dissertation bibliography; full entries on the closing reference pages.

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Literature anchors and where to find the claims

Likely question: Which prior work underlies these models and where are your detailed arguments?

[Use the relevant literature item for the question; refer to the dissertation chapter for proof or protocol details.]

Answer

Here are the main connections. For image reconstruction, Rebecq and colleagues learn an event-to-video map; Zhang, Yezzi, and Gallego pose a flow-based image inverse problem; and Antil and Sayre combine events with frames in a bilevel model. Gallego and colleagues' survey covers the sensor and the field. Horn and Schunck give the classical variational flow framework, where a data constraint is coupled to spatial smoothness. Benosman and colleagues develop the timestamp geometry behind event-based visual flow. Weickert and Schnörr establish well-posedness for classical quadratic flow. Gallego, Rebecq, and Scaramuzza introduce contrast maximization, which motivates the separate nonconvex formulation. And the MVSEC paper describes the multivehicle stereo event dataset used for evaluation.

The particular models, assumptions, and results are developed in the dissertation. Chapter 2 covers temporal reconstruction; Chapter 3, the event-derived flow models and the contrast-maximization analysis; Chapter 4, the stereo formulation and selection analysis. The appendix specifies the evaluation protocol. The full proofs and experimental definitions are in the chapters.

If asked for more

The key distinction is between a framework I build on and a result proved for the models in this dissertation. For example, Horn–Schunck supplies the data-plus-smoothness structure, while the event-derived coefficients and their integrated identifiability assumptions determine how the present theorem applies. The time-surface literature motivates the timestamp geometry, and the crossing assumptions pin down the exact constraint used here.

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. Stability controls sensitivity to the data; accuracy is a separate property.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

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Stable estimates can still have regularization bias

Likely question: If the reconstruction is unique and stable, how close is it to the true signal?

[Point to the two error terms in order: bias, then propagated observation noise.]

Answer

Stability says how much the estimate moves when the observations change. Accuracy also depends on the bias that regularization introduces. Here we assume a fixed temporal grid, a positive scalar regularization parameter, and observations equal to the exact temporal differences of a reference signal plus additive error.

Substituting that model into the reconstruction formula gives the error identity shown. The first term is regularization bias: the penalty shrinks parts of the reference signal. The second is the observation error passed through the reconstruction operator. Increasing regularization affects both, typically trading more shrinkage for better noise control. The constant component deserves special mention, because temporal differences remove it completely: the penalty selects that baseline, while the observations determine the changes.

If asked for more

For a mean-zero reference signal on a fixed grid, exact noiseless difference data recover the signal as the regularization parameter goes to zero, because every nonconstant mode's gain approaches one. With persistent observation noise, a refinement statement needs joint control of the noise level, the grid spacing, and the regularization. Any mismatch between physical events and the increment model adds another observation-error term.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate; the physical sensor adds sub-threshold ambiguity.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

BACKUP 34 / DECK 59

The physical event stream loses more than one baseline

Likely question: Does the constant nullspace describe all the information lost by an event camera?

[Point to the amplitude inequality, then compare the constant and oscillating profiles.]

Answer

No. The constant null space is exactly the ambiguity of the discrete first-difference model. The physical threshold sensor loses more, because it reports only threshold crossings. This slide gives an explicit example, with the same initial reference for both signals.

The first log intensity is constant. The second oscillates sinusoidally around that value, with amplitude strictly smaller than the contrast threshold. Its change from the reset reference stays inside the threshold the whole time, so both signals produce the same empty event stream. Exponentiating either one gives a positive physical intensity. So even time-varying signals can share an event stream. This matters for reading the reconstruction theorem: it proves uniqueness for the regularized increment surrogate. Connecting that surrogate to the physical signal also takes assumptions about the threshold measurement and about what the representation retains.

If asked for more

An empty stream makes the ambiguity easy to see. In a nonempty stream, a perturbation can shift event times by moving threshold crossings. A perturbation supported away from the crossings preserves the schedule, provided a positive threshold margin remains throughout its support. That has to be checked for the specific signal and event reference.

Integrated directional information is necessary for uniqueness

Let Ω be connected, bounded, and Lipschitz; $\alpha > 0$; $\mathbf{B} \in L^\infty$ symmetric PSD; and $b \in L^2$. Minimize over unrestricted $H^1(\Omega; \mathbb{R}^2)$ fields. Set $M_B = \int_\Omega \mathbf{B}$. If M_B is singular, choose $c \neq 0$ with $M_B c = 0$. Then $\mathbf{B}c = 0$ a.e. and

$$E(\mathbf{v} + tc) = E(\mathbf{v}) - t\, c^\top \int_\Omega b.$$

If the coefficient is nonzero, E is unbounded below; if zero, it is flat in direction c.

Takeaway. The integrated matrix condition is necessary for a unique minimizer in this unrestricted quadratic model.

Dissertation: Theorem 3.6.1; direct converse for its quadratic energy.

BACKUP 35 / DECK 60

Integrated directional information is necessary for uniqueness

Likely question: Why is positive integrated directional information necessary for unique optical flow?

[Use the displayed energy identity to distinguish a nonzero linear slope from a flat direction.]

Answer

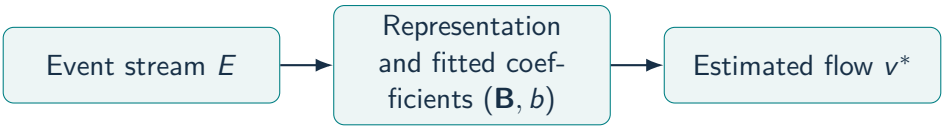
Take the quadratic energy on all H one vector fields, with a positive gradient penalty and a bounded positive semidefinite data matrix field. Suppose the integrated matrix is singular. Then it has a nonzero null vector, c, and positivity forces the data matrix to annihilate c almost everywhere.

Now add any multiple of c to a velocity field. Both quadratic terms stay the same: the gradient is unchanged because c is constant, and the data quadratic has zero curvature in that direction. Only the linear forcing term can change. If its coefficient along c is nonzero, choosing the sign of the multiple sends the energy down without bound. If that coefficient is zero, the energy is constant along the whole family. Either way there's no unique minimizer. So for this unrestricted model, uniqueness requires the integrated matrix to be positive definite.

If asked for more

The key step uses positive semidefiniteness pointwise. With B the data matrix, the nonnegative function c transpose B c has zero integral, so it's zero almost everywhere, and the square-root factorization then gives B c equal to zero. In the flat case, whether a minimizer exists at all is a separate question. Boundary constraints or a magnitude penalty can control the constant direction, and they change the uniqueness criterion accordingly. On a disconnected domain, the gradient leaves one constant per component, so each component needs its own positive-definite integrated matrix; a sum over the whole domain can hide one component's ambiguity.

Coefficient stability is one stage of event-to-flow stability



The theorem controls coefficient-to-flow sensitivity. A Lipschitz representation map could compose with that bound.

Takeaway. An end-to-end guarantee also needs a metric and stability result for event preprocessing.

Dissertation: Theorem 3.6.1, Proposition 3.3.7; supplementary scope explanation.

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Coefficient stability is one stage of event-to-flow stability

Likely question: How does your stability theorem translate into stability with respect to the raw events?

[Point first to the coefficient-to-flow arrow, then to the preprocessing arrow.]

Answer

The theorem controls the second arrow in this diagram: how the velocity changes when the fitted coefficient fields change. The first arrow covers the event representation, neighborhood selection, and any fitting used to build those coefficients. An event-to-flow theorem would connect the two arrows quantitatively.

That needs two things. First, a metric on event streams. For example, moving one event by a tiny amount of time changes its timestamp only slightly, but how far apart the two streams are depends heavily on the metric. Second, a bound on how that event perturbation changes the coefficient fields, in the norms the variational theorem uses. With a Lipschitz bound for the preprocessing and a shared positive coercivity bound, the two estimates compose. That's a concrete extension: prove representation stability in a suitable event metric, then carry it through the quadratic solve, which is already controlled.

If asked for more

Take one positive unit event and shift its time by any nonzero amount. The difference of the two point masses has total variation norm two, while their Wasserstein-one distance is just the size of the shift. Hard temporal bins can jump when an event crosses a bin boundary; smoothing with a fixed bandwidth supports a different continuity analysis. That's why the metric and the representation have to be specified before claiming stability for the whole chain from events to flow.

Refinement requires a choice of continuum scaling

Illustrative smooth, uniform-time model only: spacing h, increments $y_i = hg_i$, and target energy

$$\frac{1}{2}\int |f' - g|^2 dt + \frac{\beta_{\text{time}}^2}{2}\int |f|^2 dt \quad \rightsquigarrow \quad \frac{1}{2h}\|\mathbf{A}f - y\|^2 + \frac{\beta_{\text{time}}^2 h}{2}\|f\|^2 .$$

Multiplying the discrete objective by h gives the Chapter 2 form with

$$\lambda_h = \beta_{\text{time}} h.$$

Takeaway. The penalty's grid scaling depends on the continuum objective being approximated.

Dissertation: Eq. 2.7, Eq. 2.8; supplementary smooth-model scaling illustration.

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Refinement requires a choice of continuum scaling

Likely question: How should the temporal regularization parameter change when you refine the time grid?

[Point to the factors one over h and h before identifying lambda at spacing h.]

Answer

That depends on which continuum energy we want to approximate. This slide uses a smooth illustrative model with uniform spacing h. The observations are increments, so each one is h times a sampled derivative-like value, g, and the difference residual also carries a factor of h.

The target energy is one half the integral of the squared derivative residual, plus beta-time squared over two times the integral of the squared signal. The first integral becomes the squared increment residual over two h. The second becomes beta-time squared times h over two, times the sum of squared samples. Multiplying the whole discrete objective by h leaves its minimizer unchanged and puts it in the Chapter 2 form. Matching coefficients then gives lambda at spacing h equal to beta-time times h, where beta-time is the fixed continuum penalty strength. So for this target energy and this observation convention, the penalty scales linearly with the grid spacing.

If asked for more

This calculation matches coefficients; a convergence proof would also have to control the quadrature, the approximation spaces, and the data errors. Endpoint quadrature may add endpoint weights. Physical event data also raise separate choices: the smoothing bandwidth, and how the noise changes under refinement. Those can lead to a different scaling. The principle is to specify the continuum functional and the units of the observations first, then derive the discrete penalty from them.

Strong smoothness selects one constant flow

For fixed $\Omega, \mathbf{B}, b$ satisfying the unified theorem, with unrestricted H^1 fields and the same α^2 gradient coefficient,

$$\alpha \to \infty \quad \Longrightarrow \quad v_\alpha \to c \text{ in } H^1, \qquad c = \left(\int_\Omega \mathbf{B} \right)^{-1} \int_\Omega b.$$

Takeaway. Very strong spatial regularization removes variation and leaves the data-balanced constant field.

Dissertation: Theorem 3.6.1; direct strong-regularization limit.

BACKUP 38 / DECK 63

Strong smoothness selects one constant flow

Likely question: What does the optical-flow estimator approach when spatial regularization becomes very strong?

[Point to the limiting constant and identify the integrated matrix as the quantity controlling the mean.]

Answer

It approaches one constant vector. With the domain and coefficients fixed under the unified theorem's assumptions, increasing alpha makes spatial variation more and more expensive, because alpha squared multiplies the gradient energy. The integrated data select the constant: it's the inverse of the integrated matrix B applied to the integral of the forcing b.

That formula has a direct meaning. The data term pools all the directional constraints across the domain, and the constant balances them in aggregate. Positive definiteness of the integrated matrix makes that balance unique. The convergence is in H one, so it controls both the velocity field and its gradient. Very strong regularization therefore gives a spatially uniform compromise set by the data, with local velocity differences progressively suppressed.

If asked for more

For the proof, subtract the limiting constant and test the weak equation with the difference. The remaining forcing has zero integral, so Poincaré's inequality bounds its pairing by the gradient norm, which gives a gradient bound of order alpha to the minus two. Testing with constant vectors gives an integrated data balance. Positive definiteness of the integrated matrix then controls the mean of the difference, which completes convergence in H one.

Transporting a scalar differs from conserving a density

Scalar transport and source-free density conservation are

$$V_t + v \cdot \nabla V = 0, \qquad V_t + \nabla \cdot (Vv) = 0, \qquad \nabla \cdot (Vv) = v \cdot \nabla V + V \nabla \cdot v.$$

They coincide when $V \nabla \cdot v = 0$, in particular for divergence-free flow.

Takeaway. The continuity equation adds a density-compression term to scalar transport.

Dissertation: Section 3.4.1, Scalar transport and density conservation (p. 85); direct product-rule comparison.

BACKUP 39 / DECK 64

Transporting a scalar differs from conserving a density

Likely question: Why use scalar transport rather than a continuity equation for the event representation?

[Point to the product-rule term containing the divergence of velocity.]

Answer

The two equations make different assumptions about the represented quantity. Scalar transport says a pattern value stays constant along each moving trajectory. A continuity equation treats the quantity as a conserved density, so its value also changes when the flow compresses or expands space.

Expanding the divergence in the continuity equation gives the scalar transport terms plus the represented value times the divergence of the velocity. That extra term is the compression effect. The two equations agree when it's zero, for example in divergence-free flow. Event representations come from threshold-triggered observations and preprocessing, so treating them as a conserved physical density needs an extra model. In the dissertation, scalar transport is the constraint being analyzed. A conservation formulation would possibly need a source term for the creation or loss of events.

If asked for more

A one-dimensional example separates the two. Let the velocity equal the position, and let the represented value be the constant one. Its time and space derivatives are zero, so scalar transport holds. But the divergence of value times velocity is one, so source-free conservation has a residual of one. A source term of one would balance it. This shows exactly where the compression term enters.

If asked whether this could measure fluid flow

Event cameras are already used for particle velocimetry, for example by Willert and Klinner. Seeded particles give the flow a texture that moves with it, close to the scalar-transport model; conservation fits when the transported quantity is a density. Seeding also supplies texture in every direction, so the integrated identifiability condition is easy to meet. A known reference flow would test how well the representation describes the motion, and the new imaging lab makes those controlled acquisitions possible.

Correlated errors change the uncertainty; coercivity holds

For a centered symmetric regression design Z with $Q = Z^\top Z \succ 0$, timestamp-error covariance Σ gives

$$\mathrm{Cov}(\hat{g}) = Q^{-1} Z^\top \Sigma Z Q^{-1}.$$

This reduces to $\sigma^2 Q^{-1}$ when $\Sigma = \sigma^2 I$.

With any $\Gamma \succ 0$, the temporal system matrix is $\mathbf{A}^\top \Gamma^{-1} \mathbf{A} + \mathbf{\Lambda}^2 \succ 0$ when $\mathbf{\Lambda} \succ 0$.

Takeaway. The deterministic quadratic solve stays coercive, while statistical uncertainty depends on the error covariance.

Dissertation: Proposition 3.3.7; Eq. 2.8, Theorem 2.2.5; direct correlated-error extension.

BACKUP 40 / DECK 65

Correlated errors change the uncertainty; coercivity holds

Likely question: What changes if the timestamp errors are correlated?

[Point to the covariance matrix in the sandwich formula, then to the positive regularizer in the temporal matrix.]

Answer

Correlation changes the uncertainty, through the full error covariance matrix. For the centered symmetric regression design on this slide, the slope estimate is a linear map of the timestamps. Applying that map to the covariance gives the sandwich formula shown: inverse Q, then the design transpose, the covariance, the design, and inverse Q again. With independent errors of common variance, it reduces to the variance times inverse Q.

The reconstruction question is separate. If a positive definite covariance matrix Gamma is used in a weighted temporal fidelity term, its inverse supplies the precision weights. Adding the positive magnitude penalty keeps the temporal system matrix positive definite. So the regularized solve is still unique, with stability constants that reflect the chosen weights. Correlation affects the uncertainty and the weighting model; coercivity comes from the positive quadratic terms.

If asked for more

A common error mean cancels from the slope when the design is centered, even with correlated errors. Smoothing changes the covariance to the smoothing operator times the covariance times its transpose. That matrix can be singular, so an inverse-covariance formulation needs either a positive definite covariance or a treatment of its null space. A Gaussian likelihood reading needs a Gaussian assumption on top of that; the covariance identity itself only needs finite second moments.

Optical flow: further analysis

- Time-surface gradient estimation: bias, variance, and design.
- Contrast maximization: what it models, known-motion evidence, objective, and existence proof.

Regression quantifies coefficient uncertainty; contrast maximization changes the objective being optimized.

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Optical flow: further analysis

[Navigation only. Silent navigation cue: select the regression or contrast-maximization slide matching the question.]

Time-surface gradient estimation: bias versus variance

Fit a plane to timestamps at offsets $\mathbf{z}_j$ around a pixel.

$$Q = \sum_j \mathbf{z}_j \mathbf{z}_j^\top \succ 0.$$

For a centrally symmetric design and independent timestamp errors with common mean and finite variance σ^2:

$$\mathrm{Cov}(\widehat{\nabla T}) = \sigma^2 Q^{-1}.$$

With a Lipschitz Hessian of constant L_H on a neighborhood containing the sample disk,

$$\|\mathrm{Bias}\| \le \frac{L_H}{6} \left\|Q^{-1}\right\| \sum_j \|\mathbf{z}_j\|^4, \qquad \mathrm{MSE} = \|\mathrm{Bias}\|^2 + \sigma^2 \,\mathrm{tr}(Q^{-1}).$$

Takeaway. A larger neighborhood reduces variance but can increase curvature bias. Regression rank and velocity identifiability are different questions.

Dissertation: Proposition 3.3.7.

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Time-surface gradient estimation: bias versus variance

Likely question: What determines the accuracy of your time-surface gradient estimate?

[Point to Q for sample geometry, then to the two contributions in the mean squared error.]

Answer

Two things: the geometry of the samples, and the curvature of the time surface. The estimator fits a plane to timestamps at nearby spatial offsets, and Q summarizes the geometry of those offsets. If Q is positive definite, the samples span both spatial directions and the fitted slope is determined. With a centrally symmetric design and independent timestamp errors of common mean and variance, the slope covariance is the noise variance times inverse Q.

The bias comes from fitting a plane to a curved surface. Central symmetry cancels the quadratic Taylor term in the slope. A Lipschitz bound on the Hessian controls the remaining cubic error and gives the bias bound shown, which holds for the actual finite set of samples. The mean squared error is then the squared bias plus the trace of the covariance. A larger neighborhood can improve the geometry and reduce the variance, but it also takes in more curvature. That's the bias–variance tradeoff here.

If asked for more

The design matrix depends on the sample positions. A well-conditioned fit can estimate the time-surface gradient accurately even at a moving straight edge, and the tangential velocity is still free, because one reliable normal constraint fixes only one component of motion. So regression rank and velocity identifiability answer different questions. The uncertainty formula also assumes an explicit timestamp-error model; a model for event counts would need its own derivation.

Contrast maximization finds the motion that sharpens the events

What it models

- A moving edge leaves a smeared trail of events over the time window.
- Slide each event back along a candidate flow to one reference time. The correct flow stacks the trail into a sharp edge; image variance measures the sharpness.
- It uses individual events in continuous time, with no time surface, bins, or fitted gradients. Binned models blur when an edge moves during a bin.

Known-motion evidence

AEE (px) at window 22; a zero flow field scores 3.00.

Scene	Best other (struct. tensor)	CMAX
Idealized stripes	2.15	**0.49**
Simulated stripes (v2e)	1.92	**1.19**
Simulated checkerboard (v2e)	0.98	**0.32**

On the stripes, CMAX recovers the diagonal motion: warping the events over the whole window brings in evidence from the object's boundary.

Takeaway. CMAX has the lowest error in every known-motion scene.

Dissertation: Section 3.5; Figure 3.3 and Figure 3.4; contrast maximization [3, 55]; v2e [46].

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Contrast maximization finds the motion that sharpens the events

Likely question: What does contrast maximization model, and why is it worth pursuing?

[Describe the warp on the left, then point to the CMAX column of the table.]

Answer

An edge moving across the sensor produces events all along its path during the time window, so a plain accumulation of those events looks like a smeared trail. Contrast maximization slides each event back along a candidate flow to one reference time. When the candidate matches the true motion, the trail stacks into a sharp edge. Image variance measures that sharpness, so the method searches for the flow that makes the warped event image sharpest.

The method works directly on individual events and their own timestamps. The quadratic models first summarize events as a time surface or in temporal bins, and an edge that moves during a bin blurs that summary. On the known-motion scenes in Chapter Three, contrast maximization has the lowest endpoint error in all three: 0.49 pixels against 2.15 for the structure tensor on idealized stripes, 1.19 against 1.92 on simulated stripes, and 0.32 against 0.98 on the simulated checkerboard. A zero flow field scores 3.00. On the stripes it recovers the diagonal motion, because warping events over the whole window brings in evidence from the object's boundary.

If asked for more

These numbers come from one representative window, chosen by the rule that minimizes deviation from each method's median error, on simulated scenes with known motion. On the idealized stripes, contrast maximization also assigns flow to the background around the object. The dissertation's endpoint errors come from these known-motion scenes. The guarantee for contrast maximization is that a global minimizer exists; the solver is a local quasi-Newton method. The contrast-maximization objective and existence backups give those details.

How the known-motion comparison was evaluated

Aspect	Reported protocol
Stimulus	A 60×40 px object translates diagonally at 3 px per window; direct idealized events and v2e-simulated events [46].
Settings	Quadratic variants: $\alpha_{\text{smooth}} = 20$, no pyramid. Contrast maximization: 100 L-BFGS [71] iterations.
Selection	Window 22 minimizes summed deviation from each variant's median AEE over windows 1–31.
Display / error	Common arrow scale; AEE is mean Euclidean displacement error, in pixels per evaluation window.

Equal numerical α values give different stencil-dependent smoothing strengths.

Takeaway. These controlled examples show directional-information effects at the reported solver settings.

Dissertation: Figure 3.3 and Figure 3.4; smoothing stencils (p. 164).

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How the known-motion comparison was evaluated

Likely question: How were the flow examples selected, and how fair is the comparison?

Answer

These are controlled tests of the geometry behind the models. A sixty-by-forty-pixel object translates diagonally by three pixels per evaluation window. The chapter uses idealized events generated directly, and events from the v2e simulator. Stripes emphasize the normal-motion constraint; the checkerboard supplies edge directions along both axes.

All the panels share one arrow scale and one selected window. Window twenty-two minimizes the summed deviation of each variant's endpoint error from that variant's median over windows one through thirty-one. AEE averages the Euclidean displacement error over the scored locations, in pixels per evaluation window. A zero estimate scores three pixels here.

The quadratic variants use alpha equal to twenty, without a pyramid; contrast maximization uses one hundred L-BFGS iterations. The implementations use different spatial stencils, so the same numerical alpha gives different effective smoothing. Their computational budgets also differ.

So the comparison illustrates directional information, and how these implementations behave at these settings. A dataset-level accuracy ranking needs its own evaluation and parameter-calibration protocol.

If asked for more

VOX uses an eight-neighbor interior stencil, and the structure-tensor implementation uses a conductance-weighted four-neighbor stencil. In the isotropic interior, their continuum smoothing coefficients are alpha squared times h squared over three, and over four, respectively. That describes the stencils only; the full models differ in other ways too.

The separate MVSEC contrast-maximization configuration caps iterations at ten; the controlled figures here use one hundred. The chapter calls the AEE evaluation mask M; the experiment's evaluator defines its exact population. The convergence theorem uses simultaneous block-Jacobi updates with fixed coefficients. The evaluated time-surface implementation uses in-place updates and event-dependent reference times; the convergence backup explains the schedule that's proved.

Contrast maximization changes the optimization problem

Warp each event to a common normalized reference time:

$$\mathbf{x}_k^w = \mathbf{x}_k + (\widetilde{t}_{\mathrm{ref}} - \widetilde{t}_k)\mathbf{q}(\mathbf{x}_k), \qquad \mathbf{q} = T\mathbf{v}_{\mathrm{phys}}.$$

Represent $\mathbf{q}$ by a tensor-product cubic B-spline on a uniform knot grid ($h = 8\,\mathrm{px}$), with coefficients $\mathbf{c}$; both components share the basis.

$$F(\mathbf{c}) = -\operatorname{Var}(I_{\mathbf{c}}) + \mathbf{c}^\top \mathbf{K}_{\mathrm{reg}}\mathbf{c}, \qquad \mathbf{K}_{\mathrm{reg}} \succeq \tau_{\mathrm{ridge}}\mathbf{I}, \quad \tau_{\mathrm{ridge}} > 0.$$

- A good warp concentrates aligned events.
- A flexible incorrect warp can also concentrate events by collapsing them [55].

Takeaway. The full objective is generally nonconvex, outside the quadratic flow theorem.

Dissertation: Section 3.5, Proposition 3.5.1, Proposition 3.5.2; contrast maximization [3].

BACKUP 45 / DECK 70

Contrast maximization changes the optimization problem

Likely question: How does contrast maximization differ from your quadratic optical-flow model?

[Point to the warped locations, then identify the ridge lower bound in the regularizer.]

Answer

In contrast maximization, the event image itself changes as the motion estimate changes. Each event is warped to a common reference time, and well-aligned events produce a concentrated image. The objective rewards that concentration through image variance, and adds a quadratic regularizer on the motion coefficients.

The motion variable q is displacement over the normalized time window: physical velocity times the window duration. We represent that field with a tensor-product cubic B-spline on a uniform grid with eight-pixel knot spacing; the two flow components share the basis. The spline is continuous, so the field has a well-defined value at every event location, and the whole flow is a small coefficient vector rather than one unknown per pixel. Because the warp depends on the coefficients, the full objective is generally nonconvex. The regularizer does two separate jobs. Spatial diffusion controls variation across the field, and a positive coefficient ridge controls the overall size of the coefficients, including constant motion. So the existence argument uses finite-dimensional continuity and coercivity.

If asked for more

The basis is clamped at the image border with a Neumann boundary treatment, the natural condition for a rectangular frame. Concentration alone can also reward a wrong warp that collapses events together, which is why the regularization and the model restrictions matter. Continuous splines can be evaluated at points; a general two-dimensional H one field is an equivalence class, and needs more regularity before point values make sense.

If asked why analyze the quadratic family at all

Because the two answer different questions. The quadratic models are where I can prove what the events determine: a unique flow, stable in its coefficients, computed by an iteration that converges, under a condition you can check in the scene — edges in two orientations. That condition is what explains the stripe result. Contrast maximization is more accurate on these tests, but its objective is nonconvex: I can prove a global minimizer exists, and the solver is local. So one tells us when the data identify the motion; the other is a stronger estimator with weaker guarantees.

A global minimizer exists, even though the solver is local

For finitely many fixed event weights, let $W_{\mathrm{tot}} = \sum_k |w_k|$. The bilinear accumulation is continuous and satisfies

$$\mathrm{Var}(I_{\mathbf{c}}) \le \frac{W_{\mathrm{tot}}^2}{HW}.$$

Therefore

$$F(\mathbf{c}) \ge -\frac{W_{\mathrm{tot}}^2}{HW} + \tau_{\mathrm{ridge}} \|\mathbf{c}\|^2 \longrightarrow \infty.$$

Proof mechanism

Continuous objective + bounded focus gain + coercive ridge $\Rightarrow$ compact sublevel set $\Rightarrow$ Weierstrass.

Takeaway. A global minimizer exists by finite-dimensional coercivity; the local solver returns a local solution.

Dissertation: Proposition 3.5.2 and evaluated-solver scope.

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A global minimizer exists, even though the solver is local

Likely question: What guarantees that contrast maximization has a global minimizer?

[Follow the lower bound from bounded focus gain to the positive quadratic ridge.]

Answer

Two facts: the focus term is bounded, and the ridge grows. There are finitely many events with fixed weights. Their total absolute weight bounds how much mass bilinear accumulation can put on the image. In particular, both the sum of absolute pixel values and the largest absolute pixel value are bounded by that total weight. Bounding the variance by the second moment gives the focus bound shown.

So the negative variance term is bounded below. Meanwhile, the positive coefficient ridge grows quadratically with the coefficient norm, so the full objective goes to infinity as the coefficients grow without bound. Bilinear accumulation is continuous in the warped locations, so the finite-dimensional objective is continuous too. Its relevant sublevel sets are compact, and the extreme value theorem gives a global minimizer. That proves existence for the model; the local optimizer used in the experiments returns whatever solution its numerical search reaches.

If asked for more

The continuity holds even when events cross pixel boundaries, because the bilinear tent weights join continuously. Differentiability is only piecewise; that matters for the numerical optimizer, while the existence proof needs only continuity. Coercivity relies on a strictly positive ridge for the fixed discretization. Under refinement, that control holds in function space: the ridge is a uniform L two penalty, which Backup 68 shows. The proof gives existence; global uniqueness, or global convergence of a local solver, would need additional structure.

If asked why the knots are uniform when events cluster at edges

Uniform knots keep the stiffness matrix and the ridge scaling simple, and the proposition assumes them. Events are already adaptive samples, so knots refined where events are dense, for example hierarchical B-splines, are a natural extension. The existence proof needs only a nonnegative partition of unity. For the uniform bound, weight each coefficient in the ridge by its basis integral, so it stays a lumped L two mass term.

Stereo depth: full Chapter 4 arc

1. Coupled model and the frozen-weight inner solve.
2. Sum and difference modes; the profiled candidate cost.
3. The explicit counterexample and the correct-selection theorem.
4. Winning-margin stability and the non-identifiability example.
5. Heteroscedastic weights and IRLS fixed-point analysis.
6. MVSEC benchmark, weighting comparison, and hard fusion.

These results separate the inner temporal solve from the outer correspondence decision.

BACKUP 47 / DECK 72

Stereo depth: full Chapter 4 arc

[Navigation only. Silent navigation cue: choose the matching model, selection, IRLS, or benchmark page.]

Stereo depth in one slide: what Chapter 4 adds

At each candidate disparity d, solve two coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left(\tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

1. **Inner solve.** The Hessian is $\succeq \lambda_{\min}^2 \mathbf{I}$: the temporal proof plus a nonnegative coupling term.
2. **Profiled cost.** $C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + D_d$: the monocular term moves with d. In an exact two-sample example the perfect match costs $\frac{1}{3}$, an attenuated one $\frac{9}{40}$: a weaker right-view copy can outscore the true match.
3. **Selection.** A separation condition makes the correct candidate the unique winner; a margin Δ tolerates cost perturbations below $\Delta/2$ (e.g., noise events, threshold mismatch between sensors).
4. **Reweighting.** The IRLS map has a fixed point (Brouwer) and contracts under a sufficient condition.

MVSEC indoor flying: soft-coupled WLS is best on Flying 2 (58.35 cm) and log RMSE $\times 100$ (44.54), second elsewhere, with 2.66 M scored points versus 1.20 M for the strongest prior method.

Takeaway. Coupling keeps every inner solve well posed; the right correspondence also needs a cost margin.

Dissertation: Theorem 4.6.3, Proposition 4.6.7, Proposition 4.6.8, Proposition 4.6.13; Table A.5; MVSEC [35], Piatkowska [78], Brouwer [92].

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Stereo depth in one slide: what Chapter 4 adds

Likely question: What does Chapter 4 prove beyond the inner solve?

[Move through the numbered results deliberately.]

Answer

Three things: what the candidate cost actually compares, when the correct candidate wins, and when the reweighting converges.

The model takes one temporal profile per view and adds a soft penalty on their difference. For fixed nonnegative observation weights, positive magnitude weights, and nonnegative coupling, the temporal coercivity proof carries over directly, so every frozen inner problem has a unique solution.

The correspondence analysis looks at the unweighted cost for a single candidate, before spatial aggregation. After minimizing over the profiles, that cost has a monocular term for each view plus a coupling discrepancy. The right-view term changes with the candidate. In the exact example, the perfect match costs one third, while an attenuated candidate costs nine fortieths: a right view that's a weaker copy of the left can beat the true match, so the cost favors quieter candidates.

A separation condition guarantees that this model picks the correct candidate. Once the winner has a positive margin, cost errors each smaller than half that margin, say from noise events or mismatched thresholds between the two sensors, leave the winner unchanged.

For the reciprocal reweighting map, a fixed point exists. Under a sufficient contraction condition, it's unique and the iteration converges.

If asked for more

On the MVSEC benchmark, WLS leads on indoor flying two and narrowly on the aggregate log RMSE, while scoring more pixels than the local adaptation of Piatkowska and colleagues' method. Each method is scored on its own valid pixels. The separation results concern unweighted candidate-local costs before aggregation.

Stereo alignment after motion compensation

The model pairs synchronized, rectified observations of a scene point visible in both cameras.

Each camera fits its own translation within each temporal slice

$$\mathbf{x}'_c = \mathbf{x}_c + (t_{\mathrm{ref}} - s)\mathbf{v}_c, \qquad c \in \{L, R\}.$$

For simultaneous rectified correspondences,

$$d_{\mathrm{w}}(s) = d(s) + (t_{\mathrm{ref}} - s)(v_{x,L} - v_{x,R}).$$

- A horizontal warp mismatch changes the disparity presented to the matcher.
- A vertical warp mismatch moves a correspondence away from its scanline.
- Reported slices are 12.5 ms wide within a 0.25 s evaluation window.

Takeaway. A unique inner solve fits the supplied observations; physical correspondence also relies on compatible timing, calibration, and warps.

Dissertation: Motion-compensated accumulation (p. 108); Table A.1.

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Stereo alignment after motion compensation

Likely question: Why should two pixels' observed event signals agree when the rig and scene are moving?

Answer

Because, for a proposed correspondence, both pixels watch the same scene point once we account for the motion. The comparison assumes synchronized streams, calibrated rectification, and a scene point visible in both cameras.

The preprocessing also compensates motion within each slice. It fits one global translation separately in each camera and warps the events to the end of the slice. Subtracting the two warped horizontal coordinates gives the equation shown: the original disparity picks up a correction equal to the time to the reference, times the difference of the two horizontal warp velocities.

When the warps agree, the comparison is preserved. A horizontal mismatch changes the disparity the matcher sees; a vertical mismatch moves the correspondence off its row. The reported slices are twelve and a half milliseconds wide, within a quarter-second evaluation window.

The inner positive-definite system gives a unique fit to whatever observations it receives. Whether that fit reflects the physical correspondence also depends on the timing, calibration, and motion approximation used to build the observations.

If asked for more

For simultaneous points that start on the same row, subtracting the warped vertical coordinates gives the time to the reference, times the difference of the vertical warp velocities. A global translation approximates each slice; depth-dependent parallax, changing disparity, and independently moving objects can introduce mismatch.

These translations come from Chapter 4's own preprocessing fit; Chapter 3's dense-flow estimator is a separate study. Soft coupling allows the profiles to disagree, but its penalty still favors agreement. Occlusion removes the visible match, which needs a rejection or visibility treatment beyond the unique inner fit.

At each candidate, couple two temporal profiles

Suppress the pixel index; $\mathbf{y}_R = \mathbf{y}_R(d)$.

$$J_d(\mathbf{f}_L, \mathbf{f}_R; \mathbf{W}_L, \mathbf{W}_R) = \frac{1}{2}\left\|\mathbf{W}_L^{1/2}(\mathbf{A}\mathbf{f}_L - \mathbf{y}_L)\right\|^2 + \frac{1}{2}\left\|\mathbf{W}_R^{1/2}(\mathbf{A}\mathbf{f}_R - \mathbf{y}_R)\right\|^2 + \frac{1}{2}\|\boldsymbol{\Lambda}\mathbf{f}_L\|^2 + \frac{1}{2}\|\boldsymbol{\Lambda}\mathbf{f}_R\|^2 + \frac{\gamma}{2}\|\mathbf{f}_L - \mathbf{f}_R\|^2 .$$

- Data fidelity estimates each temporal profile from its own view.
- Magnitude penalties control temporal baselines.
- Soft coupling penalizes left–right disagreement.

Takeaway. The right latent profile is auxiliary: one per pixel and candidate.

Dissertation: Eq. 4.24; fixed-disparity model definition.

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At each candidate, couple two temporal profiles

Likely question: What are the unknowns in the stereo reconstruction model?

[Point to the two data terms, two magnitude terms, and coupling term in that order.]

Answer

For a fixed left pixel and candidate disparity, the unknowns are two temporal feature profiles, one per view. The left observation is fixed, and the right observation is sampled at the candidate disparity. Each profile has its own weighted data-fidelity term, with the same temporal first-difference operator.

There are also two magnitude penalties and a soft coupling term. The magnitude penalties control the temporal baselines, and the coupling penalizes disagreement between the profiles, with gamma setting its strength. During a frozen solve, the weights and the positive diagonal regularizer are fixed, and gamma is nonnegative. The right profile is an auxiliary variable for this left pixel and candidate; each candidate gets its own right-profile solve. Spatial context comes in later, when the candidate costs are aggregated. Because preprocessing transforms the observations, we interpret these unknowns as event-feature profiles.

If asked for more

Setting both weight matrices to the identity gives the unweighted model used in the explicit selection analysis. With finite coupling, the profiles can differ, at a cost, so the data terms keep information from each view. Absolute calibrated log intensity would need a calibrated observation model linking the processed features to physical units; the candidate-local optimization and its positive-definiteness proof concern the stated feature model.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

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Frozen weights give a unique, stable inner solve

Likely question: Why is the stereo solve unique when the weights are frozen?

[Point to the magnitude terms as the source of the uniform positive lower bound.]

Answer

Because with the weights frozen, every direction has positive curvature. For a fixed disparity, fixed positive weights, a positive diagonal regularizer, and nonnegative coupling, the objective is quadratic in the two latent profiles. Its Hessian is the block matrix shown. To show it's positive definite, evaluate its quadratic form on an arbitrary pair of perturbation profiles.

The result is a sum of two weighted difference energies, two magnitude-penalty energies, and the coupling energy of their difference. Every term is nonnegative. More than that, the magnitude terms alone are at least the smallest regularizer entry squared times the squared norm of the stacked perturbation. So every nonzero direction has strictly positive curvature. That gives exactly one global minimizer of the frozen quadratic, and an invertible linear system. Subtracting two such systems also gives stable dependence on the observations, as long as the matrix and weights stay fixed.

If asked for more

The inverse matrix has norm at most one over the smallest regularizer entry squared. To turn that into an observation-to-solution bound, also include the norm of the weighted observation map on the right-hand side. This theorem is about each inner solve. Candidate selection adds a discrete comparison, and changing the weights changes the quadratic itself; those stages have their own separation and fixed-point analyses.

Sum and difference modes expose the coupling

For equal blocks $\mathbf{K}_L = \mathbf{K}_R = \mathbf{K}$, introduce

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix}, \qquad \mathbf{U}^\top \mathbf{M} \mathbf{U} = \begin{pmatrix} \mathbf{K} & 0 \\ 0 & \mathbf{K} + 2\gamma \mathbf{I} \end{pmatrix}.$$

Mean profile
The coupling cancels. Shared information is fused by the temporal solve.

Difference profile
The coupling raises curvature and suppresses disagreement.

For $\mathbf{\Lambda} = \lambda \mathbf{I}$ in the unweighted model,

$$\kappa_2(\mathbf{M}) = 1 + \frac{g_{\max} + 2\gamma}{\lambda^2}.$$

Dissertation: Lemma 4.6.15, Theorem 4.6.16.

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Sum and difference modes expose the coupling

Likely question: What does increasing the stereo coupling strength actually do?

[Point to K in the mean block and K plus twice gamma in the difference block.]

Answer

Coupling acts only on the disagreement between the views. When the two temporal blocks are equal, an orthogonal change of coordinates separates the common profile from the difference profile. Adding the normal equations cancels the coupling term; subtracting them adds two gamma to the temporal precision. So coupling suppresses left-right disagreement and leaves the equation for the mean unchanged.

The same decomposition gives an exact condition number in the unweighted case with a scalar magnitude penalty. The smallest full-space eigenvalue is lambda squared, from the shared constant temporal mode. The largest is the largest temporal difference eigenvalue plus lambda squared plus two gamma. Their ratio is the condition number shown. So at fixed lambda, increasing gamma raises the full-space condition number, even as it pulls the views into closer agreement. The mean profile still needs the magnitude penalty to control its baseline.

If asked for more

The orthogonal transform preserves Euclidean eigenvalues and condition numbers. The simple block diagonalization needs equal temporal blocks; unequal adaptive weights generally mix the mean and difference coordinates. A mean-zero restriction removes the shared constant mode for suitably structured perturbations, and can give a sharper bound. Full-space conditioning still matters, because numerical errors can enter in any direction, so it's useful to keep the two settings separate.

Profile out the latent variables to see what disparity compares

In the unweighted model, let $\mathbf{K} = \mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}$ and

$$m(\mathbf{y}) = \frac{1}{2} \mathbf{y}^\top (\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top)\mathbf{y}.$$

The exact minimized candidate cost is

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + \frac{1}{4}\boldsymbol{\delta}_d^\top \left[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}\right] \boldsymbol{\delta}_d, \quad \boldsymbol{\delta}_d = \mathbf{A}^\top(\mathbf{y}_L - \mathbf{y}_R(d)).$$

- The final term measures left–right discrepancy.
- The right monocular cost also changes with the candidate.

Takeaway. The minimized regularized energy is more than a correspondence discrepancy.

Dissertation: Lemma 4.6.6, Proposition 4.5.1. Equal fixed weights admit a scaled analogue.

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Profile out the latent variables to see what disparity compares

Likely question: What quantity is actually compared when you choose a disparity?

[Point to the candidate-dependent right monocular term before the discrepancy term.]

Answer

In the unweighted model, we can solve out the two latent profiles exactly. The minimized cost splits into a left monocular cost, a right monocular cost, and a discrepancy term between the two observations. Here m is the minimum regularized temporal reconstruction cost for a single observation vector.

The left monocular term is the same for every candidate. The right monocular term changes with the candidate, and so does the discrepancy. The discrepancy is a nonnegative quadratic in the difference of the observations, passed through the transpose of the temporal operator. Its matrix is the inverse temporal precision minus the inverse precision with two gamma added. This exposes the selection mechanism: a candidate is rewarded both for matching the left observation and for having a low reconstruction cost of its own. Those two pulls can compete, which is why correspondence selection needs its own analysis.

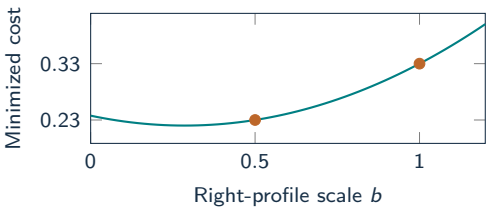
If asked for more

The derivation completes the square in the stacked latent variables, then uses the sum-and-difference transform to separate the terms. The inverse difference equals two gamma times the product of the two positive inverse matrices, so it's positive semidefinite. An exact winner of these unaggregated profiled costs minimizes the corresponding joint objective over the candidates and the latent variables. Showing that it's the physical match takes assumptions that connect that objective to the true correspondence.

An attenuated match can cost less than the perfect one

Take $\mathbf{A} = [-1, 1]$, $\mathbf{\Lambda} = \mathbf{I}$, $\gamma = 1$, $y_L = 1$, and $y_R = b$.

$$C(b) = \frac{1 + b^2}{6} + \frac{(1 - b)^2}{15}.$$



Let D denote the discrepancy term in the preceding profiled-cost formula.

$$C(1) = \frac{1}{3}, \qquad C\left(\frac{1}{2}\right) = \frac{9}{40} < \frac{1}{3}.$$

$$b^* = \frac{D(y_L)}{m(y_L) + D(y_L)} = \frac{2}{7}.$$

An attenuated wrong candidate beats the exact match if both are available.

Takeaway. Inner uniqueness, distinct observations, and correct correspondence are separate claims.

Dissertation: Proposition 4.6.7, Correspondence identifiability (p. 143). Unweighted cost before aggregation.

BACKUP 54 / DECK 79

An attenuated match can cost less than the perfect one

Likely question: Can the stereo objective favor a wrong correspondence even when the true profiles match exactly?

[Point to the exact-match point, the one-half point, and their displayed costs. Read the follow-up scalar factors slowly.]

Answer

Yes. This two-sample example shows exactly how. The temporal operator takes one difference, both magnitude penalties are the identity, and gamma equals one. The left observation is one, and the right candidate observation is a scalar, b. Solving the latent problem gives the quadratic cost in b shown here.

At the perfect match, b equals one and the cost is one third. At the attenuated candidate, b equals one half and the cost is nine fortieths, which is smaller. Over all b, the minimum is at two sevenths. Attenuation lowers the right monocular cost by more than it adds in discrepancy. Every inner solve here is still strictly convex with a unique solution. So the example isolates the selection problem: inner uniqueness gives a profile for each candidate, and whether the correct candidate wins depends on how their minimized costs compare.

If asked for more

For an observation scaled by b, the monocular part is one plus b squared, times the left monocular cost. The discrepancy part is one minus b, squared, times the discrepancy coefficient. The derivative at b equal to one is twice the positive monocular cost, so decreasing b slightly lowers the score. This concerns the unweighted local cost before aggregation; the evaluated weighted method has additional steps.

Correct disparity needs a separation condition

Write the unweighted cost as

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_d) + D(\mathbf{y}_L - \mathbf{y}_d), \qquad D(\mathbf{e}) \geq 0.$$

A sufficient correct-selection theorem

On a finite candidate set, suppose $\mathbf{y}_{d_*} = \mathbf{y}_L$, the positive regularizer is common to both views and all candidates, and $\gamma > 0$ is candidate-independent. If every competitor has

$$\mathbf{y}_d \neq \mathbf{y}_L \quad \text{and} \quad m(\mathbf{y}_d) \geq m(\mathbf{y}_L),$$

then $C(d) > C(d_*)$: the correct candidate is the unique winner.

Takeaway. The discrepancy must overcome candidate-dependent monocular bias. Stronger metric-separation criteria are also available.

Dissertation: Proposition 4.6.8(b1); additional criterion in backup.

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Correct disparity needs a separation condition

Likely question: What assumptions ensure that the correct disparity wins?

[Subtract twice the left monocular cost and point to the nonnegative monocular-cost difference and the strictly positive discrepancy.]

Answer

A sufficient condition is that every competitor costs at least as much to reconstruct on its own as the left view does. The result concerns a finite candidate set and the unweighted local cost. The positive regularizer is shared by both views and all candidates, and the coupling strength is fixed and strictly positive. At the true candidate, the right observation equals the left observation exactly.

For every competitor, assume its right observation differs from the left, and its own monocular reconstruction cost is at least the left monocular cost. Subtracting the true candidate's cost leaves two parts. The monocular-cost difference is nonnegative by assumption. The discrepancy is strictly positive for a distinct observation, because the temporal difference matrix has full row rank and gamma is positive. Their sum is positive, so every competitor costs more, and the correct candidate wins uniquely. The condition works by ruling out the cost advantage that attenuation provides in the counterexample.

If asked for more

The condition is sufficient, and it can be stronger than necessary. A competitor with a smaller monocular cost can still lose if its discrepancy is large enough. The more general metric-separation criterion covers that case. Both versions concern ideal matched profiles and candidate-local, unweighted costs. Applying the conclusion after masking, adaptive weighting, or spatial aggregation requires checking that scoring procedure and its assumptions.

A winning margin turns cost stability into selection stability

For a fixed finite set of at least two candidates with a unique winner d_*, define

$$\Delta = \min_{d \neq d_*} \left(C(d) - C(d_*) \right) > 0.$$

If $|\widetilde{C}(d) - C(d)| \leq \eta$ for all candidates, including d_*, then every $d \neq d_*$ satisfies

$$\widetilde{C}(d) - \widetilde{C}(d_*) \geq \Delta - 2\eta > 0 \qquad \text{when } \eta < \Delta/2.$$

A non-identifiability example

Two candidates with identical right-view profiles have identical local costs when the candidate procedures and parameters coincide.

Takeaway. Correspondence separation depends on the profiles' coupling and attenuation as well as temporal activity.

Dissertation: Corollary 4.6.10, Proposition 4.6.20.

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A winning margin turns cost stability into selection stability

Likely question: How much can candidate costs change before the selected disparity changes?

[Point to the two eta contributions and the strict half-margin condition.]

Answer

By just under half the winning margin. Start with a fixed finite candidate set and a unique winner. Delta is the smallest gap between the winner and any competitor. If every candidate cost changes by at most eta, the worst case lowers a competitor by eta and raises the winner by eta, so the gap shrinks by at most two eta.

So whenever eta is strictly less than half of delta, every perturbed gap stays positive and the same candidate wins. That turns a bound on continuous cost perturbations into a statement about a discrete decision. It also shows why a small margin is fragile. If two right candidates have exactly the same temporal observations, and use the same local procedure and parameters, their local costs are equal. That tie can happen even with rich temporal activity; selection stability requires a positive gap between the candidate scores.

If asked for more

The cost bound must cover the original winner as well as every competitor. At exactly half the margin, the bound allows a tie, so the strict inequality matters. An observation-based estimate can supply eta when the score parameters are fixed and the observation perturbations are controlled. Spatial aggregation can separate candidates using neighboring evidence, so the identical-profile example applies to the local score.

Depth sensitivity grows with distance

For fixed calibration and positive d and $d + \delta d$,

$$Z(d) = \frac{fB}{d}, \qquad |Z(d + \delta d) - Z(d)| = \frac{fB|\delta d|}{d(d + \delta d)}.$$

Small relative disparity error

$$|\delta Z| \approx \frac{Z^2}{fB}|\delta d|, \qquad \frac{|\delta Z|}{Z} \approx \frac{|\delta d|}{d}.$$

At the same disparity precision, absolute depth error grows quadratically with distance.

Illustrative exact calculation

	Reference d	Estimated d
Farther point	4 px	5 px
Nearer point	20 px	21 px

Depth is underestimated by 20% and 4.76%, respectively.

Takeaway. Read depth errors with the disparity precision, depth range, calibration, and scored population.

Dissertation: Eq. 4.11 and Eq. 4.12; stereo geometry [80]; example percentages are illustrative arithmetic.

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Depth sensitivity grows with distance

Likely question: What does a pixel of disparity error mean for depth accuracy?

Answer

It depends on distance: the same pixel matters much more far away. Depth is focal length times baseline divided by disparity. With fixed calibration, the exact change under a disparity perturbation is the difference formula on the slide. For a small relative disparity error, the absolute depth sensitivity is depth squared over focal length times baseline. So farther points turn the same disparity error into a larger depth error.

The examples hold calibration fixed and add one pixel of disparity. Going from four to five pixels underestimates depth by twenty percent. Going from twenty to twenty-one underestimates it by about four point seven six percent. These are illustrative calculations.

That's why we read depth error together with the depth range, the disparity precision, and the scored population. A stable winning label is one stage; interpolating it and converting it to depth each add their own sensitivity.

If asked for more

The approximation needs the disparity error to be small relative to the disparity; the exact identity holds for any positive original and perturbed disparities. If the calibration also changes, the first-order relative depth change is the relative focal-length change plus the relative baseline change, minus the relative disparity change.

On the benchmark, finite in-band estimates and ground truth define each method's scored population; the coverage slides explain that selection. This calculation explains how disparity errors propagate within those populations. Attributing a method's error tails would take a separate analysis of its returned disparities.

Heteroscedastic weights: statistical motivation and scope

As a working raw-count model,

$$N_+ \sim \text{Poisson}(\theta_+), \qquad N_- \sim \text{Poisson}(\theta_-),$$

independently. For $b = N_+ - N_-$ (Skellam [87]),

$$\mathbb{E}b = \theta_+ - \theta_-, \qquad \text{Var}(b) = \theta_+ + \theta_-.$$

$$|\theta_+ - \theta_-| \leq \theta_+ + \theta_-, \quad \text{with equality iff at least one rate is zero.}$$

The evaluated reciprocal proxy is

$$W_{ii}^{(k)} = \frac{1}{|(\mathbf{A}\mathbf{f}^{(k)})_i| + \varepsilon_{\text{noise}}}, \qquad \varepsilon_{\text{noise}} > 0.$$

Takeaway. Total variance follows the sum of the two event rates. The processed-data weights define a frozen Gaussian surrogate.

Dissertation: Definition 4.6.1 and following scope discussion.

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Heteroscedastic weights: statistical motivation and scope

Likely question: Why do you use reciprocal activity weights in stereo?

[Point to the rate sum in the variance, then to the predicted-activity proxy in the weight.]

Answer

Because the noise model treats busy intervals as noisier. A useful starting point is a raw-count model where the positive and negative event counts are independent Poisson variables. Their difference has mean equal to the difference of the two rates, and variance equal to their sum. In a high-count approximation, inverse-variance weighting then motivates a weighted least-squares fidelity term.

The implemented rule uses reciprocal predicted activity plus a positive noise floor. It's a practical variance proxy: high predicted activity gets less weight, and the floor keeps the weight finite near zero. Total and net activity differ here: opposite polarity rates can cancel in the mean, while both still add to the variance. Also, the actual observations have been warped, downsampled, and normalized. So we treat each frozen weighted solve as a Gaussian surrogate for the processed features, with the reciprocal rule supported by the reported weighting comparison.

If asked for more

The absolute difference of the two polarity rates is at most their sum, with equality exactly when at least one rate is zero. So a variance based only on the net rate adds an approximation. In the iteration, the weights are evaluated at the previous profile and held fixed during the quadratic solve. That keeps the positive-definiteness argument for each solve, and makes the statistical approximation explicit.

IRLS: fixed-point existence and conditional convergence

Fix d, $\mathbf{\Lambda}$, and γ; stack $\mathbf{y} = (\mathbf{y}_L, \mathbf{y}_R)$. Let $\Phi(\mathbf{g})$ minimize the quadratic with weights frozen at $\mathbf{g}$. For the *unclipped reciprocal map*, set $\ell = \lambda_{\min}^2 > 0$:

$$\|\Phi(\mathbf{g})\|^2 \leq \frac{\|\mathbf{y}\|^2}{\ell\, \varepsilon_{\text{noise}}}.$$

A sufficiently large closed ball is invariant; continuity and Brouwer [92] give a fixed point.

Sufficient convergence condition

$$q = \frac{\|\mathbf{A}\|_2 \, \|\mathbf{y}\|}{\sqrt{\ell}\, \varepsilon_{\text{noise}}^{3/2}} < 1 \quad \Rightarrow \quad \text{unique fixed point and geometric convergence.}$$

For fixed damping $\omega \in (0, 1]$, the bound is $q_\omega = 1 - \omega + \omega q$.

Takeaway. A fixed point exists; convergence follows under the displayed sufficient condition.

Dissertation: Lemma 4.6.12, Proposition 4.6.13, Section 4.6.1; IRLS [84].

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IRLS: fixed-point existence and conditional convergence

Likely question: Does the iteratively reweighted stereo solver converge?

[Point to the invariant-ball bound for existence, then separately to q less than one for convergence.]

Answer

Under an explicit condition, yes. The analysis gives two levels of guarantee for the unclipped reciprocal map, with the disparity, regularizer, and coupling fixed. First, a fixed point exists. Compare each frozen minimizer's objective with its value at zero. The positive magnitude penalty controls the solution's squared norm, and every reciprocal weight is at most one over the noise floor. Together, those put every update inside a sufficiently large closed ball.

The weight map is continuous, and the frozen system stays positive definite, so the update map is continuous. Brouwer's theorem then gives a fixed point. Second, the bound q shown is a sufficient contraction constant. If q is less than one, the fixed point is unique and the updates converge geometrically. That's the explicit parameter condition. All of this describes the analyzed update map; the implementation's stopping and acceptance rules decide when the numerical iteration ends.

If asked for more

With a fixed damping parameter greater than zero and at most one, the update is a convex combination of the old state and the undamped update. The fixed points stay the same, and the Lipschitz bound becomes one minus omega, plus omega times q. When q is below one, that's still a contraction. A fixed point means consistency with the frozen weights; reading it as a likelihood optimum needs a separate analysis of the objective.

Stereo benchmark: read accuracy alongside scored coverage

Method	Mean depth error (cm) ↓ Fl1	Fl2	Fl3	Fl4	log RMSE ×100 ↓	Scored points (M)
Unweighted control	140.23	115.40	142.35	113.00	96.18	1.43
Unweighted SGM [95]	83.45	72.90	85.33	71.62	50.55	2.39
Soft-coupled WLS	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20
Xie adaptation [105]	93.43	78.60	93.66	66.57	59.36	1.39
Kogler et al. [104]	123.39	99.50	125.27	78.23	80.94	0.66
Rogister et al. [103]	142.17	128.20	153.32	114.80	90.32	1.01

MVSEC [35] indoor flying, raw outputs. Each method is scored on its own estimates and ground truth in $[0.5, 8]$ m; points summed and log RMSE averaged over sequences.

Takeaway. WLS is competitive with a larger scored population.

Dissertation: Table A.5, Table A.4; Xie and Piatkowska rows are local adaptations.

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Stereo benchmark: read accuracy alongside scored coverage

Likely question: How competitive is the stereo method, and how should I interpret the coverage numbers?

[Identify the centimeter columns, the scaled log metric, and the scored-point column separately.]

Answer

It's competitive. Its mean errors are within six centimeters of the Piatkowska adaptation's on every sequence; it leads on Flying 2 and narrowly on the averaged log error; and it's scored on more than twice as many points. The table reports mean absolute depth error in centimeters on four MVSEC indoor sequences, plus an averaged log error and the total number of scored points. Soft-coupled weighted least squares has mean errors of 73.83, 58.35, 70.33, and 59.24 centimeters. Its displayed log RMSE is 44.54, on 2.66 million scored points.

The Piatkowska adaptation has the smaller mean error on the other three sequences. The other three prior methods, Xie, Kogler, and Rogister, trail weighted least squares on every sequence and on the log error. The population sizes matter: the adaptation is scored on 1.20 million points. For each method, a pixel is scored only when both its estimate and the ground truth fall inside the reporting band. So the table compares accuracy together with scored coverage. A direct accuracy comparison would score both methods on the same pixels.

If asked for more

The reporting band is 0.5 to 8 meters. Point counts are summed across sequences. The displayed log RMSE is one hundred times the base metric, averaged equally over the sequence-level values. These are complete implementations, including normalization, regularization, guidance, and activity masking, and the external methods are local adaptations.

The reference depth has known limits. MVSEC's ground truth comes from a Velodyne lidar combined with a 3D reconstruction, so it carries estimation and alignment error. Accumulating lidar points over many frames also accumulates odometry drift, which can make objects in the reference look inflated or shifted relative to the image and event data. Every method is scored against the same reference, so these errors limit how finely we can rank close results, such as differences of a few centimeters.

A weighting comparison supports the selected rule

Compare reciprocal versus rate-proportional weights while holding the remaining configuration fixed.

	Flying 1	Flying 2	Flying 3	Flying 4
Mean-error difference (cm)	4.44	8.28	2.73	5.17

The reciprocal rule has lower reported mean error on all four sequences.

Scope of this evidence

Each method is scored on its own valid pixel set. The differences subtract the reported WLS means from the WLS-Prop means; the rule was selected on these sequences.

Takeaway. The reported comparison supports the chosen rule on its scored populations; a common-set comparison requires rescoring the outputs.

Dissertation: Table A.5, Weighting comparison (p. 182).

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A weighting comparison supports the selected rule

Likely question: What empirical evidence supports the reciprocal weighting rule?

[Point to the subtraction direction before reading the four positive differences.]

Answer

A controlled swap of the weighting rule, with everything else in the configuration held fixed. Each number is the rate-proportional method's reported mean depth error minus the reciprocal-weight method's. The differences are 4.44, 8.28, 2.73, and 5.17 centimeters on Flying 1 through Flying 4. All four are positive, so the reciprocal rule has the smaller reported mean error on every sequence.

That supports the selected rule for these configurations and their scored populations. Each method's evaluation population is defined separately, so the comparison combines its errors with the set of in-band points it returns. These sequences were also used to choose the rule. So I'd call this empirical support for the design choice; a common-pixel evaluation and additional sequences would test how far the advantage extends.

If asked for more

These are differences of reported mean errors. The empirical improvement motivates the implementation choice; a maximum-likelihood interpretation would also need a justified probabilistic model for the processed observations.

Hard fusion changes both the constraint and the regularization

Joint intensity fusion (JIF) imposes $\mathbf{f}_L = \mathbf{f}_R = \mathbf{v}$ and solves

$$(2\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)\mathbf{v} = \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

For fixed equal weights, the unchanged soft model has the limit

$$\gamma \to \infty : \qquad \mathbf{f}_L, \mathbf{f}_R \to (\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)^{-1} \frac{1}{2} \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

JIF instead uses $\mathbf{A}^\top \mathbf{W}\mathbf{A} + \frac{1}{2}\Lambda^2$ after dividing its equation by two.

Takeaway. Reported mean depth errors: JIF 102.10 cm; soft WLS 65.44 cm. Hard equality, regularization, weights, iteration budgets, and scored populations differ.

Dissertation: Proposition 4.9.1, Section 4.9.3.

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Hard fusion changes both the constraint and the regularization

Likely question: Why does hard fusion perform differently from soft coupling?

[Compare the magnitude-penalty coefficients after dividing the hard-fusion equation by two.]

Answer

Hard fusion changes two things at once. It forces the two profiles to be equal and solves for one shared profile, and the implemented system also uses a different effective magnitude penalty. Dividing its normal equation by two shows half of lambda squared next to the temporal data matrix. By contrast, sending the soft model's coupling strength to infinity, with equal fixed weights and the same regularization, keeps the full lambda squared in the common-profile equation.

Averaged over the four sequences, the reported mean depth error is 102.10 centimeters for hard fusion and 65.44 for soft weighted least squares. That's the observed difference between the two evaluated configurations. But the comparison changes hard equality, effective regularization, weights, iteration budgets, and scored populations together. So it motivates controlled comparisons to find how much each choice contributes to the gap.

If asked for more

The soft model's mean equation already combines both views.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

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Motion-induced events and the aperture problem

Likely question: Can you derive the aperture problem directly from the event-generating image model?

[Point to the normal-component equation, then to the arbitrary tangential coefficient in the boxed velocity family.]

Answer

Yes. Take a straight edge under steady illumination. Its log intensity is a one-dimensional profile evaluated at the normal position minus the normal speed times time. The spatial gradient points along the edge normal, and the time derivative is minus the normal speed times the profile's derivative.

Substituting those into brightness constancy gives the normal vector dotted with the velocity equal to the edge's normal speed, wherever the profile's derivative is nonzero. Every velocity with that normal component satisfies the same equation, whatever its tangential component. These velocities produce the same local log-intensity histories, and so the same ideal threshold events when their reset references agree. That's the aperture ambiguity in this model: the edge fixes its normal motion and leaves the tangential motion free. This assumes the edge's endpoints and any other distinguishing features lie outside the observation window.

If asked for more

If the normal speed is zero, the motion is purely tangential, and the pixel values stay constant in the ideal edge interior. With nonzero normal speed, the crossing-time surface has gradient equal to the normal divided by that speed, which recovers the time-surface transport constraint. Steady illumination is assumed to isolate motion. The sensor's threshold rule also responds to changes in illumination, so treating every event as motion needs additional scene assumptions.

One complementarity lemma behind the uniqueness results

Lemma. On a Hilbert space H, let $a, r \geq 0$ be bounded symmetric forms and $N \subset H$ finite-dimensional with orthogonal projection P. If

$$r(u,u) \geq \rho \|u - Pu\|^2, \qquad a(n,n) \geq \mu \|n\|^2 \quad (n \in N), \qquad a(u,u) \leq \beta \|u\|^2,$$

then $a(u,u) + r(u,u) \geq c_0 \|u\|^2$ with $c_0 = \min\{\mu/2,\ \rho\mu/(\mu + 2\beta)\}$. When $N = \ker r$, $\mu > 0$ is also necessary.

Problem	r controls all but N	N	a must see N
Temporal (Ch. 2)	data $\|\mathbf{Af}\|^2$, spectral gap g_1	constants	penalty: $\sum_i \lambda_i^2 > 0$
Flow (Ch. 3)	$\alpha^2 \|\nabla \mathbf{v}\|_{L^2}^2$, Poincaré	constant fields	data: $\lambda_{\min}(\int_\Omega \mathbf{B}) > 0$
Discrete flow	connected graph Laplacian	constant fields	data: $\sum_i q_i B_i \succ 0$
Stereo inner (Ch. 4)	data $+\gamma \|\mathbf{f}_L - \mathbf{f}_R\|^2$; $\mathbf{W}_c \succ 0$, $\gamma > 0$	common constants	penalty: $\mathbf{\Lambda} \neq 0$

Takeaway. One term controls all but a finite-dimensional mode; the other must see it. The roles swap between temporal and flow, and c_0 falls linearly with μ.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 3.7.1, Theorem 4.6.3. The common lemma restates their proofs.

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One complementarity lemma behind the uniqueness results

Likely question: The three chapters come from separate papers. Is there one mathematical mechanism behind their uniqueness results?

[Point to the three hypotheses of the lemma, then read the table one row at a time.]

Answer

Yes. Each problem has two quadratic terms. One controls every direction except a small, finite-dimensional set of modes, and the other has to see those modes. If both happen, the sum is coercive, and the coercivity constant falls linearly with how well the second term sees the leftover modes.

In the temporal problem the data see every temporal variation, but they miss the constant baseline; the penalty fixes it. One nonzero weight is enough. Flow reverses the roles. Smoothness controls every variation, but it charges nothing for a uniform pan, so the data have to see the pan; that's the integrated matrix condition. The discrete flow theorem is the same statement on a connected graph. In the stereo inner solve, the two data terms and the coupling leave only a common constant, and the penalty fixes it.

The condition is also necessary. If a leftover mode is invisible to the second term, the quadratic form vanishes there. Backup 02 and Backup 35 are exactly those two cases.

If asked for more

The proof is one line. Split u into its part in the leftover modes, n, and the rest, w. The square root of a is a seminorm, so a of n is at most twice a of u plus twice beta times the norm of w squared, and r controls the norm of w. Adding the two bounds gives the constant.

The constant explains the scene statements: for flow it falls as m shrinks, which is why the stripe scene is nearly degenerate. Two parts of the dissertation sit outside this lemma. Correspondence selection compares finitely many candidate costs, so it needs a separation margin. Contrast maximization is nonconvex, so compactness gives existence only. This lemma and Lax–Milgram apply to quadratic forms; total variation needs separate tools.

If asked how the result extends to another regularizer

The lemma is the recipe. First find the modes the new regularizer leaves free; then check that the data see them. A fractional seminorm on a bounded domain still leaves only the constants, so the same integrated condition fixes them, with a fractional Poincaré inequality and coercivity in H s. A nonquadratic penalty such as total variation needs its own existence argument, through compactness in BV.

Adaptive weights keep the temporal estimate locally Lipschitz

Let $\lambda_i = \varphi(|dc_i|)$ with $0 < \lambda_- \le \varphi \le \lambda_+$ and φ L-Lipschitz. Subtracting the normal equations,

$$\mathbf{K}_1(\mathbf{f}_1^* - \mathbf{f}_2^*) = \mathbf{A}^\top(\mathbf{y}_1 - \mathbf{y}_2) - (\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2)\mathbf{f}_2^*.$$

With $\left\|\mathbf{K}^{-1}\mathbf{A}^\top\right\| \le \frac{1}{2\lambda_-}$, $\left\|\mathbf{K}^{-1}\right\| \le \lambda_-^{-2}$, $\left\|\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2\right\| \le 2\lambda_+ L \left\|dc_1 - dc_2\right\|_\infty$:

$$\|\mathbf{f}_1^* - \mathbf{f}_2^*\| \le \frac{\|\mathbf{y}_1 - \mathbf{y}_2\|}{2\lambda_-} + \frac{\lambda_+ L}{\lambda_-^3}\, \|\mathbf{y}_2\|\, \left\|dc_1 - dc_2\right\|_\infty .$$

Chapter 2 rule	λ_-	λ_+	L
Sigmoid	10^{-3}	5.001	2.5
Inverse proportional	10^{-3}	100	100
Exponential decay	$10^{-3} + 10e^{-10}$	10.001	10

Takeaway. Each Chapter 2 rule gives a locally Lipschitz estimator with explicit, pessimistic constants; Chapter 4's solution-dependent IRLS weights need the fixed-point analysis.

Dissertation: Theorem 2.2.5; adaptive rules (p. 38). The Lipschitz bound is a short argument beyond the chapter.

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Adaptive weights keep the temporal estimate locally Lipschitz

Likely question: The temporal theorem fixes the weights, but you compute them from the data. Is the estimator you actually run stable?

[Point to the subtracted normal equations, then to the two terms of the bound.]

Answer

Yes, it's locally Lipschitz, by a short argument. Each Chapter 2 rule maps event activity to a weight between a positive floor and a finite ceiling, and each rule is Lipschitz in the activity. The absolute value, the minimum and the maximum are all Lipschitz, so their kinks need no differentiability.

Subtract the two normal equations. The difference in the solutions equals the fixed-weight term from the change in the data, plus the change in the squared weights acting on the second solution. Three standard bounds finish it. The solution operator has norm at most one over twice the floor. The inverse system matrix has norm at most one over the floor squared. The squared weights change by at most twice the ceiling times L times the change in activity.

The second term grows with the size of the data, so the bound is Lipschitz on bounded sets of data. The table gives the floor, ceiling and Lipschitz constant for the sigmoid, inverse-proportional and exponential rules.

If asked for more

The constants are pessimistic, because the floor of ten to the minus three enters cubed. The rules decrease with activity, so for data bounded by R the sigmoid floor improves to ten to the minus three plus ten over one plus e to the R. Pixels are independent, so the bound holds pixel by pixel.

Chapter 4 is different. Its observation weights are recomputed from the current profiles, so its estimator is a fixed point and needs the contraction condition in the stereo backups. Chapter 5 lists this as a future direction. The bound covers the step from the represented data to the estimate; connecting it to raw event perturbations is the representation question on slide 24.

How the parameters were chosen

Problem	Parameters	Selection
Temporal (Ch. 2)	Scale of each $\mathbf{\Lambda}$ rule; kernel and width	Grid sweep per rule; three rules compared by fidelity and CN
Flow (Ch. 3)	Smoothness weight α	Grid sweeps on MVSEC
Stereo (Ch. 4)	Coupling γ, latent count; weighting	Grid sweep; paired weighting comparison on the four sequences

Antil–Sayre [2]: scalar weight λ_2. Chapter 2: adaptive diagonal $\mathbf{\Lambda}$, three rules evaluated.

What the theorems guarantee

The results hold for every positive value: sweeps tune bias and accuracy; uniqueness and stability hold throughout.

Principled selection rules

Discrepancy principle (Morozov 1966; Skellam noise level $\theta_+ + \theta_-$), L-curve (Hansen 1992), GCV (Golub–Heath–Wahba 1979), bilevel learning.

Takeaway. Sweeps chose the values; the theorems cover every positive choice.

Dissertation: Adaptive rules (p. 38); Table 2.7; flow sweeps (p. 164); Section 4.8; Definition 4.6.1. Parameter-choice rules: external works named above.

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How the parameters were chosen

Likely question: How did you choose the regularization parameters, and why not a standard rule such as the discrepancy principle?

[Read the table one row at a time, then point to the two blocks.]

Answer

Each was chosen by grid sweep. In Chapter 2 we swept the scale of each adaptive rule, and separately the kernel type and width, then compared the three rules by image fidelity and conditioning. The sigmoid rule has the lowest outdoor condition number, 29.60. Antil and Sayre use one scalar magnitude weight on the image; making that weight activity-adaptive, and comparing three ways to do it, is the change Chapter 2 evaluates.

For flow, the smoothness weight came from grid sweeps on MVSEC, for each solver. In stereo, the coupling weight and the number of latent images came from a grid sweep, and inverse-variance weighting from a paired comparison, on the four evaluated sequences.

The theorems hold for every positive value of these parameters. So the choice moves bias and accuracy, and the solve stays unique and stable whatever value the sweep picks.

If asked for more

The discrepancy principle chooses the strongest regularization whose residual matches the noise level, so it needs that level. For raw signed counts, the Skellam model gives the variance as the sum of the ON and OFF rates, which can be estimated from the counts; after the Chapter 4 transformation, that variance has to be propagated through it. The L-curve and generalized cross-validation need no noise level, so they apply directly.

The sweeps used the same sequences we report, so a held-out protocol is the clean next experiment. The data-driven route is bilevel learning of the weights, from the research-opportunities slide.

Learned estimators face the same free directions

Problem	Left free by the events	Model-based choice	Learned choice
Temporal	Constant offset at each pixel (Lemma 2.2.4)	Magnitude penalty $\mathbf{\Lambda}$ (Theorem 2.2.5)	Brightness learned from training images [12]
Flow	Motion along a straight edge (Figure 3.1)	H^1 smoothness, integrated data (Theorem 3.6.1)	Spatial context learned from data [39]
Stereo	Candidates with coincident right-view profiles (Proposition 4.6.20)	Spatial cost aggregation (SGM)	Learned matching cost

Combining them

Learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{\Lambda} \mapsto \mathbf{f}^*(\mathbf{\Lambda})$ is single-valued and smooth.

Data without a training set

ISS and ballistics: reconstructed from events alone, without reference images or training data.

Takeaway. The theorems separate what the events determine from what the regularizer supplies, in either form.

Dissertation: Lemma 2.2.4, Theorem 2.2.5, Theorem 3.6.1, Proposition 4.6.20; qualitative demonstrations (p. 44). External paper cited above.

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Learned estimators face the same free directions

Likely question: Why not train a neural network to do this?

[Read the "left free by the events" column first, then the two choice columns.]

Answer

A trained network is also a regularized estimator. Where the events leave the answer free, it fills the gap from its training images; my methods fill it with a penalty that is written down and analyzed.

The dissertation identifies those free directions. In the temporal problem it's the constant brightness offset at each pixel. In flow it's the motion component along a straight edge. In stereo it's the disparity candidates whose temporal profiles coincide. Every estimator supplies those from a model or from training data. That's also why event reconstruction has no agreed standard yet: scoring against frame images assumes a brightness level the events leave open.

The two approaches combine naturally. The weights can be learned in a bilevel problem, and because each fixed-weight solve is unique and smooth in the weights, the outer problem has a well-defined, smooth lower level. And for data like the ISS and ballistics sequences, the reconstruction runs from events alone, with no reference images or training set.

If asked for more

If the question is benchmark accuracy: on the public flow and stereo benchmarks, learned methods report the lowest errors. The dissertation asks a different question, what the data determine and when each estimator is stable, and its stereo comparison is against model-based event methods.

A second combination is unrolling: run a fixed number of block-Jacobi or IRLS iterations and treat each one as a network layer. The convergence results then describe what each layer does.

If asked whether the bilevel problem is well posed

The lower level is. If the weights range over a box between a positive floor and a ceiling, and the outer loss is continuous, the reduced outer objective is continuous on a compact set, so an outer minimizer exists. Uniqueness of that minimizer, generalization of the learned weights, and convergence of a particular outer algorithm each need their own analysis.

The spline regularizer is uniformly coercive under refinement

Cubic B-splines, uniform knot spacing h: $\Phi_j \geq 0$, $\sum_j \Phi_j \equiv 1$, $\int_\Omega \Phi_j \leq h^2$, and $u_h = \sum_j c_j \Phi_j$.

Step 1: coefficients control the L^2 norm

$u_h^2 \leq \sum_j c_j^2 \Phi_j$ pointwise, so $\|u_h\|_{L^2}^2 \leq h^2 \|\mathbf{c}\|^2$.

Step 2: both regularizer terms, for each flow component

$$\beta\, \mathbf{c}^\top \mathbf{K}_0 \mathbf{c} + \tau_{\text{ridge}} \|\mathbf{c}\|^2 \;\geq\; \beta \varepsilon_{\text{ell}} \|\nabla u_h\|_{L^2}^2 + \frac{2\varepsilon_{\text{ridge}}}{Mh^2} \|u_h\|_{L^2}^2 .$$

$Mh^2 \to |\Omega|$ (ratio 1.20 at $h = 8\,\text{px}$), so the H^1 coercivity constant stays bounded below as $h \to 0$; $\kappa_2(\mathbf{K}_{\text{reg}}) = O(h^{-2})$, the usual stiffness-plus-mass growth.

Takeaway. The ridge eigenvalue shrinks with refinement; the function-space penalty stays uniform.

Dissertation: Proposition 3.5.1; contrast-maximization parameters (p. 166). The uniform bound is a short argument beyond the chapter.

BACKUP 68 / DECK 93

The spline regularizer is uniformly coercive under refinement

Likely question: Your ridge, two epsilon over M, goes to zero as you refine the knots. Doesn't coercivity degrade?

[Point to step 1, then to the two terms of step 2.]

Answer

Only in coefficient space. The cubic B-splines are nonnegative and sum to one, so at every point the squared spline value is at most a weighted average of the squared coefficients, with the basis functions as weights. Integrating, and using that each basis function integrates to at most h squared, gives the L two norm of the spline at most h squared times the squared coefficient norm.

So the ridge, two epsilon over M times the squared coefficients, controls two epsilon over M h squared times the squared L two norm, and M h squared tends to the area of the domain; on our frame at h equal to eight pixels the ratio is 1.20. The eigenvalue floor in the diffusion tensor controls the gradient. Together, the discrete regularizer controls the full H one norm with a constant that holds for every knot spacing.

What does grow is the condition number of the coefficient matrix, like h to the minus two. That's the usual growth for an H one Galerkin matrix with a mass term.

If asked for more

The largest eigenvalue of the stiffness matrix stays bounded in two dimensions: its entries are of the size of the diffusion tensor, and each basis function overlaps at most 49 others, so Gershgorin bounds it. The smallest eigenvalue is the ridge, of order h squared. Knot-spacing continuation, which the implementation supports, is the usual remedy for iteration counts. I checked the mass bound numerically with the clamped cubic basis: the largest one-dimensional Gram eigenvalue divided by h is 0.981, 0.994, and 1.000 at spacings of 8, 4, and 2 pixels. With nonuniform knots, weighting each coefficient by its basis integral makes the ridge a lumped mass term again.

Weak regularization: a limit in time, the aperture line in space

Temporal, $\Lambda = \lambda \mathbf{D}$, fixed diagonal $\mathbf{D} \succ 0$

- Every $\lambda > 0$: $\mathbf{1}^\top \mathbf{D}^2 \mathbf{f}_\lambda = 0$, so the baseline rule is independent of λ.
- $\lambda \to 0$: $\mathbf{f}_\lambda \to \operatorname{argmin}\{\|\mathbf{Du}\| : \mathbf{Au} = \mathbf{y}\}$, an exact fit of the increments.
- $\mathbf{D} = \mathbf{I}$: noise gain $\left\|\mathbf{K}_\lambda^{-1}\mathbf{A}^\top\right\| \le \min\{\frac{1}{2\lambda}, \frac{1}{\sqrt{g_1}}\}$, with $\frac{1}{\sqrt{g_1}} \approx \frac{N_t}{\pi}$.

Flow, smoothness weight α^2

- If $\mathbf{B} = 0$ on an open $U \subset \Omega$: for $\mathbf{v} \in H_0^1(U; \mathbb{R}^2)$, $\mathcal{B}(\mathbf{v}, \mathbf{v}) = \alpha^2 \|\nabla \mathbf{v}\|^2 \le \alpha^2 \|\mathbf{v}\|_{H^1}^2$, so the coercivity constant is at most α^2.
- $\alpha = 0$, time surface: the minimizers are all $\mathbf{v}$ with $\nabla \mathcal{T} \cdot \mathbf{v} = 1$ wherever $\nabla \mathcal{T} \neq 0$, the aperture line at each pixel.

Takeaway. Without the temporal penalty the increments are fit exactly and noise accumulates along the chain; without spatial smoothing each pixel returns to its aperture line.

Dissertation: Regularized formulation (p. 16); Lemma 4.4.1; Proposition 4.6.23; Theorem 3.6.1; Proposition 3.3.4. The flow statements are a short argument beyond the chapter.

BACKUP 69 / DECK 94

Weak regularization: a limit in time, the aperture line in space

Likely question: What happens to your results as the regularization goes to zero?

[Left column first, then the right column.]

Answer

The two problems behave differently, and the difference shows what each penalty contributes.

In time, the solution has a clean limit. Testing the normal equations with the constant vector removes the data term at every scale, so the weighted zero-mean rule that picks the baseline is the same for every lambda. As lambda goes to zero, the solution converges to the exact fit of the increments with the smallest weighted norm; Chapter 2 states that limit. What the penalty buys is noise control. With a scalar penalty, the limit is the running sum of the increments, centered. Noise in the increments accumulates along the chain, with gain one over root g one, about N t over pi. The penalty caps that gain at one over two lambda.

In space, the smoothing supplies identifiability. An event-free region gives a test field whose energy is only alpha squared times its gradient, so the coercivity constant is at most alpha squared. At alpha equal to zero the energy is pointwise, and for the time surface its minimizers at each pixel are the whole aperture line from slide 18.

If asked for more

The full-space condition number, g max plus lambda squared over lambda squared, grows only along the constant mode, which the data never excite; Backup 17 removes that mode. Convergence to the limit sets in once lambda squared is small compared with g one, roughly pi squared over N t squared. For the structure tensor at alpha zero, pixels with a rank-two pooled block keep a unique velocity; the rest keep a line or the whole plane.

Where there are no events, smoothness fills in harmonically

Let $U \subset \Omega$ be open, connected, and Lipschitz, with $\mathbf{B} = 0$ and $b = 0$ a.e. in U, and $\Gamma = \partial U \cap \Omega \neq \emptyset$.

The weak equation inside U

For every $\boldsymbol{\eta} \in H^1_0(U; \mathbb{R}^2)$: $\int_U \nabla \mathbf{v}^* : \nabla \boldsymbol{\eta} = 0$. Each component of $\mathbf{v}^*$ is harmonic, hence smooth, in U; on U it minimizes $\int_U |\nabla \mathbf{w}|^2$ given its values on Γ.

Maximum principle

For a unit vector $\mathbf{e}$: if $\mathbf{e} \cdot \mathbf{v}^* \leq M$ on Γ, test with $\boldsymbol{\eta} = (\mathbf{e} \cdot \mathbf{v}^* - M)^+ \mathbf{e}$ on U, zero elsewhere. Then $\alpha^2 \int_U |\nabla \boldsymbol{\eta}|^2 = 0$, so $\mathbf{e} \cdot \mathbf{v}^* \leq M$ in U. Over all $\mathbf{e}$: $\sup_U |\mathbf{v}^*| \leq \sup_\Gamma |\mathbf{v}^*|$.

Takeaway. An event-free region receives the least-gradient interpolation of the motion measured around it, with no speed larger than on its boundary.

Dissertation: Theorem 3.3.9; Theorem 3.6.1. Harmonicity and the maximum principle are a short argument beyond the chapter.

BACKUP 70 / DECK 95

Where there are no events, smoothness fills in harmonically

Likely question: In regions with no events, isn't the flow simply invented?

[Point to the weak equation, then to the maximum-principle bound.]

Answer

The regularizer chooses those values, and we can say exactly how. Take a connected region U with no represented gradient, so the data matrix and the forcing vanish there. Testing the weak equation with fields supported in U leaves only the gradient term, so each velocity component is harmonic in U. Equivalently, among fields that agree with the estimate on U's boundary, it has the least gradient energy.

That gives a maximum principle. Pick any direction e, and suppose the velocity component along e is at most M on Gamma, the interface with the event-active pixels. Test with the positive part of that component minus M. The weak equation says its gradient energy is zero, and it vanishes on Gamma, so it's zero throughout U. Doing this for every direction bounds the speed inside U by the largest speed on its boundary.

So an event-free region gets the smoothest interpolation of the motion measured around it, with no new extremes. The theorem guarantees a unique regularized estimate; identifying the physical motion at each pixel needs data there.

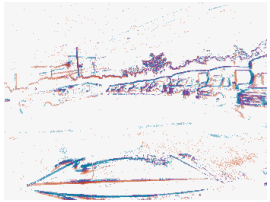
If asked for more

Where U reaches the image border, the natural Neumann condition holds and adds no constraint, because test fields are free on the outer boundary. All three representations have residual-square data terms, so the forcing vanishes wherever the data matrix does. On the pixel graph, the same statement says an event-free node's velocity is the weighted average of its neighbors'. A motion boundary inside an event-free region is smoothed across; that's the H one modeling choice from Backup 27.

What can changes tell us about a scene?



Reference image

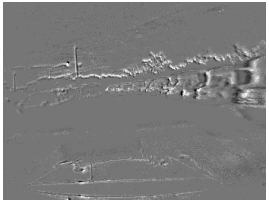


Accumulated events

50 ms sample

Orange +; blue −; purple: both.

→



Our reconstruction

What do the observations determine, and what does the model supply?

Takeaway. The observations constrain changes; the model supplies the remaining choices.

Dissertation: Chapter 2, saved MVSEC [35] outdoor_day1 reconstruction example.

BACKUP 71 / DECK 96

What can changes tell us about a scene?

Likely question: How does an event-only reconstruction compare with an ordinary frame of the same scene?

Answer

[Point to the reference image, the accumulated events, and the reconstruction.]

On the left is an ordinary camera image of an outdoor scene; we keep it there to help recognize the scene. In the middle is a fifty-millisecond sample of the events from the same recording. Orange marks pixels that got brighter, blue pixels that got darker, and purple pixels that did both. On the right is our reconstruction, computed from a longer stretch of those events, a little under a second.

Each event says only that the brightness at one pixel changed at one moment. So the events record change; the brightness level comes from the model. To get from events to the image on the right, the model has to fill in what the events leave out, starting with a reference brightness level. Regularization makes that choice and balances it against fitting the data.

That is the theme of the talk. At each step, I'll separate what the observations determine from what the model chooses. Regularization can pick out a single answer. To know what that answer means, we need to know which parts came from the data and which came from the model.

[If asked whether the baseline is all an event camera loses: Backup 34, page 59.]

Where this work fits

Image reconstruction

Rebecq et al. [12]: recurrent network, events to video.

Zhang, Yezzi & Gallego [6]: optical flow supplies a linear image inverse problem; classical or learned priors.

Antil–Sayre [2]: events and frames; bilevel estimation of an event-based blur-kernel parameter.

This work: per-pixel temporal inversion from event increments; kernel representations, activity-based penalties, and fixed-weight guarantees.

Optical flow

Horn–Schunck [16]: brightness constraint and spatial smoothness.

Benosman et al. [50]: timestamp geometry.

Weickert–Schnörr [57]: convex-regularizer framework and well-posedness theory.

This work: three event representations in one quadratic theory, with explicit identifiability and stability assumptions.

Takeaway. The contribution is the event-derived models and their precise guarantees: what the data determine and what regularization supplies.

Dissertation: Chapter 2 and Section 3.2; temporal publication: Antil, Blauvelt, and Sayre [17].

BACKUP 72 / DECK 97

Where this work fits

Likely question: How does this work relate to prior reconstruction and flow methods?

Answer

[Compare the prior approaches with the dissertation contribution in each column.]

For image reconstruction, Rebecq and colleagues train a recurrent neural network on simulated events. Zhang, Yezzi, and Gallego use optical flow to pose reconstruction as a linear inverse problem, with classical or learned regularization. Antil and Sayre combine events with conventional frames in a bilevel problem that estimates a parameter of the event-based blur kernel.

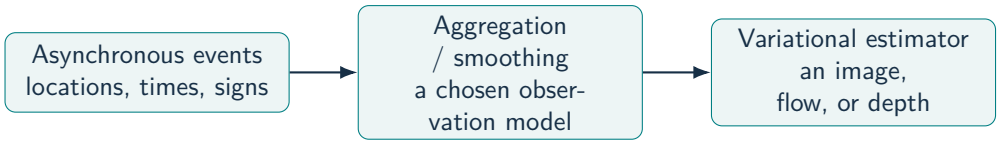
Our published temporal work reconstructs each pixel on its own from event increments. I develop the temporal kernel representation and the activity-based penalties, and then analyze the resulting quadratic solve: what the observations determine, how regularization picks the baseline, and how perturbations move the solution. That analysis shows where the reconstructed temporal structure comes from.

For flow, Horn and Schunck gave us the data-plus-smoothness formulation, and Benosman and colleagues developed the local geometry of event timestamps. Classical quadratic flow already has a well-posedness theory, for example from Weickert and Schnörr. What I add is one quadratic family that covers three ways of building coefficients from events, with explicit assumptions for when the motion is identified and how sensitive the estimate is to those coefficients.

[If asked what is mathematically new when the tools are classical: Backup 31, page 56; the mechanism shared by all three chapters is Backup 64, page 89. If asked for the literature behind each model: Backup 32, page 57. If asked why not train a neural network: Backup 67, page 92.]

A point measure needs a representation before optimization

$$\mathcal{E} = \sum_k p_k \, \delta_{(\mathbf{x}_k, s_k)}.$$



- Temporal kernels yield integrated changes at reconstruction times.
- Time surfaces and voxel grids yield coefficients for motion constraints.
- Stereo accumulation yields paired temporal event-feature observations.

Takeaway. The representation supplies usable coefficients and introduces modeling choices.

Dissertation: Eq. 2.2–Eq. 2.4; Section 3.3; Section 4.3.3.

BACKUP 73 / DECK 98

A point measure needs a representation before optimization

Likely question: Why is representing the event stream part of the model?

Answer

We can write the raw event stream as a signed point measure: a point mass at each event's location and time, weighted by its polarity.

[Follow the three boxes.]

The quadratic models in this talk need quantities that can go into a residual, so first we convert the stream. Binning, smoothing, and interpolation do that.

For temporal reconstruction, kernels spread each event's contribution across reconstruction intervals. For flow, time surfaces and voxel grids supply the coefficients of the motion constraints. For stereo, accumulated events give paired temporal features.

This conversion is part of the model. It decides what information we keep, and how measurement errors reach the coefficients.

I'll also keep track of what each variable means. Physical intensity, log intensity, represented event activity, and a fitted latent profile all play different roles. For example, the stereo preprocessing produces event-feature profiles whose scale is set by that preprocessing.

[If asked about the continuous-time meaning of the event measure: Backup 03, page 28. If asked why the kernel choice matters: Backup 05, page 30.]

Why uniqueness and stability hold

Write $\lambda_{\min} = \min_i \lambda_i$. For $\mathbf{z} \neq 0$,

$$\mathbf{z}^\top \mathbf{K}_{\text{time}} \mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2 > 0.$$

At the same fixed weights,

$$\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1} \mathbf{A}^\top \delta\mathbf{y}, \qquad \|\delta\mathbf{f}^*\| \leq \frac{\|\mathbf{A}\|_2}{\lambda_{\min}^2} \|\delta\mathbf{y}\|.$$

$$\|\mathbf{A}\|_2 \leq 2, \qquad \kappa_2(\mathbf{K}_{\text{time}}) \leq \frac{4 + \lambda_{\max}^2}{\lambda_{\min}^2}.$$

One spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$: the bound limits how far one noise event moves a pixel's history.

Takeaway. Coercivity gives a stable estimate; weakening the penalty weakens these full-space bounds.

Dissertation: Theorem 2.2.5, Proposition 2.2.7.

BACKUP 74 / DECK 99

Why uniqueness and stability hold

Likely question: Why are the estimates unique and stable, and how well conditioned is the system?

Answer

[Walk through the first inequality slowly.]

Take any nonzero direction z. The quadratic form of the normal matrix is the squared norm of A z plus the squared norm of Lambda z.

The first term is nonnegative. The second is at least the smallest weight squared times the squared norm of z. Since that weight is positive, the curvature is positive in every direction.

That gives coercivity and strict convexity, so a minimizer exists and is unique.

For stability, compare two data vectors with the same weights. Subtracting their normal equations gives the exact difference between the two solutions, shown here. Bounding the inverse matrix gives the estimate. In scene terms, one spurious or missed event changes the data by at most one in norm, because each kernel splits a single unit of polarity across the bins. So the bound limits how far one noise event can move a pixel's history, and the smallest weight sets that limit.

The last expression bounds the condition number in terms of the largest and smallest weights. Lowering the smallest weight weakens these worst-case guarantees.

All of this holds for fixed weights. If the data change and we also recompute the weights, the comparison picks up an extra term for the change in the weights.

[If asked what happens as the weights go to zero: Backup 69, page 94. If asked about recomputed weights: Backup 65, page 90. If asked how close the stable estimate is to the true signal: Backup 33, page 58. If asked about conditioning as lambda becomes small: Backup 17, page 42. If asked how the penalty treats signal and noise differently: Backup 04, page 29.]

How are the temporal weights chosen?

For activity $a_i = |dc_i|$, the selected sigmoid rule is

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}}.$$

Here dc_i is the signed event accumulation; a_i is its magnitude.

Rule	Sigmoid	Inverse proportional	Exponential
Reported outdoor CN	29.60	81.50	100.74

Selected by grid sweeps, balancing image fidelity and conditioning.
Quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.

Takeaway. Positive weights ensure a unique solve; the stability bound holds with weights fixed.

Dissertation: Adaptive temporal weights (p. 38); Table 2.7.

BACKUP 75 / DECK 100

How are the temporal weights chosen?

Likely question: How were the temporal weights chosen?

Answer

Lambda controls how strongly the reconstruction is pulled toward its reference level, and all three rules reduce that pull as the magnitude of net signed activity increases. Sigmoid gives a smooth, bounded transition; inverse-proportional weighting retains a longer tail of shrinkage; and exponential weighting drops rapidly until its input clamp is reached. At our selected scales, inverse-proportional weighting applies the strongest shrinkage.

The rule we selected uses the size of the signed event total in each bin. Positive and negative events can cancel in that total, so it measures net signed activity.

[Point to the sigmoid expression.]

As activity increases, the sigmoid rule lowers the penalty. A positive floor keeps every weight above zero. So quiet intervals get the largest weights, and through the weighted-average identity they set each pixel's reference level. Busy intervals get the smallest weights: they're free to change, and they set the worst-case noise bound from the stability result.

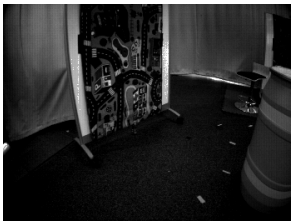
We chose this rule after grid sweeps of each rule's scale, as a compromise between reconstruction quality and conditioning. Outdoors, the sigmoid gives the lowest condition number of the three choices. Indoors, inverse-proportional weighting gives a lower condition number, so the choice trades conditioning against reconstruction quality.

The point to remember is that the weights are computed from the data and then held fixed during the solve. The theorem covers that frozen problem. Stability of the full adaptive procedure also requires controlling how the weights change.

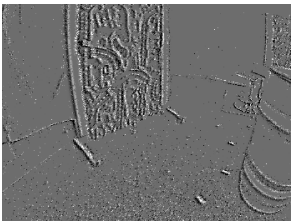
[If asked how data-computed weights fit the fixed-weight theorem: Backup 25, page 50; for the full adaptive map: Backup 65, page 90. If asked how the scales were chosen, or about the discrepancy principle: Backup 66, page 91. If asked how kernel and weight choices change conditioning: Backup 05, page 30.]

Temporal evidence: events fix structure; the penalty sets the baseline

MVSEC [35] indoor flying 2



Reference



Event-only reconstruction

MVSEC indoor aggregate results

Kernel	PSNR	CN
B-spline	8.86	22.16
Delta	7.70	1375.75
Gaussian	8.84	20.22

PSNR compares with a reference frame, so it also scores the absolute brightness that the model selects; the events constrain only changes.

Takeaway. Kernel and adaptive-weight choices affect both the reconstruction and the solved system.

Dissertation: Table 2.4; images: MVSEC indoor flying 2, Ch. 2.

BACKUP 76 / DECK 101

Temporal evidence: events fix structure; the penalty sets the baseline

Likely question: How do the temporal kernels compare on MVSEC, and what does PSNR measure here?

Answer

[Point to the images, then to the table.]

The pictures are a representative example from MVSEC indoor flying two, the dataset from slide 6: a reference image from the ordinary frames, and an event-only reconstruction with sigmoid weights. The table summarizes the kernel comparison over the indoor flying sequences.

The reconstruction makes the scene structure visible. The table reports two things: PSNR, which measures similarity to the reference image, and the condition number of the linear system. PSNR also scores the absolute brightness level, which the model selects, so it measures something different from agreement with the recorded changes.

Indoors, the spline and Gaussian kernels give similar PSNR, both above delta binning, with much smaller condition numbers. Outdoors, the ordering changes. So kernel and weight choices affect both reconstruction quality and conditioning, and the best choice depends on the scene.

The kernel changes the observations, and the adaptive weights can then change the matrix we solve. That's how representation choices reach the linear system.

[If asked why the PSNR values are low, or what the metrics measure: Backup 06, page 31. If asked about the outdoor ranking or the kernel choice: Backup 05, page 30.]

Derive the time-surface constraint by following the edge

For an ideal single crossing, define

$$\psi(\mathbf{x}, t) = t - \mathcal{T}(\mathbf{x}), \qquad \psi(\mathbf{x}(t), t) = 0.$$

If the edge translates with constant $\dot{\mathbf{x}} = \mathbf{v}$, then

$$0 = \frac{d}{dt}\psi(\mathbf{x}(t), t) = 1 - \nabla\mathcal{T} \cdot \mathbf{v}.$$

In scene terms: $|\nabla\mathcal{T}|$ is edge slowness, normal speed $1/|\nabla\mathcal{T}|$; motion along the edge leaves $\mathcal{T}$ fixed.

Model hypotheses

The edge crosses each location once, has nonzero normal speed, and its recorded timestamp equals a differentiable crossing-time map.

Takeaway. Interpolation, stale timestamps, and repeated crossings make this an approximate data constraint.

Dissertation: Proposition 3.3.4; Definition 3.3.2; time surfaces [49, 50].

BACKUP 77 / DECK 102

Derive the time-surface constraint by following the edge

Likely question: Where does the time-surface constraint come from?

Answer

Define psi as the current time minus the crossing-time map. Follow a point moving with the edge: along that path, psi stays zero, because the current time equals the crossing time there.

Differentiate along the path. The time derivative gives one. The spatial part gives minus the gradient of the crossing-time map, dotted with the velocity. Setting the sum to zero gives the time-surface constraint: the gradient of the crossing-time map, dotted with the velocity, equals one.

In scene terms, the size of that gradient is the edge's slowness. If an edge takes ten milliseconds to cross one pixel, it's moving at one hundred pixels per second perpendicular to itself. Motion along the edge leaves the crossing times unchanged, so this single equation already contains the aperture problem.

[Pause at the hypotheses box.]

The derivation assumes each location is crossed once, the normal speed is nonzero, and the crossing-time map is differentiable. In real data, a stored timestamp can be stale, or a location can be crossed more than once. Fitting and interpolation also affect the coefficients.

So we treat this relation as a model for the represented data. The analysis then shows what follows once the coefficient fields meet the stated assumptions.

[If asked how the time-surface gradient is estimated and how accurate it is: Backup 42, page 67; for the neighborhood size, Backup 15, page 40. If asked about correlated timestamp errors: Backup 40, page 65.]

Voxel projection: transport a represented event-activity field

Smooth and bin events into a spatiotemporal field $V(\mathbf{x}, t)$; project over a window, $a(\mathbf{x}, s) = \int_s^{s+T} V(\mathbf{x}, t)\, dt$.

Modeling assumption (added to the threshold model)

The activity field moves with the scene: $V_t + \mathbf{v} \cdot \nabla V = 0$, with constant $\mathbf{v}$ over the window.

Integrating over the window then gives

$$a_s + \mathbf{v} \cdot \nabla a = 0, \qquad a_s = V(\mathbf{x}, s+T) - V(\mathbf{x}, s),$$

so the voxel-projection residual is $\nabla a \cdot \mathbf{v} + a_s$.

Takeaway. The threshold model supplies the activity field; transporting it with the scene is a separate modeling choice.

Dissertation: Section 3.4.1; derivative-operator qualifications; voxel grid [39].

BACKUP 78 / DECK 103

Voxel projection: transport a represented event-activity field

Likely question: What does voxel projection assume?

Answer

A voxel grid adds a time axis to the image grid. After smoothing and binning, it gives an event-activity field, V.

Integrate V over a sliding time window to get a.

[Point to the assumption box.]

This step adds a modeling assumption, shown in the box: the activity field moves with the scene, at constant velocity over the window. That goes beyond the threshold model. A pixel can start or stop producing events as brightness, motion, and threshold conditions change, and activity is a different quantity from brightness: a steadily bright region can produce very little of it.

Integrating the transport equation gives another transport equation, now for a. Its time derivative is the activity entering at the end of the window minus the activity leaving at the start.

The residual is then the spatial gradient of a, dotted with the velocity, plus that time derivative.

The continuous identity holds under the transport assumption with consistent derivatives. Binning and the choice of numerical derivatives can introduce mismatch.

[If asked why event activity should obey transport, or why not a continuity equation: Backup 39, page 64. If asked whether the temporal slope and spatial projection must match: Backup 16, page 41.]

The structure tensor pools directional information

At a fixed evaluation time $t_\star$, set

$$\mathbf{J}(\mathbf{x}) = \int W(\mathbf{x}-\mathbf{y})\, \nabla_3 V(\mathbf{y}, t_\star) \nabla_3 V(\mathbf{y}, t_\star)^\top \, d\mathbf{y}, \qquad W \geq 0.$$

Use the local residual energy

$$\mathbf{v}_3^\top \mathbf{J} \mathbf{v}_3, \qquad \mathbf{v}_3 = (v_x, v_y, 1)^\top, \qquad \mathbf{J} = \begin{pmatrix} \widetilde{\mathbf{J}} & \mathbf{j}_t \\ \mathbf{j}_t^\top & J_{tt} \end{pmatrix}.$$

- Each gradient outer product is positive semidefinite and rank at most one.
- Pooling nonparallel spatial gradients can make $\widetilde{\mathbf{J}}$ full rank.

Takeaway. Pooling can identify both components locally; the global theorem needs only integrated identifiability.

Dissertation: Section 3.4.2, Proposition 3.4.4; spatiotemporal-surface view [49].

BACKUP 79 / DECK 104

The structure tensor pools directional information

Likely question: How does the structure tensor constrain both velocity components?

Answer

The structure tensor pools several nearby gradients of the represented event field.

Each space-time gradient contributes an outer product. With nonnegative pooling weights, the result is positive semidefinite, because its quadratic form is an integral of squares. Contracting J with the augmented velocity gives the pooled squared transport residual: for one gradient, the residual is V_x v_x plus V_y v_y plus V_t, and its square is the augmented velocity times that gradient's outer product times the augmented velocity.

[Point to the spatial block and the temporal cross term.]

The spatial block gives the quadratic dependence on the two velocity components. The temporal cross terms give the linear term. The last temporal entry adds a constant to the energy, because the third component of the augmented velocity is fixed at one.

A single spatial gradient constrains one direction. Pooling gradients that point in different directions can constrain both components at one location.

That's one way to get more directional information. The flow theorem requires edge directions to be covered somewhere in the domain, and spatial smoothness carries that information to every pixel.

The difference from voxel projection is where the averaging happens. Voxel projection averages over time, assuming constant velocity across the time window, and each pixel contributes a single constraint. The structure tensor averages over space, assuming constant velocity across a small neighborhood. The neighborhood size sets the trade-off between identifying both components and resolving motion boundaries.

[If asked when sampled gradient directions suffice: Backup 11, page 36. If asked whether integrated directional information is necessary: Backup 35, page 60.]

Proof idea: smoothness fixes variation, edges fix the drift

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

Smoothness controls variation

$$\|\mathbf{w}\|_{L^2} \le C_P \|\nabla \mathbf{v}\|_{L^2}$$

Poincaré [26]

Data control constant motion

$$m = \lambda_{\min}\Big(\int_\Omega \mathbf{B}\, d\mathbf{x}\Big) > 0$$

edges cover every direction

Together: coercivity

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) \ge c_0 \|\mathbf{v}\|_{H^1}^2$$

$$c_0 = c_0(\Omega, \alpha, \|\mathbf{B}\|_{L^\infty}, m)$$

In scene terms: smoothness leaves a camera pan free, so the edges must fix it; c_0 shrinks with m, as for nearly parallel stripes (next slide).

Takeaway. The gradient penalty controls nonconstant variation; integrated directional information controls the constant modes. Lax–Milgram [72] then gives the unique minimizer.

Dissertation: proof of Theorem 3.6.1.

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Proof idea: smoothness fixes variation, edges fix the drift

Likely question: How does the proof of the flow theorem go?

Answer

[Point to the decomposition, then move across the three columns.]

Write the velocity as its spatial mean plus a zero-mean fluctuation, w. Physically, the mean is the common drift of the whole field, like a camera pan, and the fluctuation is how motion varies across the image: rotation, zoom, or objects moving separately.

Poincaré's inequality bounds the fluctuation by the gradient, so the smoothness term controls it.

Smoothness charges nothing for a uniform pan, so the data have to fix the mean. That's the role of m, the smallest eigenvalue of the integrated matrix B. It measures how well the scene's edges, taken together, cover every direction of motion. Because m is positive, the data energy controls the mean.

Together, the energy controls the full H one norm, with a constant c zero that depends on the domain, alpha, the bound on the data matrix, and m. Lax–Milgram then gives the unique minimizer.

The constant c zero shrinks as m shrinks: nearly parallel edges weaken control of the drift. The stripe scene of slide 21 comes close to that case.

[If asked for the derivations: the full proof is Backup 09, page 34. If asked about stability: the same c zero gives a Lipschitz bound in B and b, Backup 10, page 35. If asked what happens when the integrated matrix is singular: a constant velocity direction is uncontrolled, Backup 35. If asked whether the flow is stable with respect to the raw events: Backup 36, page 61. If asked whether one mechanism runs through all three uniqueness results: Backup 64, page 89.]

Stereo on MVSEC: competitive accuracy, more scored points

At each candidate disparity d, solve two softly coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left(\tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

Mathematical result (fixed d, frozen positive weights): Hessian $\succeq \lambda_{\min}^2 \mathbf{I}$; unique inner solve.
Benchmark: the full WLS method, with weight updates, spatial cost aggregation, and candidate selection.

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
Soft-coupled WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20
Xie adaptation [105]	93.43	78.60	93.66	66.57	59.36	1.39
Kogler et al. [104]	123.39	99.50	125.27	78.23	80.94	0.66
Rogister et al. [103]	142.17	128.20	153.32	114.80	90.32	1.01

MVSEC indoor flying 1–4, raw outputs. Each method is scored on its own estimates with ground truth in $[0.5, 8]$ m.

Takeaway. WLS: best on Flying 2 and log RMSE, within 6 cm elsewhere, over twice the points.

Dissertation: Theorem 4.6.3; Table A.5, Table A.4; MVSEC [35]. Xie and Piatkowska rows are local adaptations.

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Stereo on MVSEC: competitive accuracy, more scored points

Likely question: How does the stereo method compare with all the methods on MVSEC?

Answer

[Point to the energy first, then to the WLS row and the Piatkowska row.]

At each candidate disparity, the model solves for one temporal profile per view. Each profile fits its own events, as in the temporal chapter, and a soft penalty ties the two profiles together.

[Point to the two labels.]

The mathematical result is for a fixed disparity with the weights frozen. The coupling term is nonnegative, so the temporal coercivity proof carries over, and the inner problem has a unique solution. The benchmark runs the full method: weight updates, cost aggregation, and candidate selection.

On MVSEC, here's how that performs on the indoor flying sequences, against the dataset's lidar-based ground-truth depth. The first four columns are mean depth error in centimeters, one per sequence. Then log RMSE, and the number of scored points.

Our weighted least-squares method, WLS, has the lowest error on indoor flying two and the lowest log RMSE. On the other three sequences, the local adaptation of Piatkowska and colleagues' method is lower, by between one and a half and six centimeters. WLS scores two point six six million points, more than twice the one point two million for that method. It also leads the three other prior event-based methods on every sequence.

Each method is scored on its own valid pixels, so the comparison reflects both accuracy and coverage.

[If asked why the inner solve is unique: Backup 51, page 76. If asked whether a unique solve guarantees the right disparity: Backup 54, page 79; the conditions that do are Backup 55, page 80, and Backup 56, page 81; for noise tolerance, Backup 19, page 44. If asked whether IRLS converges: Backup 59, page 84. If asked whether it is maximum likelihood or majorization–minimization: Backup 21, page 46, and Backup 28, page 53. If asked whether the comparison is fair: Backup 60, page 85, and Backup 23, page 48. If asked what supports the reciprocal weights: Backup 61, page 86. If asked what increasing the coupling does: Backup 52, page 77. If asked about hard fusion: Backup 62, page 87. For exactly which competitors beat the ideal match: Backup 18, page 43. For the IRLS contraction constant and what one IRLS call returns: Backup 20, page 45, and Backup 22, page 47. If asked about the dataset's own ground-truth limits: Backup 60, page 85.]

From event changes to well-posed estimators

Temporal: control an unobserved baseline.

Spatial: combine directional constraints globally.

Binocular: separate inner uniqueness from correspondence.

The common mathematical question is:

Which directions do the observations control, and which are selected by the model?

Thank you. Discussion and additional proofs follow.

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From event changes to well-posed estimators

[Closing summary; use it if the discussion returns to the common question.]

I began by asking what we can recover from changes alone.

Temporal reconstruction shows how a penalty selects the baseline. Flow shows how directional observations and spatial coupling together determine a motion field. Stereo shows that a unique inner solution and a reliable match need separate arguments.

Across all three, the common question is which directions the observations control, and which the model chooses.

[For questions: the discussion guide is Backup 01, page 26; flow detail starts at Backup 41, page 66; the stereo arc at Backup 47, page 72; references at page 109.]

How big is the temporal monocular problem?

Unknowns: HW independent pixel histories, each with N_t samples.

$$\underbrace{(\mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda})}_{\text{SPD, tridiagonal}} \mathbf{f}_{xy} = \mathbf{A}^\top \mathbf{y}_{xy}, \qquad \boxed{\text{solve work } O(HWN_t)}.$$

Work and storage

- Weights, right-hand side, and solve: $O(N_t)$ per pixel.
- Full output: $O(HWN_t)$ storage; solver scratch: $O(N_t)$ per pixel processed at once.
- E events, at most k bins/event: $O(Ek + HWN_t)$ with direct bin indexing.

For a low-power device

- Newest endpoint only: $O(HW)$ work per interval and $O(HW)$ state.
- Bin directly; skip histories with $\mathbf{y}_{xy} = 0$; reuse buffers.
- ROI, coarser sampling, and lower precision trade coverage, detail, or numerical accuracy for cost.

Scale example: $260 \times 346 \times 50 \approx 4.50$ million unknowns; float32 output alone ≈ 18 MB.

Dissertation: Eq. 2.7, Eq. 2.9; edge solver. Uniform bins, fixed kernel order; streaming uses a growing prefix.

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How big is the temporal monocular problem?

Likely question: How many unknowns are there, what does the solve cost, and what would you change for low-power use?

Answer

There are H times W times N_t unknowns, split into independent temporal histories. Each pixel has a positive-definite tridiagonal system. Weights, right-hand side, and solve each take O(N_t), giving O(H W N_t) total work.

With E events and at most k bins per event, direct indexing on a uniform grid gives O(E k + H W N_t) including accumulation, for fixed kernel order. Delta binning has k equal to one. The edge solver uses this structure; the main Python path uses cubic dense solves.

A 346 by 260 sensor with fifty samples has about four and a half million unknowns: eighteen megabytes of float32 output alone. Newest-endpoint streaming takes O(H W) state and work per interval. It matches the growing-prefix batch endpoint for finalized inputs; rolling windows change the boundary problem.

If asked for more

Preserve the estimator. Bin directly, reuse buffers, and skip entirely zero observation histories. Zero net sum can hide changes. Reuse factors across identical weight profiles; cache weights for repeated integer counts. Quiet pixels with earlier activity still need state updates.

Trade resources deliberately. ROI reduces coverage; coarser sampling reduces detail; shorter windows change history. Symmetric kernels add look-ahead latency. Float32, quantized weights, and fixed-point arithmetic require error checks, especially at weak regularization. Tune thread count for joules per estimate under the latency deadline. Measure peak memory too.

For detection, event counts or a leaky activity state may suffice. Asynchronous high-pass or complementary filters offer another compact estimator, including an events-only option. Their objective and baseline behavior differ from this model: benchmark task accuracy alongside cost.

Sources: Chapter 2 normal equations; edge implementation and README; Scheerlinck et al., Continuous-time Intensity Estimation Using Event Cameras (ACCV 2018). CPU timings: Backup 07, page 32.

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